Human Thought

Its Forms and Its Tasks

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Foreword

The life of thought, which it is the task of my book to describe, is intimately interwoven with the other sides of the life of the spirit. I have sought to show that thought cannot serve the life of the spirit as a whole unless it first and foremost becomes master in its own domain and unfolds itself according to its own laws as an independent side of the life of the spirit. That this unfolding, when it is rightly carried through, will bring with it great and far-reaching changes in personal life as a whole, is something of which I become more and more convinced. There are here oppositions and problems that will take on ever sharper forms, and I feel no admiration for those who here sense no sting, although they have sufficient opportunity to come to know the life of thought.

Just as I have on the whole stressed the personal factor in all thinking, the more this approaches the border-regions, so too I have not concealed that my own stance at thought's final station has a personal character, and I have not been able to refrain from stating how, in my experience, one feels at this station.

I still have much to learn. My work will need filling out and correction on many points. But beyond the framework I have given here, I cannot go. And it affords me a good place to work. May there be one or another who could also find work here, even if this work should lead him to alter both the framework and its content.

August 1910. *Harald Høffding.*

I. The Psychology of Thought (Functions)

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A. Psychic Energy

1. The task I have set myself in this book is to investigate human thought—a task that all philosophers have set themselves, whether they have been conscious of it or not. For all questions that can be raised concerning life and existence depend—both with respect to the way in which they are posed and with respect to the way in which they are answered—on the conception one has of the nature and operation of thought. It is therefore also an outline of my philosophy that I shall give here. That is to say: I shall give an account of the thoughts to which I have come back again and again, when I have occupied myself with such questions as: the presuppositions of science, the significance of the results of science for one's view of life, and the relation between scientific cognition and other sides of the human life of the spirit.

2. We know thought best as reflection. The word *Eftertanke*—"after-thought"—I use not only in its strict sense, "in one's thinking to *seek after* something or *pursue* something," but also in the sense that thought *comes after* an immediate experience. Reflection thus presupposes that we have seen, felt, experienced something. During the experience we were perhaps wholly absorbed in it; but now we seek to bring it forth anew, either to dwell once more on what was experienced, or to go through its various parts or properties or its relations to other experiences. In mere dwelling, reflection is one with a beholding, a mere representing. But such dwelling can scarcely persist for long in a wholly quiescent form. Involuntarily—perhaps precisely because of the interest that what is experienced holds—a movement of thought will be released, one that serves to bring out its inner and outer relations. Thus when the poet involuntarily reaches for similes and images in order really to let his subject come to expression. Or when perhaps

it is precisely the contrast with other experiences that serves to illuminate the peculiar character of the single experience. Both likeness and difference can thus serve to cast a clearer light on what was given in the original experience, and reflection therefore does not necessarily stand in opposition to it, but can work in its service and in its spirit, when it has itself vanished.

By the word experience we think chiefly of states in which we were essentially receptive. But reflection can also serve the involuntary life of the spirit, where this appears as urge or striving. In an instinctive or inspired manner, involuntary striving is often led to its goal without any reflection being needed, and perhaps better than could happen with the help of any reflection. When reflection comes to operate in our action, it will be because the involuntary striving met resistance. There may then be needed a deliberation upon what it really is that is being striven for, upon this goal's relation to other possible or actual endeavors, and upon the ways and means that can lead to the goal. Here too it is through the bringing-out of likenesses and differences that reflection renders its service. And here too a difference can appear between dwelling and movement. It may be that the representation of the goal is held fast without great oscillations, and that the relation to ways and means or to other possibilities plays a subordinate role, as when the helmsman stares at the compass or the seamark, and his hand involuntarily moves the rudder according to what he sees. But it may also be that the strongest movement back and forth among various possibilities takes place and must take place before the decisive concentration that is called decision sets in.

The arising of reflection is like a kind of awakening, not from sleep or from unconsciousness, but from the involuntary mental life and its experiences, from states that may provisionally be characterized as sensation, intuiting, representing, mood, emotion, urge, and striving. In its dwelling form, reflection stands closest to the involuntary mental life. In its striving to bring out likenesses and differences, it moves further away from the involuntary. Linguistic usage marks this difference rather aptly by distinguishing between *thinking of* something and *thinking over* something. It is the latter kind of reflection, the discursive, mobile kind, that we shall have most use for in what follows, since it is the one we can best observe, and likewise the one that plays the decisive role in the tasks of thought. Only for this kind of reflection do problems exist. In what follows I therefore understand by thinking this kind of reflection. And yet there is only a difference of degree between the two kinds. Only for a brief moment is a dwelling representing, a holding-fast of a memory-image, possible; precise observation shows that in greater or smaller rhythms we move toward and away from the object we are thinking of. And every return can bring with it a recognition, so that the relation of likeness comes to play a role just as in discursive reflection. Perhaps the representation was new to us at the first moment; but through repeated dwelling it becomes familiar. Apart from this too, dwelling reflection by no means excludes that manifold other representations and experiences can contribute and perhaps precisely condition the interest that makes continued holding-fast of a single representation possible. What holds for discursive reflection will therefore also hold, to a certain degree, for the dwelling kind.

It emerges from this description that the work of thought consists in comparison, in distinguishing or finding likenesses. When I think over something, I seek to determine its relation to something else, and the more relations of likeness and difference I find, the better I have thought it through. This holds of course also when it is one and the same object that I examine in different connections and under different conditions; I then compare the different ways in which the object appears in the different cases.

But comparison in turn presupposes a holding-together or combining, synthesis. I can compare two objects only when I either have them given to me simultaneously in my sensation, my memory, or my imagination, or else retain a representation of the one while the other is present to sensory observation. If the one object constantly vanished entirely when the other emerged, no comparison could take place. The work of thought consists in a combining by which likenesses and differences become determinative for the relations in which the objects are placed to one another in the connection that results from the work. Synthesis is an expression of energy, of the capacity to perform work.

The resistance that is overcome through this work is due to the objects against which reflection is directed and which are given in the involuntary mental life—in sense-sensation, memory or imagination, feelings of pleasure and displeasure, urge and striving.

The combining work must become the more difficult, the greater the multiplicity of objects that are to be brought into relation with one another. The fullness of content can be so great that the psychic energy available in the particular case cannot master it. There then threatens a dissolution of the mental life, a transition to chaos.

Yet it depends not only on the multiplicity, but also on the degree of difference that the objects present. The more they resemble one another, so that they can fuse, or so that at least the work done in the case of the one need not be repeated in the case of the other, the less work is required. Objects that stand in strong mutual opposition, or perhaps even conflict, require a great energy if they are all to be taken up into the conscious connection of the mental life. This holds both for the inquirer, who must find a point of view from which the conflicting can stand as one, and for the practically striving person, who must subordinate the conflicting impulses and interests to the thought of a great and guiding purpose.

It will, thirdly, depend on how independent a place the individual objects are to occupy in the connection that is being worked on. A schematic whole is easier to form than a whole that lets the peculiarities of the individual objects each come into their own. Here it is precisely likeness that can create a difficulty, since it easily leads to fusion and thereby to the abolition of the independence of individual objects. On the other hand, from this point of view a pronounced difference or opposition can be a relief.

Furthermore, a greater work is required the further apart the objects lie from one another in time and space. The simultaneous and the near join together involuntarily, and the connection perhaps even forces itself upon us. And if we leave the objects before us in order to immerse ourselves in the world of memory and imagination, here too the connection can often form itself with ease; "with great ease thoughts take up their abode with one another," but when it is a matter of holding together a present object with distant memories, or with ideals that have been properly to the fore in the soul only in isolated moments of enthusiasm, then special demands are made on psychic energy. To have, in the situation of the moment, the treasures of one's experiences, one's reflection, and one's ideality fully at one's disposal is perhaps the greatest testimony to mental energy.

Finally, it plays a role how close and intimate a connection the objects, despite their difference, are to enter into. There are here many degrees, from wholly external juxtaposition and grouping to the inner connection that a scientific concept, an artistically formed image, or a great decision can express.

Besides these five conditions—which might be said to form an analogy to what mass is in the case of physical energy—account must also be taken of the speed with which the synthesis proceeds. The genius and the man of will can in an instant expend a capital of energy that other people must distribute over a longer span of time. In the domain of the life of feeling, this opposition appears as an opposition between vehemence and inwardness, emotion and disposition[1], and analogous oppositions hold for the other sides of the mental life. In research, discovery often forms an analogy to emotion, the conduct of proof to disposition.

The psychic energy that expresses itself in synthesis we measure by taking all these circumstances into account. Of exact measurement there is of course no question here,

[1] See on this my *Psykologi* IV, 7; VI E, 4–5.

only of an evaluative estimate. But as soon as we seek to give grounds for why one work of thought, one intellectual orientation, one view of life can be called lower or higher than another, we must take all these circumstances into account, and we in fact do so, more or less consciously, and more or less precisely. There is less psychic energy available in the dream-state, where the changing objects each release their own train of representations, than in the waking state, where dominant purposive thoughts bring about a main direction. It is a purely psychological measure that we apply here. And even though, in forming the concept of psychic energy, we have had guidance in the analogy with the concept of physical energy, that concept is nevertheless built upon its own peculiar experiences, in which physical or physiological observations and investigations need play no decisive role. For this reason we need not fail to appreciate the significance of the attempts that have been made, from the experimental-psychological and physiological side, to measure the brain-work presupposed in certain elementary mental activities—e.g. calculating or remembering—through its relation to the muscular work that can be performed simultaneously with it. One can detect such thought-work by the diminution it produces in the muscular work performed at the same time. But this very measurement presupposes that one can, by purely psychological means, distinguish between more difficult and easier mental work; otherwise one moves in a circle. The brain-work corresponding to the mental work described above is, after all, not known directly, but only through the very mental work to which it is assumed to correspond.

3. Reflection presupposes objects on which it can exercise its combining and comparing activity. But now these objects? Does the law of synthesis hold for them as well? We stand here before a question of a peculiar difficulty, closely bound up with the whole nature of the mental life. It is only reflection that acquaints us with the objects. It is the medium through which they are accessible to us. The involuntary mental life in its immediate unfolding must not be described and judged straightway according to the way in which it shows itself to reflection. It is a life within which we are not permitted to presuppose properties and determinations that first come into being through the work of reflection. When, in isolated moments, we are able to immerse ourselves in this involuntary—but not for that reason unconscious—life in such a way that we can afterwards retain a memory of it, it stands before us as a stream, a gliding-along of manifold objects, which we designate as sensations, memories, moods, endeavors, and so on. It is not a chaos of independent objects; already in the involuntary mental life there are connections and wholes as well as differences and oppositions. Every word we use about what we can retain from the involuntary states bears witness to this. Most

striking, no doubt, is the continuity; this is why the word "stream" offers itself so readily for what we call the immediately given. But it is not a uniform stream. Differences of quality make themselves felt; otherwise we could not distinguish between the kinds of objects called above sensations, memories, and so on. And there can be eddies in the stream, or windings, so that the direction is not kept in a straight line. That is to say, strivings of different tendency, of more or less opposite direction, can make themselves felt. With this is bound up, again, that inhibitions can set in, while in other cases, where the different tendencies go in the same direction, a facilitation or an acceleration can appear.

Through the indefinite and figurative expressions that alone stand at our disposal for indicating the basis of objects that reflection presupposes, it shines through that this basis forms no absolute opposition to what emerges as a consequence of the work of reflection. There too we find both unity and multiplicity, both continuity and discontinuity, and we find many degrees and nuances of these relations. It is therefore not justified, with the older English school (*Locke*, *Hume*, *Mill*), to conceive the immediately given in the mental life in an atomistic way, as if it consisted in a multiplicity of mutually independent elements. Both in the psychic and in the physical domain, it is only reflection that leads to the formation of atomic concepts, guided by the urge to make the differences as distinct as possible, so that the connection of them can become as clear and perspicuous as possible. The unity to which reflection bears witness, and which it itself seeks to bring about, thereby becomes enigmatic. But it is also unjustified, as in our day especially *William James*[2] and *Henri Bergson*[3] have attempted, to present that involuntary life as standing in the sharpest opposition to reflection and its work. It then becomes, in the last analysis, the indescribable and inconceivable, and it would become an insoluble enigma how there can be any knowledge at all of that "immediately given" from which reflection is nonetheless constantly to draw. James and Bergson represent a significant reaction against intellectualism's misunderstanding and overvaluation of the significance of reflection, a reaction that in several respects recalls the stand taken by *Hamann*, *Herder*, and *Jacobi* against *Kant*. Not pure reason, but the immediate stream of the life of the spirit is, they maintain, the highest, and brings truth forth. The distinctions and definitions of thought are necessary in order that we may practically find our way in life. And analyzing science has carried the opposition to the immediately given still further—an

[2] *The Stream of Thought*. Mind 1884 (reprinted as Chap. 9 of *Principles of Psychology*). Later especially *A pluralistic universe* (1909), Sect. 7.

[3] *Les données immédiates de la conscience* (1888). Later especially in *Introduction à la Métaphysique*. (Revue de Métaphysique et de Morale 1903).

opposition that language itself already favors, by forming independent words for the individual objects and by taking spatial relations as the basis for our intuitive images. It is as though a Fall had taken place when reflection awoke and turned, inquiringly, toward the involuntary life from which it had itself sprung.

Without a kinship between the involuntary mental life and reflection, it would be impossible to understand how the latter could arise. But in the very descriptions that James and Bergson give of what, for them, in sharp opposition to all elaboration by thought, is the immediately given—descriptions that are masterpieces of descriptive psychology—it is nonetheless expressly stated that in this "given" there are differences and connections, qualities and continuities. James emphasizes especially the constant transitions (continuous transitions) and the difficulty of distinguishing between relation and term (relation and matter related); Bergson stresses especially the undivided and yet heterogeneous nature of the inner stream (durée hétérogène et indistincte; durée dont les moments se pénètrent). But with this an inner possibility for the arising of reflection has been indicated. The whole weight must not be placed on the external causes (practical need, the mechanics of language, and the technique of science). The motive for the awakening of reflection is given with the opposition between continuity and qualitative multiplicity, between the connection and the individual elements. Herein lies the possibility of an inhibition of the involuntary stream, an inhibition that only discursive reflection can remedy, through the clarity and definiteness it is able to bring. As a combining that is accomplished through comparison, reflection can bring about clarity concerning the relation between continuity and discontinuity, unity and multiplicity. And it is not anything wholly foreign that it brings in by its work; for already in the immediately given there appear, after all, likenesses and differences, connection and multiplicity—and likewise, what is especially significant, an involuntary striving to bring about harmony among the different tendencies. If there were not such an original striving, it would be unjustified to speak of a stream, of a continuité mouvante, a unité de direction[4]! There shows itself here, then, an analogy to what, in the case of reflection, we called synthesis—a tendency to gather and unite. And reflection awakens when there is a special urge for a fuller carrying-through of this tendency than is possible within the involuntary mental life. Already within this—not only after reflection has

[4]One of my young listeners wrote to me some years ago: "I can declare myself free of every purpose, as of every interest or belief in anything outside or inside me. I do indeed feel something willing, something pressing forward, but it lives, as it were, its own life, pushes aside everything that gets in its way, and surely never commits any act that is not to its own advantage."—Here reflection has brought no result, only dissolution; but the involuntary life has maintained itself in its unity and in its definite direction, which was all the more striking to the young psychologist the less his reflection could attain to any firm standpoint.

awakened—crises and catastrophes can arise, and chaos can threaten. A work can be required even before reflection can awaken; but the difficulties arise, are worked on, and are resolved on different terms in the involuntary mental life than where reflection has taken the helm. Perhaps some of the most significant things in our mental life take place in the regions below the domain of reflection, and it is often only the after-effects of these deep-lying processes that reflection has to deal with. It will always be impossible to demonstrate that at any point we stand before an immediately given in the absolute sense, that is, one in which absolutely no psychic work has been done. What we in each particular case call immediately given deserves this name only in relation to a certain definite psychic work, while in comparison with other objects it would perhaps have to be regarded as the result of a work.

4. By starting from reflection and seeking to become clear about its way of operating, we arrived at the concept of synthesis, and we then found that this concept could also be applied to the involuntary mental life, which forms the constant basis from which reflection springs, in order thereupon to make that basis itself an object of its activity. We must make ourselves familiar with the fact that the involuntary mental life, as it unfolds prior to the awakening of reflection, can be illuminated and designated only by way of analogy. We apprehend it only in analogy with clear reflection, and the expressions of language are throughout drawn from the world of reflection. So far is that involuntary mental life from being the thing easiest to apprehend, that it takes precisely great art to get a proper view of it in its difference from the mental life shaped by reflection. What one calls sound common sense operates, in its psychology, with concepts drawn from the world of reflection. A childlike tendency makes itself felt to regard everything in the mental life as the fruit of clearly conscious thinking, of deliberation and calculation. One personifies, so to speak, not only outer but also inner nature, placing a person behind the person. The childlike psychology explains personal life itself, in its involuntary unfolding, by supposing that behind it there is a person who consciously thinks whatever stirs in it—just as in older works on optics one can find depicted a man inside the head, looking at the image that the light-rays form on the retina!

Reflection is required in order to discover and apprehend the involuntary mental life, and yet the peculiarity of the latter consists in the fact that reflection plays no role in it. This difficulty can be resolved only by our apprehending and expressing the involuntary—insofar as any memory of it persists after the awakening of reflection—in analogy with what we know clearly from the world of reflection itself. This is what we do when we say that synthesis is the fundamental form, not only for comparing reflection, but also for the involuntary mental life, and is thus the most comprehensive psychological concept.—

But now a new difficulty arises. Synthesis means combining—but there must then be something that is combined, and this something must be a multiplicity. The objects toward which reflection directs itself in order to hold them together and compare them form a coherent multiplicity; each individual object comes into being for us through an act of distinguishing. And in a single object we can again distinguish individual sides, properties, or parts, which we may call elements. The result seems then to be that we end in an opposition between synthesis and elements, and that the multiplicity of elements is the constant presupposition for synthesis being able to operate! From this one might then draw the conclusion that what is original in the mental life was a multiplicity of elements, and that only at a later point of development did the combining activity intervene. This conception was also asserted in its day by John Locke, and it still often haunts psychology. It relies on the analogy with the formation of a mechanical whole out of elements that could be pointed out beforehand each by itself, as when a heap of sand is formed from grains of sand that previously lay each in its own place and are only now brought together. The question becomes whether this analogy is justified.

When reflection awakens, the mental life does not, however, begin with it; for then it would have no objects. The elements—that is, the parts, sides, or properties that reflection can distinguish in the object or objects, and that it can determine through their mutual likenesses and differences—are, after all, precisely the fruit of a work of thought. They are drawn out of the connection in which they involuntarily occurred, and set in a certain opposition to one another—an opposition without which they could not become objects of clear consciousness. And the characterization that each individual element, and thereby each individual object, receives is conditioned by its relation to other elements or objects. At no point is it justified to assume isolated elements in the mental life, each absolutely independent by itself—a kind of absolute psychic atoms. That is to confuse the effect of reflection with reflection's own basis—at bottom, again an example of the naive psychology that tends, after all, to confuse differences that thought first makes with originally given differences.

And insofar as we succeed in retaining a memory of the state we are in when involuntariness prevails, this state appears—and here the masters of descriptive psychology are undoubtedly right—not as a chaos of independent parts, but as a stream, a pulsation, yet not without differences, but such that there is a connection even between the lowest troughs and the highest crests of the waves. We find, as far as can be judged, unity and multiplicity so interwoven with each other that they cannot be separated in an external manner. It was, after all, precisely in the urge to gain greater clarity about their relation that we found the possibility and the motive for the awakening of reflection.

At no element can we, in the domain of the mental life (any more than in that of matter), come to a stop as absolutely isolated or absolutely simple. There constantly poses itself, for reflection, the task of finding a connection between the elements or objects that might seem to be isolated, and of finding an inner multiplicity in the elements or objects that appear as simple. It is a sign of the dullness of reflection when it comes to a stop too early in either of these respects. In other words: the law of synthesis must always be carried through anew with respect to the objects or elements themselves.

On the other hand, neither can there be any absolutely conclusive synthesis. As long as the mental life lasts, new objects are taken up into the involuntary stream and are involuntarily worked into it. They are new drops, or perhaps new brooks, which are partly reshaped by the stream and partly can also gain influence on its character and its course. To this is added the character of the new riverbeds that the stream must form for itself when the terrain changes. And as long as reflection is awake, it will constantly be able to set itself the task of bringing about a greater, more comprehensive, and more intimate synthesis than before. There is always a psychic work to be done. The synthesis that has hitherto enclosed the multiplicity of the mental life can again come to stand as object or element for a later synthesis. As soon as we become conscious of our preceding mental states and activities, such a transition to a new synthesis takes place, whether or not this signifies a progress, a more valuable state. It is a new I that becomes conscious of our earlier I's. No synthesis is definitive. Even the philosophy that spoke most of higher unity always fell into embarrassment when it had to speak of the highest unity.

It can of course be entirely justified, in a special connection, with a view to a particular investigation, to disregard these difficulties of principle and to regard elements or objects as if they were absolute and simple. Every special scientific investigation presupposes a certain abstraction. If, e.g., experimental psychology really needs to operate with a kind of psychic atoms or flashes as independent and simple, this can be just as justified as when, in natural science, one operates with material atoms. But one must avoid the dogmatism that confuses concepts we have arranged for the needs of an investigation with ultimate realities. And likewise it can be justified that in everyday speech, or in a practical—e.g. ethical or juridical—respect, we regard the individual as an absolute unity with a completed synthesis, without regard to the fact that synthesis at bottom signifies a work, an activity, not something given once and for all. There is an irrational relation between the mental life and what we call its elements. The point now is to find as many elements as possible; but it can be justified and necessary to confine oneself to a few decimal places, provided only that one does not forget that the question is then not exhausted. The irrational is here, as everywhere, both a barrier and a task.

There expresses itself here an antinomy that is closely bound up with the essence of the mental life and is peculiar to the concept of personality. Mental life and personality cannot be conceived as products of previously given elements, since the elements we know always have their properties precisely by being members of the mental life. We here come to move in a circle, unless we are clear that an inexhaustible reciprocal action prevails: the parts are determined by the whole, and the whole by the parts. In each particular case, in each particular psychological consideration, we must investigate how far the synthesis, the formation of the whole, has advanced, and on this basis observe how the emergence of a new object operates. But the very distinction between synthesis and object is always more or less artificial in relation to the constant interplay of meeting and parting, of continuity and discontinuity, of wholeness and dissolution. There is here a mental rhythmics that forms an analogy to—or rather a prototype for—the rhythms that characterize the course of organic life, a rhythmics that by no means begins with reflection, but, by all we can judge, already makes itself felt in the involuntary mental life. Indeed, this rhythmics recurs in the very relation between reflection and the involuntary course of the mental life. The difficulty of understanding the origin of the mental life rests precisely on the fact that it cannot be explained as a product of previously given elements, but from its very first moment moves in a rhythm—and that in a rhythm of the peculiar kind that none of the members in the rhythm can have absolute priority over the other.

5. In the history of the concept of synthesis this antinomy has been a stumbling-block. The concept of synthesis itself is an old idea, traceable in Plato and Aristotle, and especially in Plotinus in antiquity, and which from Leibniz's time on has always, with more or less energy, been asserted in psychology and in philosophy generally[5].

Hume cut the knot by dissolving the mental life into absolutely independent elements, and consistently he therefore found an insoluble enigma in all unity and connection in consciousness. The principle of unity (the uniting principle) was for him the unintelligible. *Kant* started from the opposite point of view, that it is precisely the unconnected, the

[5]On the history of the concept of synthesis in more recent times, see *Georges Dwelshauvers*: *La synthèse mentale* (Paris 1908), pp. 179–223 (an account of the concept of synthesis in Leibniz, Kant, Wundt, Høffding, and Pierre Janet).—In *S. Kierkegaard* this concept appears in an energetic and profound manner, especially in the measure he applies in his doctrine of the stages. See my book *Danske Filosofer*, pp. 153 f.; p. 165 f.

isolated and mutually independent, that is unintelligible to us, both in psychology and wherever else we meet it. But he still held fast to Locke's presupposition that, as the basis for thought's combining and ordering activity, there was given a multiplicity that was only now to be connected. Thereby he arrived at his fateful distinction between the form of cognition and its matter. This too was to cut the knot. And yet Kant did not lack the insight that already in the involuntary mental life there operate forms analogous to those that clear reflection employs. Synthesis is, he says, the effect of a blind but indispensable function of the soul, without which we would have no cognition at all, but of which we only rarely become conscious. It is the task of thought to form the synthesis into a concept. And this same blind function of the soul (Kant calls it imagination) is likewise what conditions the connection between the two extremes of our cognition, "sensation" and "understanding." It was surely only Kant's reluctance to give psychological considerations too large a place in his investigation that held him back from giving a fuller characterization of the involuntary course of the mental life. J. F. *Fries* developed Kant's doctrine further from this side and is even regarded by his disciples as the real discoverer of the involuntary basis[6]. But he then believed that in this basis—the spontaneous striving, the immediate cognition—he possessed a rule of truth exalted above all error. Unfortunately this rule is ineffable, since it can come to consciousness only in an imperfect and fragmentary way through reflection! A dogmatism comes to the fore in the firm belief in the validity of the involuntary mental life—a dogmatism that signifies a decided step backward in relation to Kant. Already when Fries assumes that "spontaneity" is not only continuous but also constant, he says more than any reflection could demonstrate. The irrational relation between immediacy and reflection is not strictly maintained.

This relation is, as already mentioned, very sharply emphasized by *Henri Bergson*. This thinker is not satisfied with the concept of synthesis, because it always presupposes an opposition between unity and multiplicity. Is it not an opposition—he asks—that recurs, whatever one thinks of or speaks about? What matters is to say *which* kind of unity and *which* kind of multiplicity we have in the mental life. The streaming life of the personality cannot be defined by means of concepts that always stand in a certain relation of opposition to one another. One cannot, through thesis and antithesis, arrive at a real unity. One must, conversely, immerse oneself in the immediate stream of the mental life in order to see the opposition emerge from it. Immediate beholding (intuition)

[6]Against this, see my review in *Göttinger Gelehrte Anzeiger* (1907) of "Abhandlungen der Frieseschen Schule. Neue Folge."

must take the place of reflection[7].

Bergson's standpoint recalls, viewed from one side, that of Fries, except that he gives us, with far greater mastery, an understanding of what is meant by "spontaneity." But whereas Fries and his disciples held that true cognition is attained by translating, as well as we can, that spontaneous basis into the forms of reflection, Bergson demands that we subdue reflection and, by an act of will, dive down again into the immediate, feeling ourselves one with it. It stands here as an enigma how the validity of a cognition can be demonstrated without reflection. One of the special problems of thought will later give us occasion to bring out this difficult point in the philosophy of the brilliant French thinker.

In response to the remark that the concepts we use in our reflection to characterize the mental life—synthesis, unity, multiplicity, and the like—find application to all objects, I will maintain that the reason why that group of concepts constantly finds, and must find, application to all objects is to be sought precisely in the fact that they express the forms in which the mental life, and cognition in particular, unfolds itself. Only through the application of these forms does the appropriation, the taking-up into the connection of our own being, in which all understanding, all cognition consists, become possible. No wonder that all products of cognition bear their stamp! Our cognition consists in a conscious or unconscious analogy with our own being.—

The relation of opposition that has been discussed in the foregoing under various expressions (object and reflection, matter and form, spontaneity and reflection, intuition and analysis), *Stumpf* designates as a relation of opposition between phenomena and psychic functions[8]. With respect to this relation he advances, with all possible reservation, the hypothesis that psychic phenomena are not altered because they are analyzed. When we become aware of something in an observation that we did not notice at the first moment, this is supposed to bring about no change in our observation itself, but only in the degree of explicit consciousness we apply to it. A sound rings the same whether or not I notice individual tones in it; a dish makes the same impression whether I can distinguish something sour or sweet in its taste, or whether the taste acts as a unified whole. In my judgment it is, from the outset, not probable that this is how matters stand. The differentiated and differentiating attention in which analysis consists must alter the object for us. The fact that we have discovered a spot in an otherwise beautiful picture—a spot we did not notice at the first moment—can give the whole picture a

[7] *Introduction à la Métaphysique.* (Revue de Métaphysique et de Morale 1903, pp. 15–17.)

[8] *Erscheinungen und psychische Funktionen.* (Abhdl. der preussischen Gesellschaft der Wiss. 1907), pp. 15–21.

different character. The timbre changes; an element has, quite simply, been added that was not present in the first glance. And the relation between the individual parts is in each case a different one after the analysis than before. It can cost an exertion of will to return to the immediate beholding in which the particulars have not yet become the object of special apprehension. There is here a difference between experience and reflection that can scarcely be effaced. There are of course many degrees, as we shall come to see in what follows, in the investigation of the relation between intuiting and judging. And individual differences certainly also make themselves felt. In elastic natures the transition from experience to reflection, and vice versa, will take place relatively easily; and yet I should think that even in them a certain contrast-effect will make itself felt between the two states.

Stumpf, for that matter, will only maintain the possibility of a differentiated subsistence, during the experience, of what is first differentiated with consciousness during reflection. And he concedes that there is probably no absolute independence in the relation between phenomenon and function "under the usual, complicated circumstances of the psychic process." Whether experimental investigations can, as he surmises, lead to any result here is doubtful. It is not easy to see how the difference between immediate experience and reflection could fall away. And in a psychological experiment it is difficult to obtain a wholly immediate experience.

Chemists often say that the atoms of the elements are present in compound substances—so that, e.g., carbon and oxygen are really present in carbonic acid. And Stumpf holds that the psychologist must have the same right in his domain. I think, however, that there is a difference. For the chemist can demonstrate that there are just as many parts by weight in carbonic acid as in carbon and oxygen each by itself; and at bottom that is all he means by that manner of speaking. Such an equivalence the psychologist is, for good reasons, debarred from demonstrating.—For that matter, a chemist such as *Sir William Ramsay* has expressly denied the right to take that manner of speaking literally, and *Ernst Mach* has expressed himself in the same direction. The analogy is therefore doubtful.—

We do not get beyond an antinomy between experience and reflection, and we may not, without further ado, transfer to the one what holds for the other. Experience itself we may think of neither as absolute continuity nor as a chaos of differences. The continuous and the discontinuous occur in it in all possible degrees, and it is only reflection that takes the opposition between continuity and discontinuity sharply into view, in that it—as will later be shown—receives the double task of finding differences where continuity was immediately given, and intermediate links where discontinuity was immediately given.

6. In the concept of synthesis there lies, as we have seen, an opposition between object and form, which gives rise to great difficulties that become insuperable when by object one understands an absolutely unformed matter. As already indicated, every object will pose a double task for combining thought: to determine its relation to other objects, and to find out differences and relations within the object itself. Hereby arises the constant striving to make the discontinuous continuous and the continuous discontinuous.

But when we do not separate thought from the other sides of the mental life, there is yet a third element, besides form and object, that we must bring out. When a task is posed, there is presupposed an urge to have the task solved. The work on the solution then acquires value, since we designate as valuable everything that satisfies an urge, whether this urge expressly announces itself, or whether it is only from the satisfaction we feel at the work or its result that we infer that there has been a lack that has now been filled.

Thought has biological significance, in that attentiveness to changes in the surrounding world makes it possible for dangers to be averted and advantages gained. But here too the significance of synthesis shows itself, since it is only under very elementary conditions that attentiveness to a single change is sufficient; it will be of importance that the single experience be brought into connection and reciprocal action with other, present or earlier experiences; only then can the reaction to what is experienced become entirely appropriate. Only instinct reacts to an elementary stimulus; when reflection is awakened, a more serious orientation, and thereby a richer relation to surrounding existence, is possible. An interesting example of a synthesis that is perhaps of a purely physiological kind is afforded by what *Pavlov*[9] has called conditioned reflexes. Salivation can be produced not only by direct stimulation with a substance brought into the mouth, but also by stimulation of sight, hearing, smell, or the skin-sense, on the condition that these sense-stimulations have not occurred too long beforehand together with the action of that substance. Here an associative synthesis is at work, which makes it possible for one kind of stimulation to take the place of another with which it originally had nothing to do. Already in so simple a case as this there takes place a combination of present and earlier states. What shows itself in this simple example repeats itself at the higher stages of psychic development.

Immediate value belongs from the first to the result—in the given example, salivation (which of course, as we may here disregard, itself again has its value as a member of the

[9] *The Lancet*, 6 Oct. 1906.

digestive process). The sense-sensations that indirectly produce it acquire at first only mediate value. And at more developed stages it will, similarly, be the result of reflection that has value, while reflection itself is only a means. But it is quite possible that a value can pass over from being mediate to becoming immediate. There then occurs what we in psychology call a displacement of motive.

When it is bitter need that leads to thinking, the value or motive stands in a purely external relation to thought. The value is here itself an object, since, as it cannot be immediately attained, it poses the task of finding ways and means, and thereby comes to stand as a purpose. In the definite and compelled relation between purpose and means we have man's first encounter with the necessity of thought. If he wills A (the purpose), he must also will B (the means). By this path man enters the world of reflection even where there are no more direct motives for entering it. But it can acquire an independent value, can become an object of direct interest, to investigate that necessary relation between purpose and means, which is, after all, the same as the relation between effect and cause; and the value is then bound to reflection itself and its work. It comes to stand as an activity that it is attractive to exercise, whatever result it may lead to.

When a stage has been reached where the value or motive attaches itself closely to the activity of thought itself, so that this is both end and means—when, in other words, an intellectual feeling has developed, a joy in thought itself, as it unfolds according to its own laws and according to the demands of the objects—then we need not specially recall motives and values in the case of cognition. We then desire no other satisfaction than that which is bound to the use of our faculty of thought.

It is such a stage that we presuppose in an investigation of human thought. All purposes other than that of seeing where consistent thinking leads then do not exist, and we need not constantly recall the standpoint of value; this is given once and for all together with the very urge to—observe and to infer. Even where other motives than the intellectual interest itself actually make themselves felt in a movement of thought, it will nevertheless be fortunate if such motives—in relation to which the work of thought stands only as a means—pass over, through displacement of motive, for a while into purely intellectual motives. Thereby the work is best done. And it will in any case always be, in the end, an examination undertaken from purely intellectual motives that must revise the work performed from other motives, whatever these may have been.

7. The concept of energy in the sense in which we have here taken it has its origin in natural science, and it is by way of analogy that we have attempted to transfer it to the domain of the mental life. All psychological expressions are, after all, transferred from

the material to the spiritual, from the world of space to the world of spirit and thought. But there are difficulties in carrying the analogy through. An exact measurement is possible in the psychic domain only as regards relatively elementary mental phenomena, and never in so perfect a manner as in the case of material energy. Modern physiology stands, as regards method, on the standpoint that it requires physiological states to be explained by physiological causes, insofar as physical causes from the surrounding world do not intervene. Every state of the brain or brain-activity is therefore also to be explained on the basis of the earlier states of the brain and of the organism. According to strict natural-scientific method, every material state within and outside the organism must find its explanation in being seen as having arisen through the conversion into new form of the physical energy that expressed itself in preceding material states. When two brain-states thus succeed one another, the explanation will therefore not be found until it is demonstrated that all the energy that expressed itself in the first state has, without remainder, been converted into the second state, insofar as it has not radiated out into other physiological processes. It is perhaps impossible to carry out experiments in this direction; but such experiments would mark the completion of the physiological method, which can set as its motto *Spinoza*'s words, that an explanation of a bodily state by the intervention of the soul is one with an admission that one cannot explain it. Even if now one of the experiments one might make seemed to show that, in the arising of an organic state, physical energy arose or vanished without a physical equivalent, one would, from a purely natural-scientific standpoint, sooner believe that there was an error in the experiment than rest content with having discovered a limit to the conservation of physical energy. "The principle of the conservation of energy," says *Maxwell*[10], "has acquired so great a scientific weight that no physiologist would place any confidence in an experiment that showed a considerable difference between the work performed by a living being and the sum of the energy received and expended." Since Maxwell wrote down this proposition, experiments have been carried out which—within certain limits of error—show that the exchange of matter and force during the performance of mental and bodily work presents an equivalence between what the organism took in and what it expended[11]. It will hereafter become even more difficult than before to assume that in any part of the organism something should take place that conflicts with the energy principle.

But how, under these conditions, will it be possible to maintain psychic energy as

[10] *Scientific Papers.* II, p. 759.

[11] *Atwater: Neue Versuche über Stoff- und Kraftwechsel des menschlichen Körpers* (Ergebnisse der Physiologie III, 1. 1904).

a peculiar form of energy? Is it not, then, more reasonable to assume that all energy is physical, and that the psychic energy we have spoken of in the foregoing is only a symptom of a physical energy to which we have no direct access?—For we cannot carry through the principle of equivalence within the psychic domain itself, as we can in the physical domain. Interruptions occur within one and the same individual conscious life (through sleep, faint, and so on), and between different individual consciousnesses we cannot demonstrate psychic continuity, as we can demonstrate physical continuity between different bodies—e.g. between two organisms.

Accordingly, the demand has been raised that the psychological definitions should be replaced by physiological ones. Among us, *Carl Lange* (in his "Nydelsernes Fysiologi") has asserted this demand. In strict epistemological and psychological form, *Richard Avenarius*, in his acute work *Kritik der reinen Erfahrung*, has maintained a similar view. Full scientific understanding is attained, he holds, only when the psychic states are traced back to the corresponding physiological states: hence he calls the series of psychic states the dependent vital series, the series of physiological states the independent vital series. And his justification is that only in this way can we reduce the qualitative differences that the psychic phenomena present to quantitative ones, so that an exact formulation of the relation between the changing states becomes possible. The completion of our cognition will thus consist in a reduction of the psychic to the physical[12].

In the face of the views that thus demand a reduction of psychology to physiology in order for a scientific psychology to become possible—views that thus would really abolish psychology in order to make it a science—it must be acknowledged that the actual discontinuity and the qualitative difference among the psychic phenomena will always set limits to a strict carrying-through of psychology as a science. But the question is whether the path they point out leads to the goal—whether the method they praise does not at bottom contain a palpable self-contradiction.

When it is demanded that the psychological definitions should be replaced by physiological definitions, the presupposition for this must surely be that the psychological definitions are complete and exact. But to provide these definitions is psychology's own affair, and if it cannot provide such definitions, physiology cannot know what it is that it should seek to explain in the brain. When that which is to be replaced is vague and wavering, the replacement must turn out accordingly. There will be no certainty that one has really found the brain-states corresponding to the definite psychic phenomena one has in view. The independence of psychology must therefore in any case be

[12] Cf. my characterization of *Avenarius*'s philosophy in *Moderne Filosofer*, pp. 98–106.

acknowledged, since it is to give us the symptomatology that sets physiology its tasks. Now, for every empirical science it is a long and difficult work to arrive at conclusive definitions; such are possible only when the empirical science is essentially finished; they come at the end, not at the beginning, of the investigations. All too often one operates with unfinished psychological definitions that are regarded as reliable starting-points for brain-physiological interpretation. Thus *Descartes*, and in more recent times *Lotze*, regarded the very concept "soul" as so clear and fixed that one could confidently call upon physiologists and anatomists to search for "the seat of the soul." And in the most recent times *Flechsig*[13] has regarded the concept of association of representations as so simple and clear that he could suppose he had found special tracts in the brain for processes that are supposed to bring about the connection between the physiological correlates of the individual representations. It is in reality a great abstraction that Flechsig works with when he sets the connection itself (the association) in opposition to the elements (representations) that are to be connected, so that the former could be thought of as localized in places other than the latter. He thereby quite overlooks the psychological antinomy brought out above, which warns against confusing without further ado the differences and distinctions that the investigation necessitates with oppositions in the unfolding of the mental life itself. There is no "element" in a mental life that does not stand in, and is not determined by, some connection or other, some relation or other. When a representation resists entering into a certain, seemingly natural association, the reason is to be sought in the fact that it is a member of another, more intimate association and cannot be detached from it. The conditions here are very complicated, and with a purely mechanical schema neither physiology nor psychology comes into its own.

But let us suppose that the psychological definitions are complete and perfect. Even then great, unsolved problems will remain. How can physiological states have psychological symptoms? How can qualitative differences correspond to quantitative ones, and how can discontinuous phenomena be bound to continuous processes? Far from such questions falling away through the "reduction" of psychology to physiology, they are precisely sharpened by it and appear in a more glaring form. And in any case it stands as an unsolved enigma how the qualitative differences and the discontinuities come into being. Even if one should succeed in showing that two phenomena, A and B, which one has hitherto regarded as different, are in reality identical, so that one can set A = B, one has not thereby explained how A and B could in the first place stand as different. To call this difference "subjective" does not help; it is not thereby gotten rid of out of existence. It

[13] *Gehirn und Seele*. Leipzig 1896, p. 24 f.

is, after all, precisely the fact that there is something "subjective" in the world that makes psychology both possible and necessary. It is, moreover, within this "subjective" that all cognition of material nature—physiology too, and with it the knowledge of our own body, especially of our own brain—comes into being. Cognition is itself a psychic function, both where it appears as sensation and where it appears as thinking. By regarding psychic states and activities as accidental appendages ("epiphenomena"), the natural scientist would saw off the branch on which he sits. To study this branch itself is, to be sure, not his task; but he does well, nonetheless, not to forget where he is sitting.

When *Avenarius* attempts to carry through the reduction of the doctrine of "the dependent vital series" (psychology) to the doctrine of "the independent vital series" (physiology), it is nonetheless apparent through his exposition that he constantly infers from the "dependent" to the "independent" series, not the reverse. It is on the basis of the psychological observations he has collected that he constructs the physiological schema which for him is the proper object of science, and one cannot understand what is said about the "independent" series except by means of the facts that the "dependent" series contains. In general it turns out that brain-physiology is still far from developed enough to give a thorough elucidation of the material processes to which the great, richly nuanced world of psychic processes is in one way or another bound. What one finds in this respect among physiological psychologists and experimental psychologists are mere schemata, which are supposed to "explain" what psychological experience presents. No one has more occasion to experience this than a psychiatrist who is at the same time an experienced and critical psychologist. Thus *Pierre Janet* has declared that, apart from individual observations and histological investigations, the physiological interpretations are only translations of the psychological concepts into another language, and that if one will think only anatomically, one must wholly give up thinking psychiatrically[14].

8. It is the discontinuity and the qualitative differences that cause psychology's difficulties. But greater weight is often placed on them than experience warrants. Precise reflection often leads to the discovery that even where there seemed to be a psychically empty space, there was in reality a psychic content given, although it was not grasped by attention or by recognition, or was forgotten immediately after the experience. And as regards the qualitative differences, observation will often find more and finer nuances than those one first stopped at and perhaps confidently declared to be utterly disparate. The continuity in conscious life is surely greater than is often assumed, and there can lie a scientific danger in too strongly proclaiming discontinuity and qualitative difference

[14] *Les Obsessions et la Psychasthénie*. I, p. 496; 605. Similar statements by *Wundt* (*Philos. Studien* XIII, p. 363) and by *Ramon y Cajal* (cited in *Herrlin*: *Minnet*, p. 21).

as the distinctive mark of the phenomena of consciousness. The inner connection in consciousness—the fact that the more we immerse ourselves in, and make ourselves one with, the involuntary course of the mental life, the more this appears to us as a whole, a streaming—is and remains the primal phenomenon in the psychological domain. We overlook it so easily only because in everyday speech, and on the whole in practical life, we have most use for certain single drops or waves in the gliding stream. We must perform an integration when we go back from our more or less mechanical practice to the immediately given. And even if we confine ourselves to functions that can be exercised with clear consciousness—e.g. memory and comparison—these very functions bear witness to an inner connection behind the particulars. They are forms of synthesis, of the tendency to combining and holding-together that attests the mental life's constant striving to be a whole. The unconnected is the psychologically enigmatic, that which sets psychology its problems. The task of psychology is to find connection between the elements of consciousness as well as between its different states.

Leibniz has the merit of having energetically maintained the principle of continuity in the psychic as well as in the physical domain, especially by pointing to the elements that are vanishingly small in relation to the clearly conscious states, and that only by being summed up and combined produce a result accessible to untrained self-observation. Already a comparison between modern and ancient or medieval poetry shows how many more mental nuances self-observation has now come to perceive. It is scarcely only because the nuancing in the psychic domain is now in itself greater than in earlier times that this greater richness is found. It is partly precisely the knowledge and understanding of the immediate and involuntary in the mental life itself that has grown. The childlike and primitive psychology is inclined to conceive everything that takes place in the soul as clearly conscious and all actions as undertaken with deliberation and intent. Now an explicit demonstration is required that reflection and deliberation have been present in the particular case. The transitions themselves between the unconscious and the conscious, and between the involuntary and the voluntary, can take place through many intermediate forms, many degrees. The more practiced self-observation and criticism are, the more difficult it is found to be to put one's finger on a definite place where the boundary should lie.

Leibniz was inclined to regard all psychic differences as differences of degree with respect to clarity and obscurity. This attempt was characteristic of the century of the Enlightenment, which Leibniz inaugurated. But there are other paths by which psychology can find a continuous connection despite the qualitative differences. There constantly take place fusions and compositions in the domain of conscious life—of sensations, representations, feelings, and endeavors. And displacements take place by which what at first stood in the foreground is supplanted by what was at first background—where what at first had only mediate value comes to stand as immediately valuable. And in integral characters, the highest and noblest objects of psychology, there appears to contemplation so intimate a dependence of everything that stirs in the soul on one purpose, one guiding thought, that we have here a causal connection that is every bit as firm and definite as any in the physical domain. Every single representation, every single mood, and every single impulse is, in such a character, clearly determined by its place within the whole to which it belongs and which it itself helps to shape.

Where, despite everything, a lack of connection shows itself in the psychic domain, we need not, however, give up the principle of continuity, any more than the physicist gives it up when energy seems to have vanished in a state of rest or equilibrium. Just as the physicist introduces the concept of potential physical energy to express the fact that what seemed to have vanished can be released again, so the psychologist will be justified in speaking of potential psychic energy, supporting himself especially on the fact that psychic elements can be reproduced after an interval in which they have not been in consciousness. Words such as "predisposition," "possibility," "disposition," "capacity," "trace," "tendency" express the same as the word potential energy. The psychologist is, however, worse placed in applying this concept than the physicist in applying his corresponding concept. In the first place, the physicist can directly point to the phenomenon in which the potential energy has its seat—e.g. the tensed spring or the stone resting on the roof. In the psychic domain, by contrast, complete equilibrium (rest) means the same as unconsciousness—hence something inaccessible to observation (whatever it might otherwise be in itself). And in the second place, in the physical domain one can carry out conversions that show that a quantum of energy can be released corresponding to the quantum that apparently vanished on the transition to the "potential" state. But in the psychic domain it cannot be demonstrated in this way that predisposition and development correspond to each other. If the concept of potential psychic energy nevertheless has its justification, it is because it stands as the expression of an unsolved but undeniable task, and because it contains a summons to go ever further in observation and analysis in order to find as much continuity as possible between the different states of the mental life. There is a long way to absolute rest—and it might be that it occurs as little in the psychic as in the physical domain. Possibility everywhere subsists in an actuality, whether or not this is accessible to us.

9. The only way in which psychology and physiology can work together without either of them coming to break with its guiding thoughts is provided by the psychic phenomena being regarded as symptoms of physiological states—and, conversely, physiological phenomena being regarded as symptoms of psychic states. We can then *infer* from the one series of phenomena to the other, although we cannot *derive* the one from the other.

It is the natural-scientific method that step by step leads to such a conception. The law of inertia requires that every change in the state of a material point (of rest or of rectilinear motion with a certain velocity) be derived from the influence of another material point. Thereby it is excluded that a non-material (that is, non-spatial) cause can acquire natural-scientific significance. Thereby the principle is established that a material phenomenon must have a material cause. *Maxwell* finds the experiential proof of the law of inertia in the fact that, every time we encounter a change in a body's state of motion, we can trace this change back to an action between this body and another body. This experiential proof natural science is constantly in the process of furnishing.—To the law of inertia is joined the proposition of the conservation of physical energy, which, as already mentioned, seems to have to be maintained also in the case of organic nature.

There are in the physics of our time intimations that the said propositions, despite their great importance for the whole conception of nature, are nevertheless derived from laws and relations that hold for the ether-vibrations in which one sees more and more the fundamental thing. Philosophically regarded, this would change nothing in the conception of the problem of the relation between the spiritual and the material. The vibrations of the imponderable ether are just as different from the psychic phenomena as the movements of the ponderable atoms. The problem remains the same, even if electrons take the place of atoms. It is the spatial character that, for philosophy, is the decisive property of matter; in this *Descartes* saw rightly. It must not here mislead us that physicists sometimes use the expression the immaterial in the sense of the imponderable.—

If now this is so, there can be found no place for the mental life in the series of physical, or respectively physiological, phenomena. The only possibility then becomes the conception that in the mental life we have the expression that self-observation gives us for what appears to our senses as organic processes. Soul and body are then not two different things or beings, but two languages in which existence speaks to us. We can perhaps translate from the one language into the other; but we cannot derive the one language from the other.

Since the mental life cannot be derived from the laws and forces of material nature, we must assume that in all material activity there cooperate, besides the properties whose presence natural science demonstrates, other properties that are not accessible to sensation, and in which the mental life would be able to find its explanation. Philosophically regarded, natural science's concept of matter cannot exhaust existence. But by the "predispositions" or "dispositions" that we presuppose in order to understand the possibility of the mental life's arising in nature, we do not in the least shake the application that natural science makes of what it has learned and seen.—

The conception to which we here adhere really returns to the childlike and practical view that regards the body as the outer expression of personality, accessible to the senses. And we thereby obtain at the same time the best and most durable working hypothesis, in that we are summoned to follow at once the series of psychological and of physiological phenomena, each by itself, as far as is at all possible, and to investigate the relation between them, the analogies and proportionalities, as exactly as possible. It is a functional relation, in the mathematical sense of the word, that this conception assumes between the mental and the material phenomena; we seek, and often we also find, that a variation of the mental states corresponds to a variation of the bodily states, especially the brain-states, and vice versa. And to this is in reality confined the knowledge we have of these things. Our hypothesis (for which the most fitting name is the identity hypothesis, since it does not assume that soul and body are two different beings) we regard provisionally only as a working hypothesis. Whether it gives a conclusive explanation of the relation between the spiritual and the material, between psychic and physical energy, is another question, which will be discussed in a later connection. Even thinkers who reject our hypothesis as a definitive solution nevertheless concede that it is a useful working hypothesis. Thus *H. Bergson* and *C. Stumpf*. Whether this concession is consistent from their standpoint is a question in itself. A hypothesis that has no ground at all in the nature of things will not in the long run be suited to serve as a working hypothesis.—

It has sometimes been thought that the enigma could be solved by assuming (or saying) that psychic energy is nothing other than nerve-energy. But when one does not assume that all nerve-energy is connected with psychic phenomena, a distinction must be made between conscious and unconscious nerve-energy—and the problem is then posed anew. Besides, we know as good as nothing about this so-called nerve-energy and about its relation to other forms of physical energy, and it is of no use to give the nerve-energy to which psychic phenomena are bound a special name. Thereby one only introduces a qualitas occulta and rests content with it. The natural-scientific concept of energy is, wherever one meets it, always formed on the basis of phenomena with spatial, geometrical properties. Natural science knows energy only as the expression of relations between phenomena in space. And we cannot, unfortunately, form a concept of energy from which the physical and the psychic concepts of energy could be derived as special forms. In any case, every attempt to form such a concept would fall outside the limits of psychology. In a later connection we shall return to the problem we here leave[15].

10. We have in the foregoing determined thought as psychic energy and sought to show how, provisionally and purely methodically, we must regard it in relation to physical energy. But there will perhaps arise in many the question, *who* it is that *has* or *exercises* this energy? One reasons somewhat thus: "Every activity must be exercised by someone or something. We must distinguish between that which acts and the activity itself. In particular, in the psychological domain we must distinguish between that which thinks, feels, and wills, and thinking, feeling, and willing themselves."

Such a conception, which occurs not only in everyday speech but also among philosophers, spiritualists and materialists in brotherly union, is due partly to the influence of language's mode of expression, partly to the influence of material analogies.

Language distinguishes between subject and predicate and as a rule designates each of them with its own word. There is therefore formed one word for the acting being, another for its activity. Even where it is difficult for intuition to make this distinction between the acting thing and the activity, language nevertheless often employs a formal subject, as in "impersonal" sentences of the form "it is raining." It is a grammatical schema that prevails here, a schema one has no right to give philosophical significance, either psychological or metaphysical. Long ago the acute philologist *Højsgaard*[16] impressed upon us the difference between grammar and philosophy. "Grammatically," he says, "a word or a discourse is considered when one looks not so much at the things that are understood by the words, as at the words themselves, their order and case, and at their mutual dependence or relations to one another. Whereas a philosopher considers only the things that the words call to mind." This warning is constantly needed.

Because language often needs two words to express what happens when something acts, it does not follow that we have two "things" before us. And how, really, should the relation between these two things be thought—the relation between consciousness and that which "has" it, between energy and that which "exercises" it?

In material phenomena, which have exerted precisely the greatest influence on

[15] With respect to many details concerning this problem I refer to my *Psykologi*, chaps. II and III.

[16] *Methodisk Forsøg til en fuldstændig dansk Syntax*. Copenhagen 1752. (The section "Register og Forklaringer.")

language's mode of expression, we distinguish with ease between the subject of the activity (the movement) and the activity (the movement) itself. We have often seen the body that is now in motion lying apparently still in its place, and we thus know it both as resting and as in motion. When we say that the body moves, it is really these two states that we compare. We therefore fall into embarrassment when we discover that no absolute rest can rightly be assumed. And language especially falls into embarrassment when one has to speak not of "ponderable" but of "imponderable" matter, since this latter is presupposed to be in constant wave-motion. "Ether" can therefore be defined only as that which undulates. It is now the analogy with ponderable matter, and with the apparent difference between equilibrium and motion, that operates when one distinguishes between the soul (the I) and its activity. But the soul we know (like the ether) only as active. What we know here is precisely psychic energy, not any resting "being." The soul's being or nature is activity. It is an intuiting, often a lazy, imagination, not a real thinking, that has the word when one makes that distinction between the thinking thing and the thinking. Spatial images or symbols are confused with psychological concepts. The urge for fixed, resting images, such as material nature seems to present, prevents one from holding fast the concept of activity or energy in its peculiarity and its consistency. Or one fears that the mental life will evaporate if it is not hung upon a "something," a "substance," as upon a secure hook—although one is not so consistent as to ask upon what this hook is fastened. The "substance" could just as well need a "bearer" as the activity. Just as one earlier asked who "has" the energy, so one might here ask who "has" the soul or the I.—Thinkers such as *Fichte*, *Sibbern*, and *Wundt* have the merit of having seen clearly on this point.

We can very well say what we mean by "ourselves" or "our I," without entering into any spiritualistic or materialistic hypothesis. Keeping to experience, we see the mental life unfold itself through different stages and under different forms, and we seek to describe these and to explain the transitions between them. At every given stage of development we then have the mental life before us with certain properties and dispositions, and the task is then to find out what, under certain conditions, can develop from them. (Cf. 8.) "We ourselves" or "our I" has at every stage its content in the properties and dispositions that the phenomena of consciousness make known, and its form in the degree and manner in which these are combined and ordered.

Now the more the continued mental development is conditioned by the given properties and dispositions in their mutual relation, the more energy we ascribe to the mental life and the more active it is said to be. The more, on the other hand, the continued mental development is conditioned by influences that make themselves felt independently of the individual mental life's past, the more passive it is said to be.

Just as the planets in our world-system tore themselves loose, with a certain initial velocity and initial direction, from the great nebular mass, so the mental life—at the point where it awakens from the night of unconsciousness—begins by acting according to the same laws that it later follows. It begins with peculiar dispositions that give it a certain independence in the face of outer influences. With this agrees the fact that the nervous system is one of the parts of the organism that first appear with distinctness, and that from the first it occupies a central position in relation to other parts of the organism. The functions of the nervous system, to which the mental activities correspond, are not mere effects of outer influence, but determined in form and direction from the individual's first coming into being.—

In what follows we investigate, for the closer elucidation of the nature of human thought, first the more immediate and involuntary way in which it expresses itself in intuiting, association, and comparison (B), then the more conscious and exact form it takes on in judgment (C).

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B. Intuition, Association, and Comparison

a. Sensation

11. When one often distinguishes sharply between sensation and thinking, it is because one conceives sensation as a purely passive coming-into-being of new psychic content, while in thinking one sees a working-over of such content. But one cannot at any point wholly separate passivity and activity—that is, wholly separate states whose causes lie entirely outside an individual's preceding states from such as should be wholly due to these. There always show themselves traces of psychic work, even in the simplest elements of consciousness that can be distinguished. We can be more or less active—that is, the causes of the changes that take place in us can to a greater or lesser degree lie in ourselves, in our character and our preceding experiences; but wholly passive we never are.

If we bring out the conditions for the arising of a single sense-sensation, e.g. a light-sensation, it turns out that the condition for this arising is a change that has a certain degree of strength, a certain degree of suddenness, and a certain degree of qualitative difference—all in relation to the given state of consciousness. A sensation arises only when the new, in the three respects mentioned, stands in a certain definite relation of opposition to what was previously given. Perhaps, of the three conditions, suddenness is the most important, which from a practical biological point of view is easily understandable, since it is of importance in the struggle for existence to be awakened to attention for changes in the surrounding world, even apart from the degree and character of such changes. But all three conditions point, psychologically regarded, in the same direction. The weaker the change is, the more slowly it sets in, and the more alike the new and the earlier state are, the less possibility there is for the arising of an independent sensation; hence one can rightly designate every sensation as a distinguishing.

It is, to be sure, a distinguishing of a purely elementary kind. The single sensation steps forth in consciousness with its definite degree of independence and with its definite quality, and it is only indirectly, especially by way of physical and physiological investigation, that we can here demonstrate conditions similar to—in any case analogous to—those that condition a distinguishing undertaken with reflection and intent. When we set ourselves to apprehend the difference between two things distinctly, we can do nothing further toward it than to bring these two things into immediate proximity to each other—then the difference announces itself involuntarily, springs forth as it were, since each of the two things stands involuntarily with its own peculiarity. We here produce, by express holding-together, the opposition necessary for a distinguishing; but the apprehension of the difference we call forth only indirectly. And thus too, in the arising of the simplest sensations, the given changes or oppositions are only releasing conditions. The sensation arises as if it were a result of comparison; but the two opposed terms are here not known each by itself; it is only the change in the state, released by the opposition, that appears as an element in consciousness.

It is, then, a justified analogy to call the sensation a distinguishing—and so far a thinking—and already in it to find an expression of psychic energy. Language, which has formed itself under the influence of reflection, has no expressions for functions so simple as those we here confront. Linguistic usage is inclined now to set a very great difference between the simplest psychic functions and the more developed ones, now to read “inferences” and other forms of reflection without further ado into the simplest utterances of consciousness. But by the concept of analogy both the kinship and the distance between sensation and thinking can come into their own.—How one conceives the unconscious basis from which the simplest utterances of conscious life are thus released by the power of the oppositions—whether one declares it to be merely purely physiological processes, or whether in the unconscious one will see a potential mental life—is in this connection a matter of indifference. The main thing is that we already here trace the chief form of the mental life. For that matter, it shows itself already on a purely physiological consideration how little the sense-sensation can be conceived as a passive effect of stimulations from without. The physical impression traverses, on its way from the outer sense-organ to the great brain, so many layers of nerve-cells that it can only most improperly be said to be the same impression when it reaches the brain as when it comes to the sense-organ. And in the brain itself there is released a reciprocal action of a mass of cells as a consequence of the stimulation. What emerges in consciousness as a sensation corresponds, therefore, physiologically regarded, to a multiplicity of processes; there is gathered into unity in it what physiologically stands as a plurality.

12. That the sensations are thus not simple, passive effects of stimulations from without, but can be regarded as elementary forms of psychic energy, becomes of essential importance with respect to the validity that can be ascribed to them as means to cognition. We build up our world-picture on the basis of them, and yet one became very early aware that they could not be used without further ado to form a coherent and intelligible world-picture. Reflection must at once use them and disregard them if it is to reach its goal. The reason for this is that they are not only the expression of something different from our consciousness, but also of the very nature and mode of operation of our consciousness.

In a characteristic way, reflection's two-sided relation to sensation appears already in one of *Democritus*'s fragments, in which the senses speak thus to the understanding: "You poor understanding, you fetch your grounds from us and yet want to overturn us! Thereby you prepare your own downfall."—At the founding of modern natural science this paradox appears again, in that *Galileo*, *Descartes*, and *Hobbes*, at about the same time and on the basis of the same considerations, set up the doctrine of the purely subjective validity of the sense-qualities, all while the new science was nonetheless grounded upon sense-observations. In *Kant* we meet it again in his distinction between matter and form, in that he counts all sense-sensations to the matter and regards only the form—which, however, never occurs separated from the matter—as that with which science has to do.

If sensation were purely passive, it could be assumed simply to reproduce what calls it forth from without. The subjectivity of the sense-qualities is grounded precisely on the fact that they come about as a consequence of an activity on the part of consciousness. That there exists a problem of cognition rests—as will later show itself—precisely on the fact that we are self-active in our cognition. How can we, by our own activity, cognize something different from ourselves? By our very activity we transform it, after all, according to the laws of our consciousness. Cognition is our work. And in this respect there is, according to the foregoing, no difference of principle between sensation and thinking. The paradox then disappears when one gives up the absolute difference between sensation and thinking, matter and form. The work of reflection then stands as a continuation of the psychic work that sensation has already begun.

Insofar as reflection does the work of sensation over again, it is thus not because with it a wholly new principle comes into force. It is only psychic energy operating in another manner, more appropriate for the solution of the task of cognition. The understanding can thus quite well at once use and overturn sensation. And the "overturning" takes place partly with sensation's own help, in that the one sense controls the other. To the relation between the qualities within the one sense there can correspond relations between qualities within the other sense, and it is through the analogies that can here be demonstrated that reflection reaches its result. The senses by which the forms of synthesis most clearly come to light, and by which the highest degree of continuity can be attained, are taken as basis. And it is especially the senses that permit clear apprehension of movement. One might call them the rational senses. To these belong sight, hearing, the sense of touch, and especially sensations of movement. The sensation of movement takes place distinctly by a synthesis, since it presupposes the holding-together of the successive positions and stages of the moving thing; only by such a holding-together does movement become an object of consciousness. The special qualities—color, hardness or softness, sound—that correspond to the different stages of the movement could in themselves be thought of as exchanged for others, provided only that the relation between the qualities among themselves continued to be the same. One can then disregard them, and the sensations then become, when they stand in the service of cognition, only symbols for certain stages or places in continuous processes of movement. In a sense-sensation, with its degree of independence and its peculiar quality, a concentration has taken place, a combining of what, physically and physiologically regarded, is a multiplicity. Modern physical theories interpret what for us is only the object of a single act of sense as an aggregate of small parts in constant movement. Our sensation appears here as analogous to a statistical result of a mass of particulars that are each by itself inaccessible to us and that stand in the most complicated mutual connection. The kinetic theory of gases is especially an example of this. According to it, every gas is an elastic fluid whose elements are in constant movement, with numerous collisions against one another and against the barriers, and consequent changes of direction and velocity. For our sensation it nonetheless appears as a whole, within which we distinguish nothing. And the same holds, as already remarked earlier, of the relation of sensations to the brain-processes. What in our consciousness appears as a single sensation corresponds to a very composite reciprocal action between the millions of cells of the great brain and between these and the other parts of the brain.

What takes place in the nervous system and in the brain does not resemble what stimulates the sense-organ from without. And what takes place in the brain probably does not resemble what takes place in the sense-organ, the ingoing nerve, and the lower centers that the impressions pass through on their way. The sense-sensation resembles neither any of these physiological processes nor the outer stimulations that release them.

Innumerable enigmas conceal themselves here. And yet it is a fact that sense-sensations can guide living beings in their struggle for existence. Our states can very well be decisively determined by our own nature, provided only that they lead to the release of activities by which we can maintain ourselves against the surrounding world. In instinct, which is released by definite sense-stimulations, this appears especially clearly. The answer is given in a quite different way than the question is put. In instinct, urge and capacity are so united that they are both released and operate together when a certain definite change sets in in the state of the living being. Thus when the pregnant bird builds a nest, or when the newly hatched chick scratches if it feels gravel under its feet.

Already the first thinkers who became clear about the subjective nature of the sense-qualities emphasized, in connection therewith, the practical significance of sensations as the original, the theoretical significance as the derived. Thus especially *Descartes*, *Malebranche*, and *Berkeley*. Modern evolutionary theory has again brought out this point of view. Sensation is first and foremost a means in the struggle for life, only secondarily a means to come to know the nature of existence. The connection between its practical and its theoretical significance lies in the concept of sign or symbol. The sign can be a signal for action; but it can also become a signal that announces the setting-in of a certain change, apart from the action necessitated by this change. There are many intermediate degrees between these two significations. But in neither of the two significations is there needed any likeness between the sign and what is to be signified. And the sign perhaps operates best by being as simple as possible—by presenting as a unity what is in itself a multiplicity. The main thing is that it can serve as an introduction either to an action or to a train of thought that can attest their validity—the action by leading to the satisfaction of an urge, the train of thought by leading to understanding and to agreement with new sensations. In this last case too it is a matter of the satisfaction of an urge—namely of the very urge to understanding, to the agreement of our reflection with as many sensations as possible. Motive (value), form, and object thus always operate together. (Cf. 6.)

The special qualities with which the different sense-sensations arise nevertheless contain a great, perhaps insoluble, problem. For from a biological as well as from an epistemological point of view, whatever quality a sensation appears with, sensation can perform its function in the service of life and of reflection, provided only that it is able to release activities or trains of thought that can satisfy the practical or theoretical need. Why precisely these definite colors arise for us when certain electromagnetic stimulations have propagated themselves through space to our eye, or why we hear precisely these definite tones when certain air-waves have propagated themselves to our ear, stands as a constant enigma. We stand here at mere facts and must for the present remain standing at them.

Each single one of our senses has its series of qualities, more or less continuous and systematic. Even where the continuity is greatest, as in the domain of the sense of sight, it is nonetheless interrupted at definite places, and the series of qualities is always limited. To infra-red rays the retina is not receptive, and ultra-violet rays affect it only at a particularly great degree of strength. This limitation, which probably has its ground in certain physical or physiological conditions, and perhaps has its appropriateness, entails a limitation of the possibilities of knowledge.

Each sense-organ has its own peculiar advantage. The ear possesses the fine power of discrimination by which partial tones can be recognized within the sound in which they occur. The eye's power of discrimination shows itself especially as the capacity to localize the different visual impressions. In the different living beings, according to their conditions of life, different senses show themselves especially developed. All such special degrees of development must, just like the limitation, be assumed to have arisen through reciprocal action between the inner and the outer conditions of life.

When we here speak of living beings and of a surrounding world, we make use of a knowledge that, like all other knowledge, is itself won on the basis of sensations and by the work of reflection on this basis. It is then presupposed that the psychic energy that already expresses itself in sensation has developed in the form of reflection. The biological knowledge we make use of for understanding the development of the psychic activities is itself a product of these activities.

13. It has already shown itself above that a single sensation is an abstraction. The character of every sensation rests on the relation between the conditions under which it arises and the conditions under which simultaneous or preceding sensations arise. And

there showed itself here an analogy between its arising and the discovery of a difference effected by reflection. It was for this reason that we could call it a distinguishing.

The psychic equilibrium must be abolished for a sensation to arise. But there always expresses itself a tendency to equalize this disturbance. And this tendency appears in two forms, a more elementary one, which consists in fusion, and a higher one, which consists in the formation of a sense-intuition.

Fusion sets in, as already indicated, when the change produced by the impression is slight in relation to the state we are already in, or when changes follow very rapidly upon one another. Examples of fusion we have in the timbre of tones and in the cooperation of colored and uncolored light-impressions in the coming-into-being of light-sensations. In a visual image due to seeing with one eye, it is not noticed that the blind spot on the retina gives no contribution to the filling-out of the image. When a regular alternation takes place of light-stimulation in the time a and complete non-stimulation in the time b, there is nonetheless obtained, when the time a + b is very short, a continuous sensation. Even changes that take place simultaneously in different sense-domains fuse insofar as they mutually inhibit or reinforce one another.

It would scarcely be right to see, in such a fusion—in which the individual elements cannot make themselves independently felt—the original form of conscious life, so that an "undifferentiated continuum" should be the starting-point for mental development[17]. All conscious life is scarcely ever fused; there will in any case probably stand different fused complexes over against one another. A complete state of fusedness would be a state of rest, which would be one with a state of unconsciousness. A certain breaking of oppositions, or at least of differences, is necessary for the subsistence of conscious life.

When fusion is not possible, and there nonetheless expresses itself a tendency to equalization, a sense-intuition can form itself, in that the sensations are ordered in such a way that each single one can make itself felt without swallowing up or supplanting the others. With the least possible change, the new content that a sensation brings is then incorporated into the whole connection of consciousness. The sense-intuition solves a double task, in that it at once lets the novelty and peculiarity of the single sensation and the connection of conscious life come into their own. It presents at once multiplicity and unity, while fusion abolished the multiplicity. The sense-intuition is clearest when the differences have previously appeared each by itself and only thereupon are connected in such a way that a single act of attention can survey them. The clearest form of such a surveying is space-intuition, which we therefore have a pronounced tendency to use as a

[17] *Einar Buch*: *Om Fornemmelsers Sammensmeltning, særlig ved Klangindtryk.* Copenhagen 1898, pp. 31–35; 60 f.

schema for everything we wish to make clear to ourselves. In particular, a time-intuition first becomes possible through the symbolic use of space-intuition, although we can of course quite well apprehend change and succession—e.g. of tones or of our own inner states—without using a spatial schema.

Sense-intuition denotes a higher stage of psychic energy than fusion. The multiplicity is preserved through an involuntary ordering. There will always be one or more single elements that determine the ordering, and around which the others group themselves—namely those that correspond to the dominant urge or interest. Therefore every involuntary sense-intuition will be articulated, pointed in a definite direction, that in which attention is concentrated. This determinacy of direction forms an analogy to what later stands as purpose for striving and object for reflection. Out of this concentration the other members in the sense-intuition fade off into the indefinite and obscure.—

In this description we must constantly remember what was above called the psychic antinomy. When it is said that elements fuse or become members in a sense-intuition, the meaning is not that the elements, apart from these kinds of synthesis, have the same character as they have within the syntheses. It is only in our abstract language that we can let the elements go in and out of the different connections like a kind of atoms.

b. Recognition

14. A pure sense-intuition would be one in which a multiplicity is involuntarily ordered without earlier lived experiences or experiences contributing. Apart from the most primitive states, such a pure sensation scarcely occurs. As soon as, in the multiplicity that is ordered, there are found elements that have previously been present, there is the possibility of a new form of psychic energy: recognition. The simplest example of recognition we have where a sensation, by reason of repetition, immediately appears with a peculiar quality, different from that with which it appeared when it first appeared. I use the expression quality to designate the wholly immediate difference that can obtain between the known and the new—a difference that can be as simple and indescribable as the difference between yellow and blue. The quality that a sensation can acquire through repetition I call the familiarity-quality.

While in the simple sense-intuition it is essentially the difference between the elements that operates, in recognition the relation of likeness makes itself felt. What conditions the transition from new to known is not itself something new, but is, on the contrary, precisely a repetition. How the familiarity-quality is to be explained physiologically is not decisive for its psychological and epistemological character. It is, however, most natural here to point to the influence that repetition and practice have in altering the course of processes and functions. Through repetition one and the same function, probably also the brain's function, can take place with greater ease, and recognition might then be the psychic correlate of the altered character of the functions. However this may be, the circumstance that the relation of likeness now makes itself felt marks a significant advance. One can already here speak of an understanding: the present here stands immediately as one with something earlier experienced. There is a foothold, a home for it in consciousness. *Plato* called the dog a philosophical animal, because its behavior toward men is determined by whether it knows them or not. In this sense newly hatched chicks too are philosophical animals, since they scratch only when they feel gravel under their feet, not on a carpet. The remarkable thing about instinctive actions is precisely that they are released by stimulations that are at first new to the individual and yet operate as if they were known.

In everyday speech we often use the word "understand" in the sense of recognizing. It is an understanding of the most elementary kind. I understand a sound when I know what kind of sound it is, that is, recognize it. I can in this respect understand a word when I know its sound, even if I do not know its meaning. Understanding has many degrees. I understand what kind of figure it is that approaches when I recognize the human figure in it; but there can still be much to understand, namely which person it is, and so on[18]—

Just as in the simplest apprehension of difference, so in the simplest recognition language lacks apt designations. But there is a significant analogy between such recognition and the comparison undertaken with reflection. In genuine, free comparison two independently given sensations or representations are presupposed, whereas immediate recognition takes place without the present and the earlier experienced needing to appear each by itself in consciousness. Just as a sensation can arise immediately when there is a certain opposition between the influence of a stimulation and the state at hand, so immediate recognition arises on the presupposition of a relation of likeness between the influence of a stimulation and the dispositions or after-effects that earlier stimulations have left behind. In both cases it is only the results, not the presuppositions, that appear in consciousness. The elementary distinguishing and the immediate identifying are the first intimations of what at higher stages appears as thinking in the narrower sense of

[18]It is therefore not an apt objection that *Husserl* (*Logische Untersuchungen*, II, p. 73 f.) makes against me, that unintelligible words too can sound familiar. In the wholly immediate "understanding" of a word, it is only the sound or the sign, not the meaning, that is known.

the word.

15. Distinguishing and recognizing denote two poles in our cognition, which under different forms are found again at all its stages of development, and which correspond to two necessities of life: receptivity for the new, the changing, and the capacity to take it up into oneself in as intimate a way as possible. The two capacities can stand in sharp opposition to each other. They are two different kinds of psychic energy, a centrifugal and a centripetal one. But there is a natural reciprocal action between them. Recognition makes possible a saving of force. If, in the face of changes that repeat themselves, we had to go through anew each time the experiences that the first change called forth, it would lay claim to all our energy, and we would not get anywhere. By reason of recognition, on the other hand, we can turn with fresh force toward the new, and the very opposition between the known and the new lets the new, by a contrast-effect, step forth more vividly and characteristically.

Yet recognition itself presupposes energy to perform the synthesis of the present and the earlier. In that weakening of psychic energy which *Pierre Janet* calls psychasthenia, and which essentially expresses itself as a lack of will (aboulia), the capacity to recognize (understand) the present situation is often lacking. The bond that should connect the present with the earlier is broken. On the occasion of one of his aboulic patients, Janet says: "The attention to the present, the connection (synthèse) of the new sensations with the old images, which constitutes recognition, has here, like the will, vanished on account of the mental weakening"[19]. Yet such weakening can also lead to false recognition or memory-illusion; for energy is also required to distinguish the new from the known.—

It is only an analogy that obtains between immediate recognition and reflection. One has no right to identify it without further ado with acts of thought performed with clear consciousness. It would thus be incorrect to call it a judgment, as several authors have done[20]. For a judgment there is required a transition of thought from a subject to a predicate; but in immediate recognition no transition takes place between two terms; the familiarity-quality appears as if at a single stroke. Immediate recognition can form the basis and presupposition for a judgment, but the judgment first comes into being when a distinction is made between the given, present thing and the earlier thing with which it is identified. The so-called exclamatory judgments, which we shall later come to discuss, are the first judgments that become possible on the basis of recognition; in them there is a predicate, even if there is no subject. Thus when the gypsy boy in *Blicher*

[19] *Les Obsessions et la Psychasthénie.* II, p. 29.
[20] *Meinong*: *Beiträge zur Theorie der psychologischen Analyse.* p. 374.—*Hans Cornelius*: *Versuch einer Theorie der Existenzurteile.* p. 56 f.

cries: “A dog! A hunter!” These are immediate expressions of recognitions.

16. When it is a whole, not a single property or a composite phenomenon, that is recognized, it will not always be necessary to recognize expressly all the members (the parts or properties). Recognition of a single member can be sufficient, and the familiarity-quality is then involuntarily transferred to the other members, so that the whole stands as familiar, or an expectation is awakened that the other members too are present—an expectation that can be so secure that one does not seek confirmation of it. The recognition is then partial, supporting itself merely on a part of the whole that serves as a mark of recognition.

If the expectation leads to a continued going-through of the members of the whole, and these, when they are sensed, correspond to the expectation, a mediate recognition of them takes place. A person’s gait and bearing seem familiar to us, and we then expect that other properties will correspond to them; we address him, and his expression and speech are as we expected; they are then recognized mediately, in that the representation contained in the expectation covers what we see and hear.

In immediate, in partial, and in mediate recognition alike, an involuntary interpretation is woven into apparently simple observations. It is a surplus of energy that makes us at once interpret the given and, in virtue of the fundamental law of synthesis, bring it into connection with previously given content. Later, renewed observation may perhaps correct our first interpretations and expectations. But it is only by the contribution we ourselves give that we really come to know things. If we sat still and waited, the circle of experiences would become narrow. Now, on the contrary, the urge to use our energy leads us to an involuntary experimenting, which can then confirm or correct our expectations and representations. We have here, in purely elementary form, already an intimation of the method that developed science applies: an observation (recognition) leads to a conjecture (expectation) or a hypothesis, which can be confirmed or corrected by new observations. In the simplest mediate recognition there is already described that arc from observation through reflection to new observation in which the peculiarity of all scientific method consists. This kind of recognition testifies to greater psychic energy than immediate recognition: the latter is bound to the given and present, whereas the former goes out beyond this in expectation and representation. It is from the first the urge that drives out beyond the given—partly the urge for an outlet for energy, partly the urge for the attainment of something that does not involuntarily fall to our lot.—

Some psychologists have held that there is no immediate recognition, but that all recognition takes place mediately, that is, through previously awakened expectations and representations, and perhaps even through express comparison. After the discussions conducted on this, it can nonetheless be regarded as settled that there is an immediate recognition, and that it is an interesting transitional form between sensation and reflection[21].

c. Memory- and Imagination-Intuition

17. The longer the interval becomes between the arising of the expectation and its confirmation or correction, the more independently the representations appear—the more liberated they become, or can become, from the impressions that first awakened them. There can then—on the presupposition of sufficient energy—form itself a content of consciousness with a certain independence of the supply from without, a circle of free representations.

Just as little as the sensations do the representations subsist independently of one another. They are members in a train of representations. They call one another forth and connect themselves with one another, and by reason of their mutual connection (association) a new kind of intuition becomes possible, *memory- and imagination-intuition.* In many cases one can fairly distinctly point out the process of association by which such an intuition forms itself; but the formation can also take place so rapidly that a total image appears in consciousness at a single stroke. This holds especially in inspired imagining, where nature operates "as if it were art," performs deeds that in other cases would require conscious deliberation, if they could be performed at all. Only by roundabout ways can one, in inspired production, gain an insight into how imagination-intuition has come about. And the matter becomes still more complicated by the fact that urge and interest cooperate in all memory and imagination, and that this cooperation here takes place more hidden than in sensation and recognition. The whole past of conscious life operates along with it. The psychic energy is therefore of a higher degree than in the more elementary processes.

Memory- and imagination-intuition is, like sense-intuition, a form in which the energy of consciousness solves the task of combining a multiplicity of elements. Both the unity and the multiplicity are to come into their own. In memory and imagination a more or less rich content of experience and representation is worked together into a

[21]Cf. my *Psykologiske Undersøgelser* (Vid. Selsk. Skr. 6. Række), pp. 8–26. *Katz* in *Zeitschr. für Psychologie und Physiologie der Sinnesorgane* XL., 6. 1906. p. 432. *Herrlin*: *Minnet.* pp. 128–218.—*H. Bergson* nonetheless presupposes, in his critique of the theory of an immediate recognition (*Matière et Mémoire.* p. 91 f.), that all recognition presupposes a free representation.—Besides the kinds of recognition discussed above (immediate, partial, and mediate) there are several kinds. In *Psykologiske Undersøgelser*, p. 35 f., I have attempted a systematic survey.

whole, often in such a way that a quite different ordering comes about than the original one. Such total images have their definite directions, determined by the interests or purposes that cooperate in their formation. Every time we lay a plan, we make use of the past to prepare the future and transform, often without ourselves knowing it, the given representations in such a way that they can enter as members into the new whole. When it is intellectual or aesthetic interest that determines the ordering, it is not an outer end that is aimed at, but the involuntary endeavor is directed at the intuition's becoming as clear or as characteristic as possible.

Once an intuition has been formed, there will be a striving to hold it fast as long as possible. Under the influence of new experiences, incitements, or associations of representations, individual members are perhaps shed or taken up, or else the mutual relation of the members is altered. The direction of the intuition can also be changed. Just as in quite simple organic beings, which have no proper limbs, displacements of the organic mass in certain directions can take place, so too the intuition can be articulated functionally without being replaced by analyzing and judgment. A simple example from the domain of sense-intuition is offered by the so-called Schröder staircase figure, which by changing accessory sensations and associations of representations, or by altered attention, is seen now as a staircase, now as an overhanging wall. When we stare long down into the water from a bridge, it can come to look as if we ourselves and the bridge were moving in the direction opposite to the water's real movement. According to changing moods or outer occasions, an image that stands in our memory or our imagination can thus shift without our noticing it. It can be one of our earlier experiences that is thus altered, or the image we have formed of a poetic figure. Our memories and imagination-images have their own history, whose individual stages it is not always easy to trace, any more than it is easy to find the grounds for their shifts. Their history takes place in the involuntary mental life, right down in the ground of nature within us, and reflection can here often only ascertain that changes have taken place, when it holds together two mutually divergent intuitive images of the same object from different points of time in our life. In other cases reflection can in time become aware of the change that threatens to set in, and work to prevent it. The astronomical world-pictures, which from the first formed themselves involuntarily, are an example of this. Scientific hypotheses often have their advantage in their intuitability and in the support they have in sensation's apparently immediate testimony, and there is therefore often a hard struggle before they are let go. To a still higher degree this holds of religious intuitions, in which a whole host of life-experiences—experiences of the course and value of existence—have formed themselves into great images. Here too the most important thing—both as regards the intuition's original coming-into-being and its alterations—takes place right down in the involuntary mental life, although reflection can enter the lists for the right to hold fast, despite everything, the images that have been laid down. In the relation between intuition and reflection many individual differences make themselves felt. In some it is feeling that holds back, and reflection is then forced to fight for the images in which it has found rest. In others it is the urge for intuitability that opposes reflection's intervention[22]. In others again there is such spiritual elasticity that emotional interest and the urge for intuiting relatively easily give way to reflection.

In animals the capacity to help oneself with functionally articulated sense-intuition probably has preponderance over the capacity for reflection, if the latter occurs at all except in the most highly developed animals. The dog that runs toward a man whom it "believes" to be its master will, during the running, constantly alter its intuiting, according as the man's features become more distinct, and when these features finally show themselves to be quite other than the master's, a disappointment will arise; but whether the free memory-image of the master thereby appears in its opposition to the sense-image of the stranger is impossible to decide. There is no reason to assume a proper judgment or inference.

What I call articulated intuiting, or functional articulation of the intuition, is nonetheless regarded by several authors as a judgment. But it is then in any case a judgment of a quite different kind than the proper judgments that rest on reflection. When *Gassendi* thus ascribes to the dog that "believes" it meets its master a capacity to judge, he adds: "An affirmative judgment is, after all, nothing other than the apprehension of a thing with an addition or a property." The dog sees, Gassendi holds, the man immediately as its master, and this seeing or representing (imaginari hominem herum) is—according to Gassendi—the same as forming the judgment that the man is the master (enunciare hominem esse herum); yet this judgment is passed tacitly or potentially (tacite vel virtute)[23]. In a similar way several psychologists express themselves. *Lloyd Morgan* thus distinguishes between perceptive and conceptive judgments. The perceptive judgment (perceptual judgment) is what I call articulated intuiting, in that it consists in a single feature's becoming predominant in a previously formed image. A conceptive judgment, on the other hand, presupposes a conscious distinguishing between individual features; only this, therefore, is a proper judgment. What Morgan calls a perceptive judgment, *Ribot* calls the logic of images (logique des images). This expression is more fortunate,

[22] Cf. on the opposition between intuitive and reflective natures my *Religionsfilosofi* § 95.

[23] *Gassendi*: *Opera*. Lugd. 1658. II. p. 411.

since the image-forming, intuiting process is decidedly the essential thing in the phenomena here in question. They belong under intuition, not under judgment[24]. The expression articulated intuition seems to fit best, since the intuitive image need not be dissolved because a single element is strongly brought out. If a new articulation within the intuition is not possible, a new intuition can form itself, within which the changes come into their own. Possibly the earlier intuition then vanishes entirely from consciousness. If, on the other hand, it is held fast or reproduced, a comparison and an analysis can get going, which may perhaps lead to a judgment.

The articulation stands as a preparation for, or a presage of, an analyzing and judging. Thus what at later stages of development becomes a special organ is, in embryonic development, intimated in bulging parts of the still uniform and continuous tissue.

18. In itself one might suppose that the intuition-formations, with their shifts and articulations, could be continued to infinity, without any judging conditioned by reflection being necessary. It must be taken as a simple fact that this is not so, and that—at least in human beings—a necessity for reflection and judgment sets in. The equalization and equilibrium that can be attained through intuiting does not always show itself to be sufficient. Conditions can arise that make the combining have to take place with clearer consciousness than happens in the involuntary and half or wholly unconscious process by which the intuition comes into being. In sense-intuition, recognition, memory- and imagination-intuition, it is really only the result that appears clearly in consciousness, while the formation-process itself takes place predominantly below the threshold of consciousness.

Reflection is in a certain sense not productive. It only brings to clear consciousness what is already contained in the involuntarily formed intuitions. It compares and tests, chooses and rejects, connects and separates, fills out deficiencies and determines exactly the mutual relation of the elements. The new that reflection brings and expresses in concepts, judgments, and inferences is the clear consciousness of the individual parts of the content of consciousness and of their mutual relation. Through the purgatory of reflection it can soon show itself that we know more than we thought we knew, in that there can lie in the content of consciousness unsuspected presuppositions for orderings and inferences—soon that we know less than we thought we knew, in that incompatible elements or invalid connections are discovered.

[24] *Lloyd Morgan*: *Animal Life and Intelligence.* p. 328 ff.—*Th. Ribot*: *L'évolution des idées générales.* p. 32 ff.—*Hobhouse* (*Mind in Evolution.* p. 200) ascribes to higher animals a capacity for "practical judging." But he adds cautiously that if animals have no proper [free] representations, they must have something that can perform a corresponding function.

Judgment is the form in which the content of consciousness can come into equilibrium anew, when reflection has caused a dissolution of the intuition. In a judgment different elements of consciousness are brought into definite and conscious connection with one another. Insofar as the appearance of judgment is a sign that the wholeness and equilibrium of the intuition no longer subsists, it can be regarded as a sign of an imperfection, especially since the single judgment cannot express the whole fullness that can lie in an intuition. Perhaps an infinity of judgments is required for the whole content of consciousness, with all its nuances, to come into its own. *Goethe* thus sets "das zersplitternde Urteil" in opposition to "die stille Fruchtbarkeit" in immediate intuiting, and *Wundt* even defines judgment as the dissolution of a composite representation. But judgment in reality signifies a new formation of a whole, which has become necessary because the tension within an involuntarily formed intuitive whole has become so strong that a dissolution must set in. It has a similar advantage over intuition as intuition has over fusion: both the differences and the likenesses in the content step forth more distinctly than before. The formation of judgment is therefore a higher expression of psychic energy.

There must of course be a close connection between the judgment and the intuition it builds on. One often expresses this by saying that the judgment "lies in" the intuition, or that the intuition is a potential judgment, as Gassendi said. The difference between intuiting and judgment is that the latter lets the elements step forth independently, so that a distinction can be made between "subject" and "predicate," while in the intuition a more continuous connection prevails. When I believe I know a person who approaches, I see him immediately before me as the one I take him to be; but if I feel an urge for a closer testing, I institute a comparison that distinguishes between the figure that approaches and the memory-image of my acquaintance, and thereafter there is the possibility of a positive or a negative judgment.

But the way from intuition to judgment can be long and toilsome, and it is not said that the "possible" judgments can become actual. There can be a logical possibility for a judgment without a psychological possibility being present. Thus it is said to Peer Gynt: "We are thoughts; you should have thought us!"—The logical possibility is present when the intuition contains elements that can be separated out, and whose mutual relation can receive a closer determination. But this separating-out and determination is undertaken only when there are sufficient conditions for an abolition of the whole equilibrium of consciousness that the intuition denotes. Here, as at the first arising of consciousness, there must take place an interruption of the uniform state, in order for a further-reaching mental work to get going. A special interest must attach itself to individual elements, so that an urge arises to become clearly conscious of their mutual relation. Thus, e.g., if an intuitive image contains something that could become a means to the attainment of a purpose, or serve for a better understanding of another intuitive image, or form a contrast to it.

The judgment must correspond to the intuition, since it is to express its content in another way, more appropriate under the given conditions. But this does not exclude that it is an essential change that takes place in the transition from intuition to judgment. When one does not embrace the psychological atomism that I have combated above, it must be clear that the elements of consciousness undergo essential changes by passing into new connections, and the connections of judgment can be quite other than those of intuition. The elements are not indifferent to the connection in which they stand. In the intuition an element stands in a single special connection; the judgment brings it into connection partly with elements that lie within the same intuition, but perhaps at great distance, partly with elements from other intuitions. In the intuition I dwell, e.g., on a man's action as a total image and am absorbed in the mood this awakens in me; in the judgment "he has acted nobly!" I make use of a standard with which I more or less consciously compare the action. The validity of the judgment presupposes only that from the relation established between subject and predicate one can infer back to the intuition that lies at the basis, so that the judgment can again lead to a construction of a similar intuition in those who hear the judgment without having had the intuition at first hand. There must be analogy, but not necessarily identity, between judgment and intuition.

d. Association of Ideas

19. The content to which judgment corresponds does not always appear in the resting form of intuition, and as a surveyable whole. Judgment can be formed out of a series of associations of representations, in which intuitive images of greater or lesser extent replace one another more or less rapidly. The difference between association and intuiting is only a difference of degree. The more the successive gains the upper hand over the simultaneous, the more association—which in its higher degrees becomes a flight of thought—steps in place of intuition. What intuition is in static form, association is in dynamic form. Already in the formation of memory- and thought-intuitions associations cooperate, and no less is this the case in shifts and articulations, as well as in the dissolution of the intuition.—But we now regard association as an independent form of

the involuntary conscious life.

Just as involuntarily as an intuitive image can emerge in consciousness, just as involuntarily can one representation draw another after it, this a third, and so on. In both cases laws operate that do not themselves come to consciousness. The articulating intuition stands closest to association; but in it too the dynamic is still subordinate to the static, since the shift quickly leads to a resting image.

The transition from association to judgment takes place through the relation between the members of the association-series becoming the object of distinct consciousness. The urge for this will already arise from practical interest, as when the preceding members stand as means in relation to the following ones. Where merely practical interest prevails, one will attend to the preceding members only insofar as they indicate means to the attainment of an end, and the whole order in which the members are brought will be determined no longer by their involuntary sequence, but by their relation to the end. It is, as *Hobbes* remarks, the thought of a purpose (cogitatio finis) that makes us, in the waking state, order our representations in a definite and coherent way, while in dreams, where the firm thought of purpose is lacking, they can unfold more chaotically. For a thoroughgoing ordering there will be required holding-together and comparison with other association-series and intuitions than those involuntarily setting in, and the more such series and images thus interact, the greater the psychic energy that comes to light.

Every attempt to order, by way of reflection, the members in the involuntarily arisen association-series consists in a testing of the members' relations of likeness and difference. Even if the objects a and b have occurred together ever so often in my experience, so that a has an irresistible tendency to call forth the representation of b, the justification of expecting b when a has set in will nonetheless depend on whether the a that earlier occurred as predecessor of b is identical with the a that has now set in, and further on whether it can be shown that a cannot set in without being followed by b. The greater the continuity between a and b, so that they really form a whole, the more justified the expectation will be. If it is an association by likeness that we have before us, the transition from the one member (a_1) to the other (a_2) will be the more justified the more the likeness approaches identity; it is then merely a transition from one example to another. Thus association approaches judgment the more the transition from object to object is determined by continuity and identity. In the genius, the formation of judgment can proceed as if it were an involuntary association-process, since the psychic energy is released with great ease and in great fullness. Reflection is here limited to a minimum. In everyone, practice in a definite train of thought can operate in the same direction.

Where reflection can be spared, it is spared.

20. Association of representations is, just as much as reflection, an expression of psychic energy, and it is wrongly that one has regarded it as a purely passive process and set it in absolute opposition to "thinking." What happens in every association is really that a whole, a larger connection, is built up, constructed out of a single given object that occurs in a certain isolation. There are indeed always several objects or elements that cooperate at the springing-forth of an association-process; but a single one will be the decisive one, and for simplicity's sake we can keep to it. Consciousness has both an urge and a capacity to go out beyond the single object or element and connect it with others. In this instinctive tendency the psychic energy expresses itself. All association can be traced back to a transition from part to whole; what we call association by likeness and by contiguity are only limiting cases. In every association relations of likeness and of contiguity operate together, in such a way that now the one, now the other has preponderance[25]. The urge for the release of energy is, in the domain of the life of representation, so strong that a minimal occasion can operate as a release. There is here an analogy to the urge for movement. Now the incompletely given is filled out (by predominant association by contiguity), now one circles around it through dwelling with kindred representations (by predominant association by likeness). In both ways the single element can be incorporated into the whole psychic world, and its relative isolation can be abolished. Which of the different paths is taken in the particular case—and which special relations of likeness or contiguity come to operate—nonetheless does not depend on purely intellectual conditions. Like a hidden weight, interest can press the scale-pan down for a single definite relation of likeness or contiguity that in itself would not be decisive. It is ultimately man's striving that determines his associations of representations, even though these, once they have got going, can in turn alter the strength and direction of the original striving.

When now association is replaced by judgment, the relation is reversed. It is no longer a transition from part to whole that takes place, but a transition from whole to part, that is, an analysis. Association is, like intuition, an integration, but judgment presupposes a differentiation. In the association-process it holds that a little tussock can upset a great load; in the formation of judgment a needle is found in the great load. The associations can bring forth the whole content of consciousness, but the formation of judgment draws out a single thread from the rich web. Association is productive, the formation of judgment is critical.

[25]See more closely my *Psykologi* V B., 8.

Yet neither is this opposition complete. The associations are—as already remarked—not determined exclusively by a single element. This is seen already from the fact that one and the same representation does not, under all conditions, awaken the same associations in one and the same individual. And on the other hand the formation of judgment presupposes that attention is directed toward a single element's relation to other elements (in or outside one and the same association-series), and so far it too can be said to begin at a single point, with the little tussock.

Yet another opposition has been found between association and judgment, in that the former is supposed always to be able to be continued, since there is no inner limitation for it, while the latter is supposed to denote a definite conclusion. "Association," says *W. Jerusalem*[26], "is never concluded in itself. Ever new members attach themselves, ever new deviations from the original direction take place, and nowhere is there found a natural conclusion lying in the very nature of association ... In judgment a process is released from its connection with the rest of the content of consciousness; it is isolated and, as it were, concluded for consciousness. We are for the present done with it, and precisely this concluding moment in the act of judgment forbids regarding it as an association." Against this it must be objected that an association-series can very well find its natural conclusion, namely when the motive that has called it forth is fully satisfied. One can also reach a representation that perhaps lies far from the starting-point, but that offers an independent interest and therefore takes attention captive. When one takes into account the interest, which is never lacking in the association of representations, an inner conclusion of this can very well be given, not to mention that the relations of likeness and contiguity that are at one's disposal in the particular case will always be limited. On the other hand, judgment does indeed denote a settling of accounts, a decision conditioned by reflection, but only a provisional decision, since new reflection can correct it or else bring it into connection with other judgments and thereby unfold what "lies in" it.

e. Comparison

21. The transition from intuition or association to judgment takes place through reflection, which consists in an analysis, a successive turning of attention to the individual objects and elements in the intuition or in the association-series. Thereby free comparison becomes possible, in distinction from the elementary comparison in sensation and the bound comparison in recognition.

[26] *Die Urteilsfunktion.* p. 79; 82.

There lie two objects before me on the table, which I can see at a single glance. But although they are both within my field of vision, I can fix my attention now on the one, now on the other, move back and forth between them without necessarily moving my eyes. The one object does not fall out of the field of vision because attention is directed toward the other. By such alternation of attention an experiment is made, whose result is that the two objects stand either as similar or as different. The likeness or the difference springs forth for us as a peculiar quality, an element that was not present as long as we regarded the objects each by itself without holding them together. It is this holding-together that brings it about that the new property we call likeness or difference comes forth. The older English psychology (especially *James Mill* and *Stuart Mill*), which laid the whole weight on the individual representations each by itself, could not explain how comparison was possible. For them, relation-representations differed from other representations only in that they presupposed two "things" instead of one.

The active element in comparison—apart from the energy that holds together the elements that are compared—is one with attention. This is a successive concentration toward the different members of the intuition or the association, a concentration that precisely by reason of its alternation lets the content between the elements step forth distinctly. Experience shows that we, all else being equal, apprehend a successive relation more distinctly than a simultaneous one.

If the objects we compare are not given simultaneously, the comparison comes about through the one member being reproduced while the other is directly observed. Memory thus provides the simultaneity within which attention can then swing.

If, at the passage of attention from the one object to the other, no difference at all appears—apart from the exertion involved in the alternation of attention itself—the relation between the two objects stands as a *relation of identity*. Identity in the strictest sense is the likeness an object has with itself. When different objects are said to be identical, the meaning is that it is as if attention returned to the same object; and conversely, that an object is the same means that attention can return to it without any difference appearing.

If it is emphasized that the two objects, despite their lack of difference, nonetheless occur at different times or in different places or both, their relation is called *Sameness* (*Congruence*).

If some of the two objects' parts or properties appear as identical or the same, others as different, we get a composite *Likeness.* It has sometimes rightly been held that all likeness—apart from identity—was of this kind.

If a likeness is present without there being identity or sameness at any point, we can call the relation *Qualitative Likeness*. It is a peculiar kind of qualitative difference. The remotest degree of likeness is *Relational Likeness* (*Analogy*), where the likeness does not concern any single property or part in the two objects, but the relation between properties or parts in the two objects.

22. Comparison can take place involuntarily or voluntarily. Comparison is involuntary when no intention, no preceding representation, no wish to find out likeness or difference makes itself felt. Voluntary comparison presupposes that one has set oneself as a task, as a purpose, to find whether there is likeness or difference between two objects. The transition from involuntary to voluntary comparison is formed by such cases in which the one object is reproduced, so that it is merely held together with the other. It is a transition reminiscent of the transition from immediate to mediate recognition. In mediate recognition a representation is released, which thereafter covers a following sensation. An analogy in the domain of will is formed by the transition from urge or instinct to drive (when by drive we understand an urge connected with a representation of its goal). And here there is really more than analogy; for there is an urge to compare just as to act, and in both cases it can be of importance that one becomes conscious of what one feels an urge for.

Now it is not always the case that both the objects to be compared are given or can without further ado be called forth in memory. Often only the one object is at one's disposal, and the task is to find another object that stands in a definite relation of likeness or difference to it. The more clearly and distinctly this relation can be represented, the greater the possibility that the new object can be found. The most exact indication of that relation we have where the task can be put into an equation. We then have a definite standard to decide whether the sought object appears or not, and by a methodical procedure we can perhaps force out the object that satisfies the demand. In a more indefinite way something similar happens when we wish to call forth something in our memory and know of it only that it stands in a certain connection with a representation already present in us; by concentration of attention on this, or, as it has been called, by a boring, the sought thing can then spring forth. The psychic energy appears quite especially in such cases. The point then is not merely to connect given members, but to find a member that is perhaps quite new, but that stands in a familiar relation to a given member. Where we can indeed determine the relation to the given member, but on account of the limitation of our experience cannot find the new member, there we end in a problem. Thus we are placed, e.g., when we assume that in the psychic domain there is something corresponding to the difference between potential and actual energy in the physical domain; for of potential psychic energy we can form no positive representation; we can only say that it is related to actual psychic energy as potential energy is related to actual energy in the physical domain. (Cf. above 8–9.)

In all comparison an urge makes itself felt, even if the finding-out of relations of likeness and difference is not directly set up as a purpose. It is an urge for clarity and distinctness, for the use of the combining capacity that constitutes the innermost essence of consciousness. Often, and especially under the most elementary spiritual conditions, it is a downright urge for self-preservation that leads to thinking. When something that has value for the subsistence of life does not immediately fall to our lot, we must make it a purpose and strive to find the means for its realization. In the relation between end and means, man meets for the first time with necessity: if he wills the end, he must also will the means. The urge for the end begets the urge for the means. Will and thought here go together. This meeting has not merely practical significance. Practically regarded, it is a roundabout way that man here comes to make; but this roundabout way points to a definite unavoidable connection that can become the object of an independent interest, a joy of cognition. In this way the intellectual interest can, through displacement, develop out of the practical one (Cf. 6).

When it is a matter of determining the relation between two objects or of finding an object that stands in a definite relation to a given object, several possibilities will often make themselves felt. Several different possible relations or objects emerge, and a choice must then be made, which is determined by which of them best satisfies the definite conditions. The choice is determined by a comparison between what is required and what offers itself. If the two objects are given, the point is the fixing of the relation that takes place between them; if only the one object and the relation to the sought object are given, the point is the fixing of which object stands in the definite relation to the given object.

As far as possible, comparison leads to an Either—Or, so that there is a choice between two possibilities. The other possibilities are rejected by a preliminary comparison, or they show themselves able to be subsumed under one of the two main possibilities. We stand here before a fundamental peculiarity of our thinking. This moves forward through two-membered relations. A predicate must either be ascribed or not ascribed to a subject. No concept can at once contain and exclude a certain mark. A grounding of this peculiarity of our thinking cannot be given. It is evidently bound up with the psychological fact that attention, during reflection, can be concentrated only on one element at a time, and that comparison takes place through the passage from one element to another. *George Boole* has called the logical principle springing from this necessity *the Law of Duality*. Before it even the most pronounced practical interest must bow when it wishes to take reflection into its service.

Boole shows in an interesting way the purely factual character of the said feature of our thinking. Starting from the fact that the combination of a concept with itself has no logical significance, he finds the nature of our thinking expressed in the equation $x^2 = x$, which is satisfied only by $x = 1$ or $x = 0$. Here 1 means All, 0 Nothing. The derived equation $x(1 - x) = 0$ expresses the Law of Duality. A combination that should at once contain a concept and its negation is invalid. But if now the fundamental equation had not been of the second, but of the third degree, our thought-activity would be otherwise constituted than it in fact is. We would then not work our way forward through two-membered relations, holding together two elements at a time and affirming or rejecting their likeness (or difference), but through three-membered relations, trichotomies instead of dichotomies—a procedure of which we can form no real representation. Boole by no means wishes, by this presentation, to teach that the fundamental method of our thinking is accidental or arbitrary, but only that it must be taken as a simple fact[27].

Already earlier *Salomon Maimon* had shown that, on account of the unity of our consciousness, what is to be connected by our thought in a definite way can immediately consist only of two members. Every immediate comparison that leads to a definite connection concerns two immediately consecutive members. A mediate connection can take place only after a series of immediate connections; thus a mediate connection comes about between a and d when there has successively taken place a connection of a and b, b and c, c and d[28].

By Maimon's and Boole's remarks, the law of the excluded middle between two mutually negating assumptions—already set up by *Aristotle*—is illuminated with respect to its connection with the whole nature of our reflection.

23. Besides the object there are, as already said several times, in all reflection two moments that stand in a constant relation of reciprocal action: the form of the thought-work and the motive that awakens reflection and drives it forward.

A motive or an interest that could not be satisfied by the laws and forms that human thinking must by its nature follow will in the long run be without intellectual significance. Such a motive can perhaps become fruitful for imagination, for the capacity to form images and symbols, but it would be a misunderstanding to attempt to force thought

[27] *Boole*: *An investigation of the laws of thought.* London 1854. pp. 48–50.
[28] S. *Maimon*: *Versuch einer neuen Logik oder Theorie des Denkens.* Berlin 1794. p. 73 f.

onto paths it cannot by its nature walk. Fortunately, "reason" cannot be taken captive forever; it will with inner necessity burst its prison, and its captivity will not be of benefit to the interests that were to be served.

The laws and forms of reflection are the same, whether the thought-work takes place involuntarily or voluntarily, half unconsciously or with full clarity about goal and ways. These laws and forms operate, as we have seen, in analogous forms already in sensation, in memory and imagination, and in association of representations. That the thought-work, the comparison, takes place "with will" entails merely greater definiteness and firmness in the synthesis and greater clarity in the transitions from member to member. However significant the transition from the involuntary to the voluntary can also be, it nonetheless does not mean that a wholly new "capacity" should come into force.

One of the older psychologists who has laid most weight on the will-element in thinking, *Charles Bonnet*, has stated the proposition: L'attention est la mère du génie. But however decisive concentration in a single direction is, it nonetheless gives only the introduction, the impulse. No less important is it that, as a consequence of this impulse, a series of intuitions and associations is involuntarily released, which can then become objects for reflection. In the content of consciousness thus emerging, new elements and new connections can come forth that show themselves able to pass the purgatory of reflection, and genius often consists precisely in the richness and depth of the involuntary intuitions and associations. *Kepler* and *Fechner* are examples of inquirers whose guiding thoughts first came forth in the form of imaginative combinations, which through critical testing led to significant results. Such minds unite the freedom and freshness of the involuntary with the exactness of attention and comparison. Genius can finally also express itself in the indefatigability and carefulness with which the working-over of the objects at hand takes place. Regarded from this side, *Buffon* was right when he defined genius as great patience.

We easily come, in our psychological theories, to emphasize the difference between the involuntary and the voluntary more strongly than close observation warrants. The transition itself from involuntary to voluntary comparison of course takes place involuntarily: the representation of a goal that is the condition for voluntary striving—the one that *Dante* called "the will's first thought"—must itself first arise by reason of involuntary association. Here can lie a significant turning-point in the life of thought and in the whole development of the mental life; but it does not justify assuming a break in the continuity.

C. Judging

a. The Formation of Judgment

24. It has already been mentioned several times, provisionally, how intuition and association can be replaced by judgment. Judgment is a higher solution of the task that is solved in more elementary forms through fusion, sense-intuition, memory- and imagination-intuition, and association. Such a higher form becomes necessary when neither the likeness nor the difference between the objects fully comes into its own in those forms.

Judgment is a synthesis, a connection of objects that, prior to the judgment, have occurred in one or more of those simpler forms. The connection can only be clear and definite when the objects are each by itself clear and definite. When we become clearly conscious of an object (or a group of objects), so that our representation of it does not vary in the different cases in which it appears, we have formed a concept. So far it might seem that concept-formation would have to precede the formation of judgment. But a closer consideration will show that there is a constant reciprocal relation between the formation of judgment and the formation of concepts, or rather, that these two processes are at bottom not different.

Concepts as well as judgments are formed by analysis of intuitions or associations. When we become clearly conscious of an object that is given in a single state, through attention's successive turning to its different elements (all that can be distinguished in the object), we form a concrete individual-concept; when we become clearly conscious of an object that has appeared in a series of different states, we form a typical individual-concept, in that a single state comes to stand as an example of the whole series; when we hold together and compare a whole series of objects and can regard a single object as an example of the whole series, we have formed a general concept. Between the different examples of one and the same typical or general concept there need not be identity, only analogy. It is only in certain definite respects (indicated in the concept's definition) that the examples can take the place of one another. In place of the example there can step a schema, a symbol, or a word. They are directions for finding examples, when one wishes to save the time that the more detailed representation of such an example takes. The main thing is that one keeps, in one's attention, to the elements that can approximately recur in the different objects or in the different states of the single object, even though such elements by themselves alone would not be sufficient to constitute a content of representation. But it is then really by a series of judgments that concept-formation takes place. In order to form concepts we must connect elements in one or more objects in a clear and definite way. It is by judgments that we establish that an object presents certain elements—that the same object occurs in different states—and that different objects present the same (analogous) elements.

We seem then to move in a circle: judgment presupposes concepts, and concepts are formed by judgments! But the matter is that concept-formation and the formation of judgment are really one and the same process—reflection's analysis of what is given in intuition or association. The difference lies merely in the purpose and direction of the analysis. Now it is a chief interest to find out the elements that the objects contain and gather them into a whole, and then we speak of concept-formation; now our interest is essentially directed at becoming conscious of the relation between different objects or states of the same object, and then we speak of the formation of judgment.

Insofar as a difference can be made between the two processes, they stand in constant reciprocal action. The clearer the concepts can become, the clearer too can their connection become in the judgment; and the clearer a connection can be shown through judgments between a concept and other concepts, the more perfect the concept itself becomes.

The difference that can nonetheless always be made between concept and judgment is a new example of the antinomy that—as earlier shown—lies in the nature of consciousness: synthesis presupposes elements, but these are only what they are by being comprehended by the synthesis. Hereby is explained the constant unrest in the life of thought, and in the mental life as a whole; ever new tasks are set: now that of expanding the content of consciousness, now that of bringing about a more intimate connection within the content.

25. The connection that judgment brings about is due to a recognition. Two concepts can be connected into a judgment only when the one can be found in the other. The one concept then denotes a point of view from which the other can be seen, a closer determination of it. Here again it is seen that thinking is comparison. When I say "A is B," I must have found B in A. This re-finding can have different degrees or forms. It can show itself that A and B have wholly the same content, so that it is only due to limited or inexact apprehension when they earlier stood as different concepts. Or B can be a single one of A's marks, hence be a means of recognizing A (by a kind of partial recognition). Or the connection of A and B presupposes that account is taken of a larger connection within which A and B belong, so that only under certain conditions, which cannot be derived from A or B alone, can one find a relation of identity; it is then not A and B that

are identified, but A_x and B_x[29]. The two first-named relations might seem to belong to Kant's "analytic" judgments, the third to his "synthetic" judgments. But the critique of this Kantian division of judgments has shown that no real boundary subsists between them. Even in absolute identity (A = B) there will always be certain presuppositions and limits for the identity. The difference between the three named kinds of relation will in particular cases depend on how far back one goes in the series of presuppositions.

When, with *Leibniz*, one conceives judgment as the expression of a relation of identity, its kinship shows itself clearly with the forms and processes that form its psychological basis. Like these, it is a form in which consciousness seeks unity and harmony in its content, and it is, like them, a stage in the constant breaking of likenesses and differences. It lets both the relation of likeness and that of difference come into their own to a greater degree than the more elementary forms were able to.

Judgment itself can enter as a member in reflection's continued work. Inference is related to judgment as judgment is to the concept. Judgments can be connected into inferences when identical concepts can be inserted from the one judgment into the other. The relation between premises and conclusion is analogous to that between subject and predicate in the judgment, and that between intuition and concept. At all stages reflection can lead only to a higher degree of consciousness of what is contained in its objects. In the concept it becomes conscious of the intuition's content, in the predicate of the subject's, in the conclusion of the premises'.

b. Obstacles to the Formation of Judgment

26. No judgment is formed when two intuitions or two members in an association-series replace one another in such a way that the first object has vanished when the second steps up. If the dog mentioned earlier, which "believed" it was going to meet its master, had wholly forgotten this "belief" when it turned out that it was a stranger who came, it would feel no disappointment and would have no occasion for any judgment. And even when the conflicting objects, the representation of the master and the sight of the stranger, meet and suddenly replace one another, so that a disappointment is felt, it is not said that this leads to a judgment. (It would have been the negating judgment: "it was not my master.") The contrast, the collision, can constitute the whole experience. Experience shows that one must to a certain degree have freed oneself from an experience in order to be able to convert it into a judgment. It is more than rhetoric when *Shakespeare*, both in a comedy and in a tragedy, puts utterances in this direction into the mouths of his

[29]This tripartite division is due to *Jevons*, *Principles of Science* (1874), Chap. 3.

characters. "Silence is the perfectest herald of joy: I were but little happy, if I could say how much," says Claudio (Much Ado About Nothing II, 1). "The worst is not so long as we can say, 'This is the worst,"' exclaims Edgar (King Lear IV, 1). What in memory, or in the mind of the outside spectator, forms itself into a judgment need not, in the experience itself, be formed so. It is a frequent psychological error that one reads judgments into where the immediate experience laid claim to the whole energy of the mind. We have met this error already in immediate recognition, which one has also wished to conceive as a judgment. Disappointment can, like every relation of contrast, be formulated as a judgment, but need not entail one.

Far sooner than entailing a judgment that expresses what has happened, disappointment will—when it does not lead to complete slackening, but when there is energy still to spare—lead to a more or less impatient demand, a claim that the expected shall come. The collision between expectation's rich image and impoverished reality begets the demand. In children the demand appears not only before the judgment, but also before the question. Or rather, the question is at first a demand. The purely theoretical question always presupposes a certain development—a cooling-down.

The question can be a determination-question, in that a whole stands before consciousness within which there is a part that is not filled in. There is lacking, e.g., a time- or place-determination for an event. Here the answer consists in a factual indication, the bringing forward of an object by which the deficiency is filled. In the decision-question, which presupposes a higher stage of development than the determination-question, the opposition between two mutually exclusive total images makes itself felt, and the answer must be either Yes or No. (Is he dead? Is it ten o'clock?) Yet the difference between the two kinds of question is not absolute; for in the decision-question too it is a matter of the filling-out of a whole, namely a whole that comprises both of the two mutually exclusive wholes. In the question "Is he dead?" it is a matter of whether we are to think of the world of the living with him or without him. Only, this larger horizon here stands more in the background, and that comprehensive world can in turn have different extent.

Negation too appears first in practical form; it has its first germ in aversion, anger, and fear, and appears as rejection. A dim urge lies at the basis both of negation and of the demand. A transition between practical and theoretical negation is formed by the triumphant or despairing ascertainment of what is lacking.

Negation itself, like disappointment, presupposes a transition from an object with a manifold content to another object in which something of this content is lacking, and also a holding-fast of the first object after the second has set in. AB is replaced, e.g., by

A, and AB and A are held together. The negating judgment "B is not here" ascertains the difference. It springs immediately from the opposition between the two objects. In simplest form the negative judgment, like the positive, can be expressed by a mere exclamation: "Gone!"

Probably positive judgments arise before negative ones. Consciousness will at first be more occupied with an increase of its content than with a diminution of it. The transition from A to AB brings a fullness that can be enjoyed or become a motive for action. For a judgment to appear, it is not necessary that the preceding object (A alone) be preserved in memory; consciousness revels in B's entrance, the new positive thing. If, on the other hand, the transition from AB to A is to beget a negating judgment, AB must be held fast in memory, since it is B's absence that is to be ascertained. All else being equal, a negative judgment therefore requires greater energy than a positive one. Energy is required to miss and to grieve, and to demand and to wish. There is lacking here the help that the positive B provides when A is replaced by AB.—The relation becomes of course more complicated when, after the transition from AB to A, A gets a new companion, so that AX appears. What the consequence will be will depend on what relation X stands in to the earlier content of consciousness—whether AX can become an equivalent for AB or not.

We have here an example of a small series in which the order of the members is not indifferent. The transition from A to AB is of another kind than the transition from AB to A. The two processes that lead to positive and to negative judgment can be equally simple, but the psychological motives for positive judging can be present earlier than those that lead to negative judging.

The state or experience that gets its expression in a negative judgment can very well in itself have a positive character. Lack is a positive state; it is only its cause that we express by a negation. There are no negative states at all. The negative numbers denote, after all, merely a quantifying in the opposite direction from the quantifying that the positive numbers denote. The Hindus denoted what we call the positive numbers by a word meaning fortune, the negative numbers by a word meaning debt. The representation of fortune must, from the nature of the case, precede the representation of debt. Life leads to positing before it leads to negating. A logical parallel to this psychological fact is offered by the circumstance that a negative judgment, as regards content, presupposes a corresponding positive judgment, just as a negative concept presupposes a corresponding positive concept. "Alle Begriffe der Negation sind abgeleitet," says *Kant.* The person born blind does not know what darkness is, the ignorant person does not know what ignorance is[30]. A negating judgment gives us no new content, but rejects a positive content, which is thus presupposed known. A negative judgment can therefore, logically regarded, not be placed on a par with the corresponding positive judgment, although it can, as we have seen, build on just as simple a relation of opposition as the latter. It stands, on the other hand, on a level with that kind of positive judgment which rejects a possible negative judgment, and which is often expressed by means of more or less solemnly asseverating words ("truly," "really") or by strong emphasis on the copula. Such affirmative judgments are a peculiar kind of positive judgments. *Hegel*[31] wrongly held that narrative sentences which do not reject a doubt about the correctness of what is narrated do not deserve the name of judgments. A narrative judgment will, when it is more than idle repetition, be the fruit of an analysis of events or reports, hence of a work of thought, even though it does not stand in opposition to a doubt.

Therefore *Sigwart* and *Lotze* (and in antiquity *Menedemus of Eretria*) have rightly denied the negative judgments a place among the logically simple or primary judgments. Sigwart calls the negative judgment a judgment about a judgment, namely a judgment whose content is that another judgment does not hold[32]. When *Hermann Cohen* does not agree with this[33], it is because he overlooks the difference between positive judgments in their simplest form and the special kind that can be called affirmative judgments; with these, not with the former, the negative judgments must, logically regarded, be paralleled.

27. The formation of judgment can, as we have seen, be hindered by the memory of one of the objects falling away, or by not all the objects necessary for the completion of the judgment being given, so that only demand or question are possible. It can further happen that a judgment is formed only to be rejected, so that negation prevails. There is yet another way in which a judgment can be hindered from asserting itself in consciousness, namely when there also appears another judgment with which it shows itself incompatible. A problem then arises. The word problem really means "that which is laid before," "task." In this general sense a problem can already be said to be set when closer determinations are lacking. The demand and the question thus already set problems, in that they set the task of finding determination and decision. In a narrower and sharper sense the problem arises when two judgments—either two positive judgments, or a positive and a negative, or two negative—stand as incompatible.

[30] *Kritik der reinen Vernunft*[2], p. 603.

[31] *Wissenschaft der Logik*. II. p. 76 f.

[32] *Sigwart*: *Logik*[1] I. p. 119; 260.—*Lotze*: *Logik*. p. 61.—Cf. also *Bergson*: *L'évolution créatrice*. p. 312.—On Menedemus see *Diog. Laërt.* II., 135.

[33] *Logik der reinen Erkenntnis*. Berlin 1902. p. 89.

One might call the one kind of problems completion-problems, the other kind liberation-problems. In the first kind the sting of the problem lies in the incompleteness of our world of thought; in the second kind the sting lies in a discord within our world of thought. Common to both kinds is the relation of tension (vital difference) described by *Avenarius* in his natural history of problems[34], which arises through a psychic equilibrium or harmony being abolished.

Psychic energy is required to have problems. Both lack and discord subsist only when there is a capacity to hold oppositions together. If one lets go the one opposed term, neither the incompleteness nor the contradiction is felt. There are problems in all spiritual domains, and it is through their purgatory that spirits are tested.

In this connection we keep to the second kind of problems, that due to the collision between judgments. Where a simple positive and the corresponding simple negative judgment stand over against each other, the rejection of the one will, according to the Law of Duality, be one with the acceptance of the other, and vice versa. B must either be a mark of A or not be one. Where two positive or two negative judgments stand against each other, the relation will not always be so simple. The relation can be many-membered, so that both judgments can be false and it is a third that holds. The rejection of a hypothesis thus does not necessarily lead to the acceptance of a definite other hypothesis, when more than two hypotheses are possible. The task must in such cases be to reduce the different possibilities to two, between which the choice must then be made.

Often the choice is made on negative grounds. One becomes clear that one cannot go the one way, and only therefore does one go the other. Necessity is then expressed negatively. In the history of logic this shows itself in that the negative logical principles, the principle of contradiction and the principle of the excluded middle, were set up before the positive one (the principle of identity). Our criterion of reality often gets a negative expression: we cannot but believe this or that! The primitive ethical commandments have a predominantly negative form: "Thou shalt not—." It has been the task to set up barriers within which life could unfold. The commandments are the tribe's reaction against those who deviate from the paths that have shown themselves salutary. Indignation and disapproval express themselves before recognition and approval. It was above all a matter of preventing encroachments. The prohibitions were of course motivated by positive purposes that were to be promoted; but the negative measures stood as most necessary.

[34]See my book *Moderne Filosofer.* p. 100–104.

Often one uses, out of politeness, considerateness, or melancholy, negative expressions to cover over the crass and sharp character with which a positive reality can announce itself. One then denies the opposite of what one means. It is the so-called litotes. "It is no joke." "I cannot praise him." "It is not going well."[35] When *Luther* at Worms said: "I can do no other," he thereby not only indicated that he had tried all possibilities, but at the same time expressed his resolve in the most considerate form. His conviction received at once the strongest and the gentlest expression.

28. Only those negative judgments are fruitful which are necessary intermediate links for arriving at a positive judgment. But it follows precisely from the conditions of our cognition that we can often reach a positive determination only through a series of negative determinations. "Empty" space was at first a purely negative concept; one could ascribe no physical properties to it. Later such properties, especially electrical and magnetic ones, have been demonstrated. When one says that an organ dies away through "non-use," it is because one stops at a preliminary observation. What in reality takes place is that when the function falls away, the chemical processes in the tissues stop, and the organ then becomes an inactive mass that is absorbed by the neighboring tissues.

The history of classification shows us interesting examples of how we often have to content ourselves with staking off a domain, in that we can say nothing else about it than that it is different from another, definitely and positively characterized domain. It then becomes the task of continued inquiry to find positive marks for the domain thus staked off. *Aristotle* divided animals into rational and irrational, the irrational again into those that had (red) blood and those that did not have (red) blood[36]. *Linnaeus*'s classification of animals seems to support itself on positive marks. He has six classes: Mammals, Birds, Amphibians, Fish, Insects, and Worms. But *Cuvier* pointed out that the sixth class is characterized only negatively. "It is well known," he says, "that Linnaeus united under the designation *Worms* extraordinarily numerous and differently formed animals, for which it was difficult to indicate any common mark. While I was working on my first treatises on comparative anatomy, I came to face the impossibility of saying anything universally valid about the worms' nervous system, or about their circulation, or about their organs of respiration and reproduction, indeed not even about their organs of digestion. Thereby it became clear to me that this class was not, like the others, founded

[35] *Kr. Nyrop*: *Ordenes Liv.* p. 47.

[36] Yet Aristotle did not base his classification on a single mark. He precisely combats dichotomy. *Gomperz*: *Griechische Denker.* III. p. 117.

on positive marks."[37]

The inductive method often moves, like classification, forward through negations, in that different possibilities or hypotheses are set up and tested, and only those that are not rejected under the testing are held fast. *Kepler* arrived at the assumption of the elliptical form of the planetary orbits first after he had rejected various other forms. It was the "method of exclusion" that led to his result.

Between the significance of negation under induction and under classification there is the difference that the negative paths under induction mean paths that are barred, while under classification they mean paths that are left for later inquiry to travel. Perhaps our cognition ends finally at a crossroads where it can get no further, neither as classification nor as induction. We would then stand at a conclusion of the process that began with the first, primitive distinguishing. Through the forward-driving force of differences and oppositions, lacks and contradictions, ever higher forms of psychic energy have been developed, until the limit is reached where perhaps pure speculation or poetic analogizing still develops thoughts about the nature and worth of existence, but where experiential confirmation of our judgments is no longer possible.

That negation plays so great a role during this whole development comes from the fact that we have to orient ourselves in existence from a definite standpoint in time and space, and from definite presuppositions that are given with the very nature of thought. The richness of existence is too great for it to be surveyed at a glance or by a single progressing series of intuitions. On the other hand, thought can see more possibilities than experience shows us realized in the existence we know. Now it is our all too limited points of view, now our all too comprehensive ideas, that must be negated or at least corrected, before true cognition can be attained.

c. Subject and Predicate

29. A wholly definite point where judgment replaces intuition or association cannot be demonstrated. There is a multiplicity of transitions in the degree of attention that is directed toward the relation between the elements in an intuition or in an association-series. The boundary is formed on the one side by the immediate and involuntary streaming-along, in which neither objects nor processes are distinguished, and on the other side by hypnotic or ecstatic absorption in a single element. Between these two boundaries lie a series of states that contain more or less favorable conditions for the arising of a judgment. The decisive thing is that, however much attention is directed

[37] *Cuvier*: *Sur un nouveau rapprochement à établir entre les classes qui composent le regne animal* (1812).

toward a single member, the connection with the whole intuition or the whole series in which it occurs is nonetheless preserved, so that its relation to the whole can stand clear. Continuity is the constant presupposition for a fruitful application of attention in a single definite direction. It does not go, as older presentations of logic might mislead one to believe, that one first knows the individual members each by itself and then knows the relation in which they must stand to one another when they are to constitute a whole. If there were not beforehand a whole that enclosed the members, it would be inexplicable how thought could come to connect them. What thinking can do is only to bring the relation between the members of a given whole to clear consciousness, and this may perhaps require a rearrangement of the immediately given relation between them.

One has often thought to find a definite mark of distinction between judgment and the simpler mental processes we call intuition and association in this, that judgment is expressed in words. But the significance of the word for thinking consists essentially in the help it provides toward holding fast the typical and general in its difference from the concrete and the individual. Herein it is presupposed that a comparison of the objects in question has beforehand been instituted. And such a comparison is also necessary in order to decide the right to transfer a word from one object to another. The judgment is expressed by the word, but does not owe its coming-into-being to the word.

The word is, in its simplest form, an exclamation in which the state of mind called forth by an object gets air. An exclamation common to several individuals can become a means of communication. And this means of communication can in turn become a means of thought, a means of holding fast a thought in its limitation, in particular of holding fast the difference between points of likeness and of difference within a circle of objects. Hereby the word comes into so close a connection with thinking that it is no wonder it has been difficult to keep the grammatical, the psychological, and the logical side of the nature of judgment apart from one another. The psychology of cognition and logic have developed under the constant influence of grammar and its forms. Especially as regards judgment, one has transferred relations and forms from the nature of the grammatical sentence, determined by the laws of language, to hold immediately for the thought-activity itself. This has especially had influence on the conception of the relation between subject and predicate in judgment, a relation I shall now attempt to illuminate.

30. The logical predicate is the most important member in the judgment. It denotes the new that comes to be added and that is pointed out. The subject is always presupposed. All words therefore begin as predicates, and it is only later that they are used to express

subjects. It is at first the regard for the hearers that leads the speaker to express the subject in words: for the speaker himself his object, the subject, the thing at hand, the situation, stands as given; he sees in each case the object that is the starting-point for his thought and speech before him in spirit; but this is not always the case with the hearers, and therefore he must indicate that about which he wishes to say something.[38] There are, however, sentences in which the initial representation, the logical subject, is not expressed at all. This does not mean that the subject is lacking in the speaker's consciousness, but only that there is no urge or time to express it. The interest rests on the predicate. The logical subject is often hidden, and yet is that about which everything turns. It is the pivot on which the door turns;—but it is the turning, not the pivot in itself, that is the new thing.

Such sentences, in which the logical subject is not expressed, we can call predicate-judgments. Different kinds of them can be distinguished. The exclamation is the shortest and most forceful linguistic expression for the fact that something new in relation to the thing at hand or preceding has set in. No special initial representation (terminus a quo) is formed at all, if consciousness is immediately occupied by the final representation (terminus ad quem). Small children are especially occupied with what moves. A little boy of a year and a half said, when he saw an animal coming, "Bow" (from bow-wow), and the same word he used when he saw something move, e.g. when the tall grass on the lawn where he played moved in the wind. It was clearly enough the movement that laid claim to his attention, in contrast to the preceding, relatively resting state. The resting and unchanged called forth no outburst[39]. The exclamation "A star!" when a star is suddenly discovered in the bright evening sky is a predicate-judgment. Likewise the exclamation "A good idea!" when a happy inspiration puts an end to an embarrassment. In naive narration the initial representation is often presupposed known, in that the narrator, absorbed in his subject, hastens straight to the predicate and forgets to acquaint readers or hearers with the situation. In *Herodotus* several delightful examples of this are found (I, 144; V, 35; VIII, 21).

Often what one feels an urge to express is only the value of what happens. It can

[38]Cf. on this *Ph. Wegener*: *Untersuchungen über die Grundfragen des Sprachbaus.* Halle 1883. p. 54.

[39]W. *Stern* (*Beiträge zur Psychologie der Aussage. Erste Folge.* Leipzig 1903. p. 52 f; 124 ff), by showing children of seven years pictures, has found that what first interests them are persons and things, later activities and actions, last relations and properties. These experiments have, like so many psychological experiments, the defect that they do not show us the purely involuntary. Furthermore, they concern children who already have a certain development and command many memories. And finally the striking opposition between rest and movement does not come forth. "Activity and action" cannot be presented as strikingly in a picture as they are in the experience. "Substance" is not, as Stern holds, the child's first "category."

be done by a single word ("Splendid!" "Bravo!"). What is splendid stands as given and presupposed, and one has neither desire nor time to express it. In such exclamations, which contain the germ of the proper value-judgments, is expressed an object's intervention in the whole inner state of the speaker, and the logical predicate is here the property ascribed to the object as a consequence of this intervention; the object itself is not expressed.

Different from the mere exclamations are the so-called impersonal sentences. Here too only the logical predicate is expressed. There are two kinds of such.

There are impersonal, subjectless sentences in which the subject is either named beforehand or emerges from the connection in such a way that one at least knows of what kind it is, and can construct it in more or less definite form. In *Kaalund*'s poem "Nissernes Vandring" it is described how the nisses leave a region and in great throngs swarm over a bridge; ever new flocks come: "it was as if it would never come to an end." What seemed as if it would never come to an end was the swarming of the nisses. The indefinite impression of the endless is emphasized by the sentence's having no definite grammatical subject. In *Schiller*'s poem "Die Erwartung" the lover waits for his beloved; every sound and every sight can announce her coming; it is only a question of whether the interpretation of what is heard and seen is correct. "Hör' ich nicht Tritte erschallen? Rauscht's nicht den Laubgang daher? . . . Glänzt's nicht wie seidnes Gewand?" By the indefiniteness of the grammatical subject, the phenomenon comes to step forth in all its immediacy. And yet there is an initial representation; for only on the basis of the definite expectation do the rustling of the leaves and the bright gleam have interest. In other cases the initial representation is more indefinite. Only its kind can be represented. E.g. "There is dancing," "Someone is knocking." Often one here uses the indefinite grammatical subjects "one" or "they." "One says" = "it is said."

In another class of subjectless sentences it is not grammatically possible to indicate any definite subject. A wholly indefinite representation lies at the basis. These are the impersonal or subjectless sentences in the narrower sense. They are used to express natural phenomena, indefinite states, the course of fate. E.g. "It is cold." "It is spring." "Things are going backward." "It goes up and down." The chaotic and total character of the initial representation shows itself in that one can often, instead of "it," put "everything." E.g. "It is so still" = "Everything is so still." The word "it" is, in the opinion of the philologists, used in impersonal sentences only to fill an empty place in analogy with complete sentences—a grammatical counterpart to the way in which one often reads the forms and functions of developed consciousness into the elementary psychic phenomena.

Psychologically regarded, an initial element is never lacking (even if it does not become a distinct initial representation), and logically regarded one will always be able to construct such a terminus a quo and thereby translate the most primitive exclamations into logical judgments. But the initial element can be vanishing in relation to the final element. Hypnotic and ecstatic states denote the culmination in this respect.

If one defines a judgment as a conscious and definite connection of concepts, this definition does not fit predicate-judgments, in which the subject-element lies wholly in the background of consciousness and is not represented distinctly according to its whole content, as we require of a concept. But that definition can hold as an ideal. Logic can say that it really has to do only with the fully developed judgment, in which both subject and predicate are completely determined. The predicate-judgment and other judgments in which this is not the case then become only approximations to judgments. And of such approximations there can be extraordinarily many, since there can be many degrees in distinctness and definiteness.—

Predicate-judgments express acts of recognition. The application of the predicate contains a referring of the object to other objects, a kind of classification. The simplest cases of recognition permit no special designation. The object can stand as "known"—but this is also the only predicate at one's disposal. When immediate recognition is translated into the form of judgment, the classification consists in the object's being reckoned to the known in contrast to the new. In contrast to this there can be predicate-judgments where "unknown" or "new" becomes the only predicate, in that a closer description would already presuppose recognition of certain features in the new. In the very simplest cases one can only point.—

The more many-sided a relation the predicate-content is brought into with the subject-content, the greater the role played not only by acts of attention and recognition, but also by further-reaching associations and comparisons. If the predicate's relation to the subject is to be determined exhaustively, the consideration must not be limited to the immediately given, but this must be held together and compared with as many other objects as possible. Greater and greater demands are then made on the psychic energy.

31. Also in judgments in which both subject and predicate are expressed, the logical predicate continues to be the most prominent element, while the subject can stand in different degrees of obscurity. It is on this relation between the obscure or indefinite and the clear or definite that one must fix attention when it is to be decided what is logical subject and what is logical predicate. One cannot always see it from the single sentence, but must often infer it from the whole connection.

Often the logical predicate is placed first in a sentence for emphasis' sake, while the subject follows toneless afterward: "Great is Diana of the Ephesians!" "Now the time is up!" "Felix qui potuit rerum cognoscere causas!"—The logical predicate is often the grammatical subject or an adjective belonging to it: "*Thou* art the man!" "*All* the guests have come."—One will always be able to recognize the logical predicate by the emphasis, whatever place it otherwise occupies in a grammatical respect: "The king is *not* coming!" "He *has* gone."—In questions—both determination- and decision-questions—it is always a logical predicate that is lacking and sought. One has the terminus a quo and seeks the terminus ad quem. When we go back to the chief questions that lie at the basis of all other questions, we come to the fundamental predicates, the so-called categories. *Kant* aptly called the categories, the fundamental concepts of thinking, "predicates of possible judgments." To think, he says, is cognition by concepts, and concepts determine, as predicates for possible judgments, something that hitherto stood undetermined[40].

One and the same subject can be successively determined by a series of predicates. Indeed, in sentences of a describing or summarizing kind almost every single word can express a logical predicate. E.g. "A poor peasant went in to his neighbor to ask him to stand godfather to his newborn son" (the beginning of *Tolstoy*'s tale "Gudsønnen"). We here get a whole series of new things to know at once.—

32. If now a predicate is attached to a subject as a closer determination of it, then a new, more complete concept has been formed than the subject-concept one began with. The earlier predicate has now become an attribute. The attributes of existence (fundamental properties or fundamental forms) are found by analysis of experience and of the objects it presents. It is also this path that *Spinoza* has gone in his doctrine of attributes, despite his system's apparently deductive character. He finds in experience objects of spiritual nature and objects of material (spatial) nature, and since these two characteristic marks are irreducible, he regards them both as attributes or predicates of that which lies at the basis of all existence ("Substance")[41]. And so it goes in each single case; we constantly make a transition from a predicate found by analysis to an attribute. If I discover that a man is just, then I can for the future operate with the concept "this just man." The new concept thus formed can then in turn become the initial representation for a new formation of judgment, and so on.

Of special interest here are such sentences in which the grammatical subject appears with a quantitative determination attached, whether this determination is expressed by a number or in words (all, some, one). If the emphasis lies on this determination, it is in

[40] *Kritik der reinen Vernunft*[2] p. 94; 105 ff.

[41] *Ethica.* II. Ax. 5.

reality the logical predicate. In the sentence already used above, "*All* the guests have come," the new thing, hence the logical predicate, is "all"; the formally logical formulation would thus be: "the guests who have come are all who are expected." So it goes with all quantitative judgments, where the quantity is the main point. *Maimon*, *Herbart*, and *Sigwart* have therefore rightly maintained that there is no ground for setting up such judgments as a separate kind of judgments.—Something corresponding holds for the hypothetical judgments, just as we have already seen it for the negating judgments.

Benno Erdmann polemicizes against this and maintains that in a judgment such as the one cited the quantitative attribute belongs to the logical subject and is not predicate; one could otherwise, he holds, with the same right make all attributes predicates, which would be a forced reinterpretation of the linguistic mode of expression[42]. But precisely the linguistic mode of expression distinguishes between two different cases here. Where the tone lies on the quantity-determination, this is terminus ad quem, the final representation. And where the quantity-determination is not emphasized, it must have been won by an earlier judgment, so that we in any case have a secondary judgment before us.

Husserl concedes that all attributive designations mediately or immediately have sprung from judgments. But he holds that with these earlier judgments we have nothing more to do[43]. This is now in a certain sense correct. But the main thing is whether a certain kind of attributes (e.g. the quantitative) entail acts of judgment of a certain peculiar kind. And it is this that must be denied. When Husserl praises Benno Erdmann because he has "the courage to follow the natural meaning of the words" (Wortsinn), —and when he adds: "Ich halte es mit ihm für gewiss, dass 'alle S sind P' ein rein affirmatives Urteil ist ... das 'alle' gehört zum Subjekt,"—then it must be answered that one cannot at all pronounce on the sentence "all S are P" without knowing the emphasis; for this too belongs to the "word-meaning" (Wortsinn)—in any case it belongs to the meaning (Sinn)!

It is remarkable that two such acute thinkers as Erdmann and Husserl can still discuss the logical significance of the sentence without taking into account either the emphasis or the whole connection in which the single sentence occurs, and to which one must go back in order to see where the tone is to be laid, that is, where the logical predicate lies. Every skilled reader-aloud knows this. And it was long ago stated in the old *Art de penser* (Port-Royal's Logic) that only by attending to the connection in which a sentence occurs can one decide what is the logical subject and what is the logical predicate: L'unique et véritable règle est de *regarder par le sens* ce dont on affirme, et ce qu'on affirme. Car le premier est toujours le sujet, et le dernier l'attribut [i.e. le prédicat], en quelque ordre

[42] *Logik* I. p. 325–329.

[43] *Archiv für systematische Philosophie.* 1903. p. 114.—*Logische Untersuchungen.* II. p. 438.

qu'ils se trouvent (II, 9).

It makes no difference in this connection that judgments with quantitatively determined subjects acquire a quite special methodical significance on account of the great importance that counting, measuring, and weighing have for research. This has nothing to do with the judgments' purely logical and psychological character. Here it is a matter of not letting oneself be misled by differences in the linguistic expressions into assuming differences of kind with respect to the logical significance of acts of judgment as well as of judgments.

33. It will now be time to bring out the difference between the psychological and the logical consideration of judgment. Logically regarded, it is really only the relation between the judgment's two members that matters, while psychologically regarded it is the process by which thought reaches from the initial representation to the final representation that matters. Logically regarded, it is ultimately indifferent with which of the two members we begin, provided only that the relation between them stands clear and definite. But psychologically regarded, we in each particular case start from a certain representation and pass from it to another; the former becomes in the judgment the logical subject, the latter the logical predicate, however the judgment may be expressed purely grammatically.

No object can become logical subject without an interest attaching to it. If there are wholly interestless representations, they glide without effect through consciousness. The interest attached to an object makes the representation of it into an initial representation and operates as a spur to finding a closer determination of it. A certain tension arises before the final representation is reached, and this tension is replaced by the interest in the predicate found, which is therefore emphasized in the linguistic utterance. The interest in the subject begets the interest in the predicate. The whole formation of judgment would be superfluous if the interest in the subject were not inhibited by its indefiniteness. The mystic, who revels in his one, constant thought, forms no judgment; and his object would—on account of its unity and lack of difference—only be violated by any predicate whatever. And he who is absorbed in intuiting or in associative imagining will also seek to keep judgment at bay. *Goethe* thus says in his "Italienische Reise": "Ich halte die Augen nur immer offen und drücke mir die Gegenstände recht ein. Urteilen möchte ich gar nicht, wenn es nur möglich wäre."

In the completed logical judgment subject and predicate appear clearly and definitely connected. Yet even under the psychological process in which the formation of judgment consists, they do not at any point fall wholly outside each other. If that were the case, it would be unintelligible how they could be connected into a judgment. While the judgment successively develops out of the intuition or the association, the whole from which the judgment's members are fetched stands constantly before consciousness. It is within this whole that attention's transition takes place from the initial representation, which becomes the judgment's subject, to the final representation, which becomes its predicate. And the movement will be repeated several times back and forth. Two currents will form, a forward-going and a backward-going. What was at first initial representation becomes, by the backward current, final representation: in the judgment the subject is therefore just as much determined by the predicate as the latter by the former.

In logical theories one has now had more regard for the one determination-relation, now more for the other. The most common conception has been that the subject is, in the judgment, subordinated (subsumed) under the predicate, whose extension was thus greater than the former's. "The man is good" will here mean: "The man belongs to the good beings." The predicate is continens, the subject contentum. Here only the subject gets a determination, the predicate none—beyond this, that it contains, among other things, the subject in itself. In contrast to this, *Benno Erdmann*[44] has maintained that it is the subject that determines the predicate, in that the latter's content is subordinated to the former. "The man is good" will thus here mean: "The property goodness is a part of the man's being." These apparently conflicting conceptions are easily reconciled when one remembers attention's double movement during the formation of judgment. When Galileo, according to the legend, came to think of the motion of falling by seeing the church lamps swing, his attention—before the judgment "to swing is to fall"—moved back and forth between the two representations; he saw the swinging as a kind of falling-motion, but at the same time saw a falling-motion contained in the swinging. The intuition of the swinging lamps became clearer through the final representation found, of falling. The judgment's clarity is attained only by a going-back from terminus ad quem to terminus a quo.

In the impersonal sentences the initial representation still stands in indefiniteness. One notices a rustling in the foliage and a gleam (Schiller's "Erwartung"); but it is still uncertain who causes these sensations. If it turns out to be the awaited one, the impersonal sentence can be replaced by a new judgment;—but here the subject then becomes what in the impersonal sentence was predicate, and the predicate becomes the subject that in the impersonal sentence stood indefinite: "the one whose steps make the foliage rustle, and whose white gown shines through the leaves—it is *she*!"—In an

[44] *Logik* I. p. 251 f; 261 f.

example like this the two currents are clearly seen, before the proper judgment is formed.

Logically, the time it takes before the relation between subject and predicate can be determined plays no role, and the stages the judgment-process must run through have also more psychological than logical interest. Discovery and its winding ways interest the psychologist and the historian; but the logician as such asks for grounding and proof. In the last instance, therefore, it really does not concern logic what, during the formation of judgment, is initial representation and what is final representation. The difference between subject and predicate loses its significance when one formulates the judgment as exactly as possible, which is attained by expressing it in the form of an equation, as the designation of a relation of identity of a certain kind. This path Leibniz entered upon in some treatises that were first printed in 1840. Later, by various paths, William Hamilton, Morgan, Boole, Jevons, and their successors have worked in the same direction.

If the concept A shows itself by analysis to contain the concept B, one can with Leibniz express this by: $A = A + B$. For one can everywhere, where A is found, put $A + B$, since B is given when A is given. B has here, during the formation of judgment, been the final representation, terminus ad quem, and in the judgment it is predicate (A is B). But when that logical equation is formed, there is no ground to distinguish between start and conclusion. We can now keep to the result won, which can be read from right to left as well as from left to right. That distinction is only a memory from the preceding psychological process, insofar as it is not due to an influence from grammar.

Leibniz's mode of expression at the same time reminds us aptly that we do not let go of the subject when we pass over to the predicate; A occurs, after all, on both sides of the equals sign. Yet every mode of expression will more or less distinctly show the mutual determination of subject and predicate. Even according to the usual manner of presentation, according to which the judgment means that the subject's extension belongs under the predicate's, so that the judgment can be expressed by $A < B$—even there not only is A determined by B, but also, conversely, B by A, insofar as it can serve to characterize B that it comprises A in itself. When I, e.g., learn that Amphioxus belongs to the vertebrates, I learn not only something about Amphioxus, but also about the vertebrates, e.g. that a brain is not a necessary organ in them. In the languages in which the predicate-word conforms, in gender, number, and case, to the subject, the mutual connection between the two members is also seen; this mode of expression reminds one of Leibniz's.

In a logical equation the relation between the judgment's members stands before us at a single stroke, wholly clear and precise. Here has been attained, in a perfect way, what intuition attained at earlier stages of development: the abolition of equilibrium that took place at the dissolution of the intuition is now replaced by a new form of unity. The judgment expresses a higher form of psychic energy, insofar as the whole is here attained through a more composite work, and insofar as the whole that is attained is more exactly determined. But if the whole fullness of objects that an intuition (or an association-series) can contain is to be expressed in the form of judgment, a great mass of judgments will be needed for it, and it will always be only a matter of an approximation.—

What holds of the relation between subject and predicate in the judgment—namely that the time it takes to get from the former to the latter plays no role when the judgment is finished—holds analogously of the relation between premises and conclusion: there is no time-relation between them. The conclusion "lies in" the premises, just as the predicate "lies in" the subject, and just as the judgment as a whole "lies in" the intuition (or the association). Inference is really only a continuation of judgment: we infer when we discover that the predicate we have ascribed to a subject is itself subject to another predicate (*Aristotle*'s first figure), or that a subject has two different predicates (his third figure). Thus inferences grow out of judgments. But the more judgments that must be connected for an inference to be drawn, and especially when they must be fetched from highly different connections, the higher degree of psychic energy the inference-process presupposes.

Just as little as all intuitions can be exhausted in judgments, just as little can all judgments enter as members in inferences. And even where judgments and inferences can be formed, the original forms of unity—intuiting and associating—do not lose their significance. They form the involuntary basis that progressing reflection cannot do without. And perhaps, with the help of the judgments and inferences that have been formed, a higher intuition can be built, an intuition in *Spinoza*'s sense, a new comprehension of the whole won through reflection, in which the single thing could be intuited in its connection with the larger whole to which it has shown itself to belong. What value such an intuiting can have will be discussed at a later place in our investigation.

34. Historically there appears, right from *Aristotle* on, a tendency to lay the chief weight on the logical subject, although this is always given prior to the judgment, and although reflection's interest is attached to the closer determination that can be won by the predicate. According to Aristotle[45] there are concepts that can be used only as subjects, never as predicates: these are concepts of individual beings ("substances"). In Aristotle subject and substance are denoted by the same name (ὑποκείμενον); the predicate is in

[45] *Analyt. post.* 1. 22.—*De categ.* c. 5.

the judgment added to the subject, just as in reality actions, states, and properties exist only in the beings that perform the actions and live in the states. The judgment's subject thereby gets a higher dignity than the predicate: the subject corresponds, after all, to the one who performs the action, has the state, or is the bearer of the property—to "that which lies at the basis." This conception has, under various forms, maintained itself to the most recent times. One did not heed the lesson that language itself gave through emphasis, perhaps because one studied the written more than the living language.

The inclination to lay the chief weight on the judgment's subject expressed itself in a characteristic way in the first attempts to form a Danish terminology in logic. *Eilschow*[46] translated "Subject" by "Hoved-Sag" (principal matter) and "Predicate" by "Bi-Sag" (subordinate matter), and *Jens Kraft*[47] used in his logic the expressions "Sag" (thing) and "Tillæg" (addition). Even *Højsgaard*[48], who otherwise sharply distinguishes between the grammatical and the philosophical significance of the sentence, does not let his fine linguistic apprehension gain sufficient influence on the philosophical conception. He aptly emphasizes the predicate's predominant significance by translating "Verbum" by "Hoved-Ord" (head-word) and adds: "There is no kind of word that has so important a post in any discourse as the verb has. Every verb is in its proposition (sentence) like a captain for his company, and is not unlike the head, which enlivens all the limbs, and by whose nod they must all direct themselves." But he nonetheless starts from the assumption that, philosophically regarded, the judgment's subject expresses "the efficient cause," which quite conflicts with the fact that this subject first gets its determination by the predicate, whether or not this determination is to be a cause.

In more recent times *Wundt*[49] finds in the relation between subject and predicate an expression for the relation between thought-activity and its special content; *Benno Erdmann*[50] finds in that relation an analogy to the inherence of properties in the thing; *W. Jerusalem*[51] sees in every judgment an expression for the relation between will and action. But all this is analogies, due to language's tendency to personify—analogies that can have their significance with respect to making things intuitable, but are not necessary for understanding the logical nature of judgment. The analogy between the subject's relation to the predicate and the thing's to its properties, or the will's to its actions, can do a similar service to that which other logicians have found in the use of geometrical

[46] *Cogitationes de scientiis vernacula lingva discendis.* Hafniæ 1747. p. 36.
[47] *Logik.* København 1761. p. 141.
[48] *Methodisk Forsøg til en fuldstændig dansk Syntax.* Kbhvn. 1752. p. 256.
[49] *Logik*^{3} I. p. 149 f.
[50] *Logik* I. p. 261.
[51] *Die Urteilsfunktion.* Wien 1895. p. 92–96.

figures, especially for making the doctrine of inferences intuitable. The proof never lies in the analogies or the schemata. One must first show one's logical right to connect two concepts as subject and predicate before one can apply those analogies. Even though we perhaps always involuntarily use certain analogies, schemata, or symbols when we think, there are nonetheless great individual differences both with respect to which intuitive forms one uses, and with respect to the more or less predominant role they play in individuals. Logic must therefore distinguish definitely between the thought-relation itself and its being made intuitable. Signs, as well as words, are always only invitations or directions to carry out a certain act of thought. We must therefore always subject our images, analogies, and schemata, as well as our words, to a thorough critique, by testing whether we can draw all the consequences of them, carry them through in all details. Where this is not possible, there is a limit to the validity of the sign in question. We can perhaps not think at all without images or signs; but thought shows its independence of them in the constant critique to which it subjects them. *Leibniz* has clearly illuminated this relation, especially in the following utterance: "Even though the signs are arbitrary, there is nonetheless in the use of them and in the manner in which they are connected something that is not arbitrary"[52].

In the use of examples the relation between thought and image appears clearly. We draw only those inferences from the example that we could just as well draw from any other example. And if we do not keep to this rule, we misuse the example, just as when we extend a simile beyond the tertium comparationis.

35. Until the judgment is finished, the subject lies, as we have seen, in a certain indefiniteness. And even when the judgment lies formed, it is a question whether the whole content of the initial representation has been taken up into the predicate (or the predicates). This content must always undergo a limitation in order to enter into a judgment. So far every determination is a negation. To define is to delimit. Beneath our clear judgments there therefore always lies an obscure basis that is not exhausted, and for which predicates may perhaps later be sought. The limitation that is necessary for the formation of the judgment may perhaps later be abolished. There is an irrational relation between judgment and intuiting (or association). There is always the possibility of more decimals.

A first subject can therefore not be found. But just as little can a last, concluding predicate be found. Every predicate can in turn become subject, or the subject determined by a predicate can in turn become the basis for a new formation of judgment. Our thought

[52] *Dialogus de connexione inter res et verba.* (Opera philos. ed. Erdmann) p. 77.—Cf. my treatise on *Analogien og dens filosofiske Betydning.* (Mindre Arbejder. Anden Række). p. 35.

moves constantly from a more or less indefinite basis toward a complete determination, which nonetheless constantly sets new tasks and stands as an ideal. Existence, which announces itself in our experience, cannot be exhausted in any series of predicates, or in ever so many series of predicates. New initial representations can always emerge, and new final representations must be sought.

d. The Validity of Judgment

36. Since judgment denotes a new state of wholeness or equilibrium, it will be connected with an assurance of its validity, even if this assurance does not specially come to expression. But this is not to be understood as if it were first the judgment that brought the assurance or the presupposition of validity at all. Consciousness operates from the first with an involuntary, instinctive trust in everything that emerges in it and for it, both in its individual elements and in their connections. We are all born in faith; doubt comes only afterward—if it comes. And that faith is there is seen from the involuntary way in which it shows itself in action. A sense-impression releases a movement just as quickly as it releases a sensation, and perhaps quite as quickly, as often in reflexes and instincts. The single impression operates like the spark that falls in the powder; it releases an energy that can seem wholly disproportionate to the stimulation. And this energy can get full outlet only through movement and action. Faith shows itself in action before it appears as a special state of consciousness. It expresses itself in the immediate absorption in, or devotion to, that with which it enters into relation. Drive and expectation precede the deliberating thought, the calm contemplation.

English psychologists have strongly emphasized this original and involuntary faith. *Hume* laid the weight especially on the expansion by which an arousal, a lively movement from a single point in consciousness, can spread to the whole state; under the influence of a strong sensation or feeling we are inclined to place trust in all representations that hang together with them; they get a share in the latter's life and strength. *Bain* has laid weight especially on the original motor tendency, the urge for movement, that makes us often act on representations before we have tested them. The single representation's connection with the urge for movement is from the first closer than its connection with other representations that could determine and limit, perhaps even inhibit, it. It is the urge for movement that makes us involuntarily test our representations by applying them in action and thus apply a kind of experimental method.

This original faith or expectation makes itself felt, as we have seen above (16), already in mediate recognition. It operates in simplest form already before proper, free representations have really developed. It is by no means the case that we first develop our representations completely and clearly, and then afterward ask about their validity.

But apart from the influence of expansion and the urge for movement, it follows from what one might call psychic inertia that sensations and representations win full trust and assent as long as no conflicting elements appear. We have no ground to doubt their validity as long as nothing inhibits them. Just as a movement changes speed and direction only when an outer force intervenes, so the development of our consciousness in a certain direction changes only when motives arise that lead in another direction. An urge or a drive that has once been awakened can (when it does not die from lack of nourishment) be inhibited only by another drive. And thus men's first assurances and expectations, whether they are due to their own observations or to traditions, change only when opposing experiences or assertions exert their influence. Doubt is always secondary.

Every intuiting, every association, and every judgment therefore has an existential character. And the same holds for the inferences we draw from our judgments. Dogma precedes critique. The history of thought bears witness to this by the tendency that expresses itself to regard points of view, concepts, and forms of intuition as absolute realities. The Pythagoreans' numbers, Plato's Ideas, Spinoza's Substance, Newton's absolute space, materialism's atoms, Kant's "Ding an sich"—are examples of this. There stirs here in the thinkers the same tendency that gives the ordinary consciousness so secure an assurance of the reality of the sensory surrounding world, despite all subjectivist objections and despite all the disappointments produced by sense-illusions.

This primitive assurance, which in practice bears our whole life and stirs in our cognition from its first germ, has its ground in the instinct of self-preservation and is of great significance in the struggle for life. But it does not have quite the same character as the conviction that can be attained through the purgatory of doubt and contradictions (cf. above 26–27). It does not appear as an independent element different from the intuition, the association, or the judgment itself, and does not itself become the content of a predicate. Precisely because it is wholly involuntary, it does not become the object of a seeking and does not come to stand as final representation, terminus ad quem. It determines the whole character of the state, but is not itself object or subject-matter. This wholly immediate character of the assurance one could express by ascribing to our intuitions, associations, and judgments an original existence-quality, which first becomes the object of explicit consciousness when an opposite quality, non-existence, can appear over against it. This happens first through disappointment and lack. Then the sharp difference between being and not-being is experienced. It goes with the existence-quality as with innocence; it is first brought forth before consciousness when it comes to stand over against its opposite—is first experienced when it is threatened or abolished.

I here use the word quality, as where the talk was of the simplest kind of recognition, to designate the wholly immediate and involuntary, ultimately indescribable feature of a state or an object, and in order to exclude the tendency that, both as regards recognition and existence-assurance, makes itself felt to read relations that first come forth at more developed stages of consciousness into the purely elementary phenomena of consciousness. Just as one has held that all recognition is due to a judgment, so one has also held that the difference that at later stages can be found between existence-conviction and mere representation must be present from the first. But what one can find out by analysis of the grown European's consciousness one has no right to presuppose in the child or the Patagonian. Analytic and genetic psychology here enter into decided opposition to each other. When one, with *F. Brentano* and his disciples, starts from the assumption that a judgment must be presupposed for there to be existence-assurance, one naturally comes to read judgments into the most involuntary utterances of the mental life. It becomes scholasticism, neither psychology nor epistemology[53].

37. The involuntary experimenting that the original presence of the existence-quality calls forth can lead to experiences by whose help critique can be applied to the immediate existence-quality. If the urge to act in a certain way or in a certain direction meets disproportionate resistance or entails unbearable suffering, it will fall away by degrees, if such an energy cannot be mustered—perhaps by the very resistance or suffering—that the inhibitions are overcome. Or it can happen that the original secure expectation, by reason of the opposition to what actual experience shows, is formed into a more distinct intuition or into a judgment, so that a clear comparison between the expected and the occurred can take place. Here one of two things can then happen. Either the content of the expectation can lose the existential character and pass over to becoming "mere" representational content, a representation of something "merely" possible, indeed perhaps of something impossible, or at least of something whose time is past, if it has ever been. Or the original expectation can be altered in a similar way to that in which we have seen the immediate intuition can be altered. A lesson is then drawn from the disappointment met, and the expectation is limited or reshaped in such a way that it is

[53]Cf. my critique of *Meinong*, one of Brentano's most eminent disciples, in *Göttinger Gelehrter Anzeiger* 1896. p. 299 f (where I have for the first time used the expression existence-quality).—*Stumpf*, who in his *Tonpsychologie* (1883) joined Brentano's conception and regarded every observing and noticing as a judgment, now distinguishes definitely between these more elementary functions and judgment. *Erscheinungen und psychische Funktionen.* (Abhdl. der preuss. Academie der Wissenschaften. 1906). p. 16.

no longer shaken by conflicting experiences.

The chief task for the psychology of cognition becomes, after the whole consideration we have here made, not to explain how our sensations and representations can come to stand as the expression of something real; but the task becomes, on the contrary, to show how representations of something merely possible and of something unreal or impossible can arise. And only when this has happened can there be a question of how we distinguish between the real and the unreal—what criterion of reality we have—and what reality really is.

The criterion of reality now shows itself to be merely a use, clarified through reflection, of the involuntary experimenting to which the existence-quality and the original expectation already lead. The expectation must be formulated as a definite question. Definite inferences must be drawn from the objects at hand and the involuntary assumptions they have led to, and it must then be investigated how far these inferences are confirmed by continued lived experiences and experiences. The expectation is thus formulated as a hypothesis that seeks confirmation. The criterion of reality consists in agreement and law-governed connection between as many sensations and representations as possible. The testimonies of the different senses, the different memories, the different inferences, and the different individuals must agree and show themselves compatible, preferably in such a way that a law-governed connection shows itself between them, so that one can infer from the one to the other. All that exists must ultimately constitute a great totality whose members join together in a natural, or even necessary, way. Illusions and hallucinations, dreams and fantasies, speculations and intimations must, even where they appear with the most vivid existence-quality, give way to the great, consistent connection that is our last refuge in cases of doubt, the firm fortress from which we judge what is to be called real and what is to be called mere representation. If the world of waking and sober life could just as well be subsumed under and understood from the world of dreams as the latter from the former, it would be impossible to draw any boundary between reality and unreality.

Reality is real truth. Formal truth a circle of representations can be said to have when it is won by consistent reflection from a definite point of view. And since the firm, consistent connection is an essential element in the concept of reality, the concept of reality presupposes the concept of formal truth, which it is logic's affair to develop. The concept of reality comprises, besides formal truth, also the law-governed connection and agreement between as many different points of view as possible. It requires a working-together of all the points of view from which we can orient ourselves in existence[54].

That the criterion of reality cannot consist in anything wholly simple was already impressed in antiquity by *Carneades*[55], in that he pointed out that one cannot rely on any single and momentary sensation or representation. It is not the immediately persuasive force of a representation that is decisive; what matters is whether it can be held fast and carried through by exhaustive testing. (The representation must not only be πιθανή, but above all ἀπερίστατος and διεξωδευμένη.) One cannot rely on a single mark, but on the agreement (συνδρομή) of all marks.

The law-governed connection is the criterion of reality of the more recent science. Thus *Galileo* says: "The mark of what is natural and true (la proprietà e condizione delle cose naturale e vere) is not merely that no self-contradiction is present; nor merely that an easy transition can be made from the one to the other; but it consists in the transition's taking place with necessity, and in its being impossible that it can proceed in another way"[56]. And the same thought recurs in all the philosophers of the more recent time from Descartes *(sixth Meditation)* to *Kant* (*Kritik der reinen Vernunft*, p. 236–247). In the period of Romanticism it became a chief thought that truth or reality must be a totality, outside of which nothing can be understood or have reality. And the fundamental features of this totality one held one could construct by way of pure speculation.—In a work that on many points seeks to hold fast Hegel's fundamental thoughts, F. H. *Bradley*'s *Appearance and Reality* (1893), the idea of reality as a totality of all possible experience has taken on the character of a standard: a reality is the higher the greater the multiplicity it encloses, and the greater the unity that makes itself felt in this multiplicity. But Bradley has given up the Romantics' hope of being able to construct the absolute totality (the one absorbing experience)[57].

38. How great the totality of experiences must be, into which the single phenomenon must fit in order to be able to be regarded as real—that cannot be decided once and for all. The world of experience expands constantly and cannot be concluded. Therefore the concept of reality or existence is, however strange it sounds, an ideal concept. For an absolutely secured assumption of reality it would really be required that the single object had been held together with all possible observations, and this is of course impossible. We can only strive to work with as large a horizon as possible. Only a provisional assurance can ever be won. The concept of reality is constantly in its becoming. But this

[54]Cf. with respect to this whole consideration my *Psykologi* V D (The Apprehension of the Real).
[55]See *Sextus Empiricus.* ed. Bekker. p. 229–232.
[56]*Dialogo sopra i duo massimi systemi.* Giornata quarta, (Ed. Fiorenza 1710. p. 417.
[57]Cf. on Bradley's philosophy my book *Moderne Filosofer.* p. 44–54.

does not exclude that this concept, having emerged from the application of the criterion of reality that is possible at a given time, can now become a predicate-concept, in that reflection investigates how far it can find application to the single object at hand. Only now do proper existence-judgments become possible. The concept existence is formed in a way analogous to that in which the concept likeness is formed. Sensations and representations must often have been placed together on account of their likeness before the concept likeness could be formed and become predicate in judgments. The association by likeness operates long before the concept likeness is set up. Thus the existence-quality can long have been present without reflection having turned against it and investigated its conditions and its justification. Existential judgments, such judgments in which the concept existence is predicate, cannot be primary, but presuppose a whole development. Behind the immediate assurance they express, e.g. "There *are* good people!" "There *are* flying mammals!"—there really lies the presupposition that the judgment's subject belongs within the great connection of our experience as a single member. The presupposition for the existential judgment is a combining and comparing process, whose possibility is maintained, even if the process is not carried through explicitly.

It is by way of reflection, not by way of mere sensation or the involuntary train of representations, that we reach reality. It is so far of ideal nature, is an object of thought. In many ways, as the history of science shows, thought must reshape the sensuously and involuntarily given in order to be able to ground a conviction of reality. Naive, secure realism has another concept of reality than the constructive realism that seeks to carry through a justified concept of reality, and that it is the affair of the cognition of nature, of historical research, and of philosophy to separate out. Slowly but surely human thought moves in this direction.

Yet it will always show itself that there are certain definite objects that form starting-points for reflection in its striving to apply the criterion of reality. All objects do not stand on the same line; reflection must begin by presupposing the reality of certain objects and testing the others in relation to them. Our thought must have a place to stand, and only at a later stage of reflection can this place itself be drawn into the discussion, namely when another starting-point has been won. The beginning-point is in a certain sense accidental; it is perhaps historically and psychologically explicable; but history and psychology are themselves part of the cognition whose validity is to be tested, become themselves objects for epistemology. From this side too it shows itself that the concept of reality is an ideal. It is not only the content of our apprehension of reality, it is also the very standpoint from which we have found this content, that must pass through the critical purgatory. And without a standpoint no reflection can, as said, be exercised.

39. Value-judgments offer an analogy to existential judgments. In their simplest, most involuntary form they appear as mere exclamatory judgments (see 30). When doubt arises about the validity of the involuntary valuation—when, e.g., the one cries "Abominable!" where the other cries "Splendid!"—a criterion is needed here too. And here too the only criterion will show itself to be agreement between the individual valuations. The two conflicting parties must seek out something about whose valuation they agree, and then investigate whether one of them did not commit an inconsistency in the valuation that became the occasion of the dispute. If they can find no such point at all, the dispute is hopeless. Each one must then seek to preserve agreement with himself in his valuations. But whether or not agreement can now be reached, it will show itself that, just as the criterion of reality in the individual cases presupposes the reality of certain objects as given, so there is in fact, in the value-judgments that are passed, presupposed one or more values. Their validity is not—at least provisionally—drawn into the discussion, but the relation to them conditions the validity of all other values. Such values we can call fundamental values. A fundamental value denotes, for each individual personality and for each individual historically given group of men, the presupposition on which valuation in the particular case, the fixing of the individual values each by itself, rests. Only in reflection's further development can the fundamental value itself be drawn into the discussion—but this will presuppose that a new fundamental value has developed and consolidated itself. Reflection can here, just as little as in the purely intellectual domain, hover in the air; but it can successively investigate its earlier standpoints, and it can become conscious of the very standpoint from which it makes its concluding valuation.

The two considerations we have here distinguished between, that which concerns reality and that which concerns value, stand in a very close relation to each other. For on the one hand the cognition of reality itself has its value, a value that for many minds stands as the fundamental value, and on the other hand the values, especially the fundamental values, must be so constituted that they can be held fast and developed within the world whose reality is shown by the greatest possible connection between the greatest possible number of experiences. This possibility belongs to the conditions of the value's validity. Finally, the reality that a purely intellectual interest leads to discover gets a peculiar value if it shows itself to afford possibilities for the development of a world of values. In the special discussion of the tasks of thought we shall return to what is here intimated.

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II. The History of Thought

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A. Animism, Platonism, and Positivism

40. The psychology of thought presupposes its history. The functions arising step by step, which have been described in the foregoing, have developed under definite conditions of life, physical and social, and their character is determined by a constant reciprocal action between inner and outer conditions. It falls to sociology and comparative psychology to unravel more closely this relation, which in the foregoing could only be intimated. In *Hobhouse*'s book *Mind in Evolution* (1901) a comparative-psychological treatment of the development of the life of thought is given. And it is here shown at the same time how social conditions enter into reciprocal action with intellectual development, in that the life of thought is developed through the living-together of individuals, while on the other hand social instincts and feelings presuppose a certain intellectual development, without which recognition of other individuals and understanding of their conditions would not be possible. Already language, which is the work of the race, not of the single individual, shows the life of thought's dependence on social development. The individuals do indeed give their contribution to the growth of language, more or less significant and more or less conscious contributions. But essentially language is nonetheless a social inheritance into which the single individual lives himself. In it many representations and points of view are already arranged. It has, as has already shown itself in the foregoing, its own metaphysics, which is a result of development that now promotes, now inhibits the individual life of thought. And language, as well as the thought-content deposited in it and the thought-forms developed through it, has itself developed in connection with the whole social life in family, tribe, religion, and state. The biological-psychological consideration thus leads over into the sociological, which shows the development of the life of thought under the influence of social institutions. It has even been maintained from the sociological side that, just as little as obligation can be

understood from the standpoint of the individual will, just as little can the necessity with which certain fundamental concepts (categories) make themselves felt be understood from the individual's intellectual development; in both places the explanation lies in social and historical development. The categories are of social origin, are tools of thought that human societies have laboriously formed through centuries, and in which they have gathered the best of their intellectual capital. *Émile Durkheim*, who has especially asserted this pronounced sociological point of view, does not, however, thereby wish to dispute the significance of a direct psychological investigation of the life of thought. To a complete investigation of the human fundamental concepts belongs, according to him, also an investigation of the germs of thought-life that can be found by direct analysis of the individual consciousness[58]—such an investigation, then, as I have attempted in the foregoing. It must at the same time, in the face of the strong emphasis on the sociological point of view, be definitely maintained that the overwhelming social influence on the life of thought is easiest to demonstrate at more primitive stages of human development. During the development of culture the individual thought frees itself more and more. Already comparative inquiry in the domain of psychology and sociology shows this. Sociology itself would not be possible if such a liberation had not taken place, which of course can never be absolute, any more than, for that matter, the social influence can ever be absolute. When it is so easy to demonstrate the social influence on spiritual development in earlier periods, this has for no small part its simple explanation in the fact that the individual influences are forgotten. One has the products left, but the factors have vanished, and one then lets "society" or "the group" bring forth itself in a mystical way. At every given stage "society" consists of definite individuals, each with his dispositions, with his urge and his circle of experiences. "Society" easily becomes an asylum ignorantiæ. It is a convenient word to use where one does not know or cannot survey the operative individual forces in their many differences and in their infinite reciprocal action. The historical point of view must not be made without further ado one with the social point of view. Every individual lives himself into a historical connection, also as regards the life of thought; but this historical connection is itself the result of manifold individual forces' cooperation through the ages, of giving and receiving between individuals in constant reciprocal action. The social is what makes the reciprocal action possible; but the reciprocal action itself presupposes the individual forces.

41. That the individual life of thought always unfolds on a historical basis was

[58] *É. Durckheim*: *Sociologie religieuse et théorie de la connaissance.* (Revue de Métaphysique et de Morale. 1909) p. 750.

strongly emphasized by both the two great main directions in the domain of philosophy in the nineteenth century, Romanticism and Positivism. Both directions had—in common opposition to the eighteenth century—a clear understanding of the social and historical atmosphere within which the spiritual development of individuals takes place. They could think of no spiritual development on bare ground. By *Hegel* and *Comte*, two eminent representatives of the two main directions, the history of thought has therefore also been written in broad strokes. In his *Phänomenologie des Geistes* (1807) Hegel showed how cognition begins with the single and isolated thing, but is from there driven forward step by step through the contradictions in which it becomes entangled. Through step-by-step development thought unfolds its essence and shows itself to be an ideal totality within which everything single can find its place, at once as a peculiar member and yet "sublated" in the "higher unity" that encloses all oppositions in itself. The individual stages in this development are illuminated by examples and types, partly of individual, partly of social kind. But it is Hegel's view that only the thought that can unfold on the basis of what the history of spirit has brought about will have the conditions for exercising a significant work. The "Spirit" whose history of manifestation (Phenomenology) he describes in his work is the spirit of the human race, in which the individual spirit is only a single moment. It is from the "higher unity"—won at the given stage—of formerly struggling oppositions that the individual thought-work is to be exercised.

Comte too distinguishes (*Plan des Travaux.* 1822) between different stages in the development of the life of thought. But with him it is not the spurring force of oppositions and contradictions that leads forward. His chief thought is that at all stages a relation makes itself felt between what is given in experience (the objects) and the representations by which it is interpreted. The difference between the stages rests partly on how large a circle of objects is at one's disposal, partly on which representations prevail in man, so that they can more or less involuntarily be taken as basis in the interpretation. The narrower the circle of experience is, the greater the role the ruling representations will play; the more comprehensive it is, the more the thought-work will come merely to aim at bringing about connection within the given, so that this interprets itself.

The points of view that the two philosophers take up supplement each other. For when experience necessitates a new mode of interpretation, or when the difficulty of carrying through an interpretation leads to seeking out new experiences, it is evidently because it discovers contradictions or oppositions in the hitherto given, or between the prevailing interpretation and the given. Thought is shaken in its accustomed interpretation by new experiences, or it hopes that its difficulties will fall away through new experiences[59]. Whether by these paths "higher unities," more comprehensive unified intuitions, can always be reached, as Hegel held, is a matter in itself, which will be discussed later. And whether Comte is right in the sharp difference he sets up between three stages in cognition, the theological, the metaphysical, and the positive, we will here also not discuss more closely[60]. But with some change in the points of view one will rightly be able to preserve Comte's tripartite division, as I shall now attempt to show. The change entails a corresponding change of the designation; one will suitably be able to distinguish between Animism, Platonism, and Positivism as three main stages, in that these three words are each taken in an extended sense.

42. In its simplest form the difference between object and interpretation appears in the opposition between distinguishing and recognition, the two poles in our cognition, which correspond to the two great conditions of life: openness for the new and application of the old to the new (Cf. 14). But only when judgments are formed, and especially when questions and problems are set (26–27), is there a matter of interpretation proper. The presupposition for this is that one does not immediately proceed to action. For when the emerging cognition-elements immediately release reactions in a reflex and instinctive manner, there is neither urge, time, nor force for interpretation. Interpretation presupposes a certain degree of intellectual interest and, in connection therewith, experiences that there are limits to our capacity to intervene in the course of things, so that we must come to know the essence and laws of things in order to be able to place them in the service of our purposes.

Frazer's hypothesis about the relation between magic and religion[61] is of interest in this connection. Prior to religion, which always presupposes experience of dependence, there goes, according to Frazer, a period in which man has trust in his will's power to such a degree that he believes he can bend nature under himself, not only where he really intervenes actively, but also by mere wishing, made known by word or ceremonies. Or rather, the difference between action, word, and ceremony does not exist for man at this stage. It is experience of the will's limitation that leads beyond this stage, and it is likewise this that leads into the world of cognition. The magical art does not always lead to the goal. Events often take place independently of or in defiance of the human will. Through disappointment and the feeling of impotence, man learns that in existence there

[59] These two sides were emphasized by Danish philosophers in the eighteenth century, the first by *Holberg*, the second by *J. S. Schneedorff*. (*Danske Filosofer*. p. 5; 18.)

[60] See on this *Den nyere Filosofis Historie*[2] II. p. 334–338 and a treatise by *Hobhouse* in *Sociological Review*. 1908. p. 262–279.

[61] *The Golden Bough*[2] I. p. 61–78; III. 458 f.—*Adonis—Attis—Osiris*. p. 3–4.

prevail powers that are stronger than he. These powers he conceives as personal beings, and this conception—however it has arisen—afforded the first means of interpretation, the first "categories." Man did not give up magic; but it was now exercised toward the overmighty beings whose will it was a matter of bending, or whose power it was a matter of supporting by the performance of certain ritual actions. Already the magician at the pre-religious stage performs his ceremonies in a certain definite way, since he otherwise has no trust in their operation, and he so far recognizes a certain order of things; but this recognition becomes more penetrating and comprehensive when magic is replaced by, or at least subordinated to, religion.

I here disregard the question whether the arising of religion can be explained merely by the disappointments to which autonomous magic led. There must, after all, have been many other paths by which man could experience his dependence. What in this connection is of interest is the indisputable opposition between a standpoint where man has no occasion to recognize an order of things, and another standpoint where such a recognition is alive. From the one standpoint to the other man can only have come by way of experience, a way that has probably often led through hard disappointments. Only when this way has been traversed is there a matter of interpretation. And history teaches that the first attempts at interpretation have the character of *Animism*, in that the explanation of natural phenomena, at least the most striking and far-reaching ones, was sought in the intervention of personal, man-like beings.

Animism appears in a series of forms—from momentary and special gods to polytheism's individualized god-figures, and from these to a more or less thoroughgoing monotheism[62]. These transitions are not only of significance for the development of the religious life, but also for the development of the life of thought. The mythological imagination's individualization of a god-figure—which corresponds to the psychological transition from concrete to typical individual-representations (24)—leads naturally to a more definite distinction than before between the natural phenomena that depend on the god's intervention and the god himself. This distance-relation makes a more thorough investigation of the natural phenomena themselves possible; the god withdraws and hides himself more behind his work than before. Human experience gets time and space to spread out, since it does not everywhere or immediately strike upon the god. And the god-figure itself now presents an inner connection, a more or less consistent characterization, so that the representation of it denotes an advance in psychological insight. A typical individual-representation presupposes that one regards all the individual

[62]See my *Religionsfilosofi* § 46–53.

utterances and situations in which a being lays itself bare as examples of its character. It indicates, where it is distinct, the law for the development of the being in question, la loi du changement (Leibniz). Both physically and psychologically, then, polytheism denotes a great advance. Comte has even called its development the most important advance within the “theological stage.” The transition from polytheism to monotheism too has of course its great significance. An absolute monotheism is scarcely ever carried through. But the fewer gods there are, the more the divine intervention in the course of natural phenomena can approach a minimum. On the other hand, the strictest monotheism can get a likeness to the belief in momentary gods, when the one god does all too many miracles.

During this whole process there is a reciprocal action between religious and intellectual elements, or, if one will, between religious experiences and other experiences. When by religious experience one understands the way in which the human life of feeling is attuned through what is experienced with respect to the fate of human purposes and ideals under the course of things, then this experience will contribute much to the way in which the powers of existence are conceived; religious experience has a spurring and forward-driving influence on the whole world-view. One need only compare Homer’s world-view with Hesiod’s, Plato’s with Marcus Aurelius’s, the Vedic poems’ with Buddha’s, older Judaism’s with that of the prophets. But by degrees experience and understanding of the natural connection of things, hence physical insight, is also gathered. And the more social life and the personal life in individuals develops, the more connection the human life of personality presents, the less can one think of the gods’ operation as guided by momentary and changing motives. Psychological and physical experience thus meet with religious experience.

43. Animism (in the widest sense) has not yet disappeared as a principle of interpretation because strict science develops. It is limited to “the creation of the world,” or to certain extraordinary phenomena strongly intervening in human destinies, or one finds in it the ultimate explanation of the fact that there prevails law-governedness in nature at all. In some of the greatest inquirers of the more recent time the animistic principle of interpretation is found in one or more of these forms. *Kepler* and *Descartes* ground the conviction of the universe’s natural law-governedness on God’s perfection. According to *Leibniz*, all monads, existence’s ultimate elements, “radiate” from a primal monad, by whose choice the best of all possible worlds has come to be, a world whose course is determined by strict natural laws, but in such a way that these natural laws themselves are determined by regard for the greatest possible “perfection.” The solar system’s har-

monious structure and other purposive natural arrangements are, according to *Newton*, due to God's intervention and are not subjected to natural-scientific explanation; the infinite space itself that mathematics presupposes is God's sense-organ, the organ for his omnipresence. *Linnaeus* derived the different species of organic beings from just as many different divine acts of creation, and *Kant* found in the concept of time the only possible solution of the great problem set by the opposition between moral ideality and nature's actual course.

Yet during the development of the life of thought there takes place a constant elimination of animism. This is used less and less for the interpretation of the objects at hand. It is by this indirect way, by non-use, that animism disappears, not by direct disproof. It really cannot be disproved. *Kepler* saw a disproof of the assumption that the planets were moved by souls in the fact that the moving force decreases with distance[63]. But how could he really know whether a soul's operation did not also decrease with distance?

Animism has, despite everything, exercised a significant function during the development of the life of thought. It was the first coherent train of thought in which man could set his scattered experiences and lived experiences. And if it had been possible to develop and carry through the concept of personal beings in such a way that it could afford frame and basis for all understanding of material and spiritual nature, and at the same time guide toward setting research new tasks, then its time would not be past. Theology would then be both the highest and the fundamental science. The history of the life of thought shows that it has not been possible.

To this is added that the personal life itself shows itself to be subject to definite laws and is itself a part of the objects that are to be interpreted. Animism consists really in a part of our experience, namely our experience of the mental life, being taken as basis in the understanding of all other parts of our experience. One cannot, with Comte, distinguish absolutely between the given and its interpretation, since every interpretation ultimately builds on something that is itself given. Interpretation consists at all stages in the one object's coming to illuminate the other. There is thus a constant reciprocal action between experience and interpretation. In each single case there are some objects or experiences that are to be interpreted, and others that serve for interpretation. This was, in Comte's view, to be the characteristic of the "positive" stage, but it is in reality characteristic of all stages. The difference is only the choice that is made of interpreting objects, and the exactness with which the interpretation is undertaken; and both these things hang closely together, when it shows itself that not all sides or parts of our

[63]Cf. *Den nyere Filosofis Historie*[2] I. 165.

experience are equally well suited as basis for the understanding of the others.

44. Animism always had to undertake a certain ordering, in any case have a capacity for recognition, in order to be able to know under which divine being's domain of power the single object belonged. In "special gods" (departmental gods) this shows itself especially clearly; but it is a relation that is found again under one form or another in all religions. But the ordering or classification that here took place was very external; there was no inner relation between frame and content. And yet an ordering survey of the objects that are to be interpreted is the first condition for their understanding. The point is then to order them according to likenesses and differences. Reflection's essence as comparison expresses itself here. It soon showed itself that there are many degrees of likeness (and difference), and it dawned on thought that it is possible to bring about an ordering of the content of our experience from this point of view. One had then to try to discover as many and as essential likenesses between the objects as possible, and order them accordingly. The significance of Greek philosophy lies for a great part in its having impressed the importance of this task and made the first attempts at its treatment, and *Plato* can here be regarded as its typical representative. For Plato thought's essence laid itself bare in its capacity to form concepts that contained the points of likeness between large groups of objects and thereby comprehended these in themselves. The discussion of objects that led to sharp determination of their likenesses and differences Plato called dialectic, and the dialectician had to have a capacity for comprehending survey (ὁ μὲν γὰρ συνοπτικὸς διαλεκτικός, ὁ δὲ μὴ οὔ. Republic 536 C). There could by this way be formed a world of Ideas, in which the nature of the objects found its highest expression. The individual things and events arose, changed, and vanished, but this changed nothing in the concepts in which thought had comprehended their essence. All that is beautiful, e.g., is perishable, but the Idea of the beautiful is imperishable.

The ascent from the world of differences and changes to that of Ideas, in which thinking for Plato essentially consisted, was now for him not merely a formal systematic ordering, a classification, but in the Ideas or general concepts that held for a group of objects he found at the same time the ground that such objects existed at all. The doctrine of Ideas is not merely a doctrine of method, an attempt at a conceptual ordering of all thought-content. For Plato's aesthetic-religious conception the Ideas also came to stand as plastic figures, as constituting the true reality, of which the individual special objects are only imperfect copies. The value of the copies rests on their nearer or remoter relation to the prototypes, and of these prototypes again the Idea of the Good is the highest[64].

[64] *Natorp*, in his work *Plato's Ideenlehre* (1903), keeps only to the methodological significance of the doctrine of Ideas, and interprets Plato rather too much from modern points of view. Thus especially when

Interesting here is Platonism's relation to Animism. They meet in the presupposition that what lies at the basis of the phenomena must also be something that can be the object of men's urge and longing, contain values or be the cause of values. For Animism it was the god that was both the ground of things and the source of values. Plato gives up the personification, and the highest value he found in the very relation to the Ideas, in the happiness of beholding them. And this holds for gods as for men. The advantage the gods have is only that they stand nearer to the world of Ideas. In the great procession that in the "Phaedrus" sets out to admire the Ideas, the eternal stars in the heaven of thought, the gods drive first. Herewith is denoted a decisive turning-point in the relation of religious representations to scientific thoughts.

By the projection that let the Ideas stand not merely as thoughts, but as things, there came forth in Platonism a great problem: how do the Ideas relate to the phenomena? It was a reflection of Animism's problem, how the gods relate to the phenomena. A critique that possibly derives from *Aristotle* emphasized that consistently there would have to be set up an Idea under which both the individual Ideas and the phenomena belonging under them were to be comprehended. The justification of this critique lies in the fact that reflection cannot end in any principle that should "have its ground in itself," but must always anew discuss the relation of the Ideas or principles won to experience—and that (what Aristotle did not see) not merely to the experiences hitherto won, the objects hitherto at hand, but also to the new ones that can emerge or perhaps even be called forth.

At another point too Platonism shows its insufficiency. In that it makes classification, tracing-back to Ideas, one with causal explanation, it comes to assume just as many causes as there are kinds of objects to explain. That the Ideas are again to form a systematic connection, and finally be subordinated to the Idea of the Good, makes no difference in this connection. Here too Platonism reminds one of Animism; just as the latter has a god for each group of things or events it is able to distinguish, so Platonism has an Idea for each such group. Thereby a way was paved for the kind of pseudo-explanations such as that life is derived from a life-force, gravity from a force of gravity, and so on. It was this peculiarity that Comte especially laid weight on for the characterization of the second of his three stages, the one he called the "metaphysical."

Yet it denotes a great advance that the "Idea" or the force stands as something that operates constantly, always in the same way under the same conditions, something that

he interprets the phenomena's "participation" in the Idea by the relation of single events to the law (p. 151).—J. A. *Stewart* (*Plato's doctrine of Ideas* 1908) distinguishes sharply between the methodological and the contemplative significance, but concedes that Plato constantly confuses these two significations.

could not be said of Animism's gods. But the advance lies especially in Platonism's weight on the ordering thought-work that systematizes according to likenesses and differences. This will always continue to be the first step toward gaining penetrating understanding. From the mere catalogue with its external ordering to the "natural" systems resting on deeper-lying principles, such work appears in its great significance. Platonism's time will therefore never be past, as long as series-formations, type-arrangements, and overview (synopsis) continue to be of necessity. And there are domains, such as the psychological and the historical, where research perhaps never gets beyond this kind of work, at least as regards the most significant questions.

45. *Plato*'s Ideas express relations of likeness between different objects. When it is asked why the individual objects appear precisely with their definite properties and at a definite time and place, Platonism gives—just as little as Animism—any answer to this. Its method can only lead to an ordering that is grounded on comparison between the objects with the properties they happen to have, but cannot say anything about the origin of the objects and their properties. It sees existence in equilibrium; the comparing and ordering reflection stops the stream of time in order to fix the types that the objects present. But since the objects and their properties arise each by itself at definite stages of the stream, the task naturally comes up of investigating the relation between the coming-into-being of the different objects and properties. Does the one object arise independently of all others? Can the single property appear without other properties appearing or vanishing? Answers to these questions are not obtained by a merely resting ordering, but by finding out an ordering of the succession in time according to definite rules, so that the ground for the arising of the following objects and properties can be found in the presence of the preceding ones.

We can use the word Positivism to denote the direction in the life of thought that sets new tasks in the way just intimated. Here again the point is to understand objects through comparison. The comparison concerns the definite conditions under which an object comes to be or a property appears, the conditions for its localization in time or in space or in both respects, and the connection is to give the unavoidable historical succession as determined by a law. Every event is thus attached to another event, and it is at this that the endeavor aims, not merely at finding the likeness and difference between the different objects and properties. And the explanation is now sought not outside the series of events, in a being or an Idea, but in the very law according to which the members of the series follow one another. What is called cause is thus itself to be demonstrated as object, as member in experience, just as well as the object that was to be explained. It is the standpoint of the more recent empirical science, as it was indicated by *Kepler* and *Galileo* through the demand for a cause demonstrable in experience (vera causa).

The differences stood unexplained in Animism and Platonism. How could very different, perhaps mutually conflicting, objects and properties have their ground in the same god or in the same Idea? The more rich in content and comprehensive experience became, the more the sting of this question had to be sharpened, and attention had to turn toward the mutual connection of the differences. The urge for recognition can nonetheless constantly find satisfaction, if it succeeds in re-finding the same events when the same conditions are present. The earlier experiences comprised in a certain sense the new ones, if these are due to the same law as the former.

Yet it is not merely the urge to overcome in thought the multiplicity and change of objects that operates at the transition to Positivism: it is also the wish for power—the wish that in magic expressed itself in immediate acting, in Animism more mediately and on the presupposition of a certain resignation, and in Platonism faded into intellectual contemplation. Power over nature is now won by understanding of the laws according to which nature operates and by attempts, in the single case, to bring about the law-determined conditions.—

Is now this form the definitive one, or will there be able to come new stages in thought's great history of development? To this, from the nature of the case, no answer can be given. But we can see—and it is an insight of the very greatest significance—that the positivistic tendency itself, the standpoint of strict empirical science itself, rests on certain presuppositions concerning thought's relation to existence, so that, even if this standpoint should become the definitive one, problems remain with respect to the ultimate grounding of it. It is at this point that philosophy in the more recent time has taken up its work. The seventeenth century's great systems sought to find out what world-view resulted when the new standpoint, which I have here for brevity's sake called the positivistic, was strictly carried through. The eighteenth century's philosophy then went back to the presuppositions for the whole standpoint and subjected them to a critical testing. The nineteenth century's philosophy made partly an attempt to return to Platonism, partly to stop the thought-work there where the positivistic principle did not lead us further. It will be a chief task for the continued thought-work to carry further the treatment of the problems that through the purgatory of the great discussion have shown themselves unavoidable for every waking reflection. A contribution to this it will be attempted to give through a closer investigation of thought's modes of operation and tasks. But first there is yet another opposition to illuminate within the history of thought, an opposition that indeed reminds one of the opposition between Platonism and Positivism, but does not quite coincide with it and in any case illuminates it from a new side.

B. Ancient and Modern Thinking

46. The opposition between the second and the third stage, as it has been described in the foregoing, coincides in part with the opposition between ancient and modern thinking. But it will serve for the closer illumination of the nature of human thought if we give a more thorough characterization of this opposition. It can be summed up under two points of view. Ancient thinking has a predominantly formal character; on this rests in part its great and lasting significance. And next, and in connection therewith, it is characterized by the presupposition that the unchangeable stands, not only in a logical but also in an ethical respect, in value above the changeable.

47. The opposition between formal and real knowledge appears already in *Parmenides* of Elea, and so sharply that he recognizes only formal knowledge as real knowledge. Only those propositions that can be formed as pure identities can be the object of true conviction, and only such propositions can express what truly is. Being must be unborn, imperishable, unchangeable, indivisible, concluded in itself. Of the manifold, changeable, and perishable no knowledge can be had, only "opinions of mortals."

Here appears, with the thought-energy that distinguished the Greeks, the principle of identity, the demand for thought's unity with itself, the strict recognition, as the first principle of all thinking, as the formal ideal to which thought seeks to bring all cognition as near as possible. Parmenides would not even concede that differences and changes could set serious problems, whether such problems can be solved or not. Only conjectures, not true science, are here possible[65].

Two other Greek thinkers, the one Parmenides' predecessor, the other his successor, made, on the contrary, attempts to bring identity, the highest form and norm, into closer connection with the objects' difference and change. *Heraclitus* maintained that in all the constant streaming and under all the conflicting oppositions there is nonetheless something that is constant, namely the law, existence's eternal fundamental thought (λόγος), which subsists under the constant change of all phenomena. And *Democritus* set up the proposition that one reaches a true understanding only when one disregards

[65]Cf. my treatise on Parmenides in *Mindre Arbejder.* Anden Række, p. 141–149.

all the qualitative differences that the senses show us, and explains everything by the reciprocal action of homogeneous parts. His atomic theory is Parmenides' doctrine of identity applied to each single one of existence's smallest parts.

While Parmenides denied the reality of what falls outside pure identity, Heraclitus and Democritus set themselves the task of tracing the given differences and changes back to identity, partly the law's identity under different conditions, partly the elements' identity in different connections.

In *Plato* appears the same conviction as in Parmenides of the world of identity as the true world, only that he finds the identical in the general concepts or Ideas that hold for the sense-world's multiplicity and change, and that for him are not merely human thought-products, but the expression and content of what truly is. The logical point of view, which more hiddenly made itself felt in his predecessors, and which he had in Socrates' school been trained to use, came in him to clear consciousness. What filled his mind with ever new enthusiasm was the fact that concepts can be formed which, in their unchangeableness, their identity with themselves, hold for large groups of differences. The victory of likeness over the differences in human thought was for him a testimony that the world of Ideas was the true world. All cognition was recognition of the Ideas in the sensory world, and the highest spiritual capacity was that by which, from the points of likeness within the sense-world, a world of thought is formed, so that an ascent to the true reality becomes possible.

The Greek concept of science is predominantly formal. True science was for the Greeks logic and mathematics; their empirical science for the most part did not get beyond the collection of more or less exact observations. Democritus and Plato are the classical Greek thinkers, and against them both there was in antiquity directed the objection that they set aside and rejected what the senses teach (τὰ αἰσθητά), and kept merely to what pure thought grasps (μόνοις ἑπόμενοι τοῖς νοητοῖς)[66]. Both Democritus' atoms and Plato's Ideas were pure objects of thought, constructed from the demand for identity. Both for the founder of the atomic theory and for the founder of the doctrine of Ideas the sense-qualities conflicted with thought's pure identity.

Aristotle maintains—in opposition to Plato's doctrine of Ideas and Democritus' atomic theory—the reality of quality and of individual existences. It was thereby established (without Aristotle's fault, for millennia) that the science of empirical objects could consist only in classification. Aristotle himself did not fail to appreciate formal science; he is, after all, himself the organizer of formal logic, probably influenced by the development

[66] *Sextus Empiricus* ed. Bekker. p. 299.—Cf. my treatise on *Forholdet mellem Platon og Demokritos* in Nordisk Tidsskrift for Filologi. 1909.

that mathematics had gained; but logical thinking and qualitative comparison continue in him to stand side by side. He saw well that movement, change of place, lay at the basis of all other change in nature—of qualitative change, of coming-to-be and passing-away, and of increase and decrease; but he made no attempt to carry through this thought, which would have led him to the point where modern natural science begins.

It was two different, but both deep-lying, presuppositions that hindered the Greeks from letting formal and real thinking cooperate in a fruitful way.

From the Orient the Greeks had inherited the presupposition that the star-world's movements took place with absolute regularity. It was this unchangeableness, in opposition to the earthly (sublunary) world's mutability, that first opened the view for the possibility of science. Here one found the self-identical being realized. And one found then, in thought's firm and constant forms, expressed in the general concepts, a counterpart to it. Both the visible and the invisible unchangeableness were embraced with religious awe, as a refuge under the change of things. A peculiar religious type makes itself felt here[67]—but in remarkable connection with a physical and a logical ideal. The presupposition that the unchangeable is better or more perfect than the changeable the Greeks did not give up, any more than the Indians gave it up.

In a peculiar opposition to this tendency in Greek spiritual life stands the Greek sense for the concrete, the individual, and the intuitable, the living urge for plastic formation of the spiritual content. Not only mythology's gods and art's figures, but also thought's Ideas stood as clear images, whose self-resting firmness neither thought's analysis nor experience's increasing fullness could dissolve. The single observation and the single individual stood as something apart, both in opposition to conceptual thinking. In Aristotle this opposition culminates; for just as convinced as he is that only the universal is object for thought and science, just as convinced is he that only the individual has existence or being. The individual is that which is the object of all assertions, but which can never itself be used as predicate. (Cf. 34.)

48. The Renaissance's thinking led beyond the opposition at which Greek science had stopped. The multiplicity and mutability that life and experience present were now conceived precisely as means to come to know nature rightly. *Nicolaus Cusanus* and *Giordano Bruno* showed that thought itself works its way forward through oppositions, operates by constantly setting limits and overstepping them in order to find new limits, so that no opposition, no barrier, can be established once and for all. And experience itself now led to the discovery that one had drawn in far too external a way the boundary

[67]Cf. my *Religionsfilosofi*[2] p. 51–55; 125–128.

between the world of unchangeableness and that of changeableness. *Tycho Brahe*'s "new star," which so suddenly emerged in the sphere of unchangeableness, moreover presented changes both in brightness and color, and his investigations of the comets showed that these apparently so capricious beings did not move about in the sublunary world, but on the contrary traveled through the ether-world higher up. And Galileo stated outright that the earth appeared to him precisely "most noble and admirable" (nobilissima e ammirabile) on account of the many changes that take place on it. He cannot in itself find any perfection in the unchangeable. The standard for value becomes another. It is a matter not merely of ascending to the unchangeable world of eternal Ideas, but of finding the laws according to which the changes in the world of experience take place. As one had found changes in the world of "unchangeableness," so one also sought to find unchangeable laws in the world of change. In antiquity *Archimedes* had made the doctrine of equilibrium an exact science. Galileo began to subject the doctrine of movement to an equally exact investigation. Ancient science had essentially been statics and morphology; now dynamics gradually came into its own in the different domains.

If one was to find an unchangeable law through the multiplicity of changes, one had to keep to the properties of things and events that make quantitative determination possible. Counting, measuring, and weighing had now to become means of thought to a far higher degree than before. One had then to keep to properties such as magnitude, figure, and movement, and try to determine the appearance and changes of the qualitative properties through their relation to them. The opposition between formal and real thinking now became an opposition between quantitative and qualitative knowledge, and in this form a more intimate cooperation between the two kinds of thinking was possible. Speculation and observation, deduction and induction, could now work into each other's hands, in that the inferences drawn from general propositions could, by reason of the quantitative formulation, be tested in an exact way by their relation to what experience showed, while conversely, from experience, ever more exact general propositions could be formulated. Thinking could now describe an arc from experience through inference to new experience. Ancient thinking really knew only the first part of this arc. When it had found a starting-point for its abstraction, it ascended to the world of Ideas, the world of logical and mathematical forms, and shrank from descending again to apply and test the thought-results won in the world of experience. It believed in apotheoses, but not in incarnations.

It was now no longer the ideal to order qualities under general concepts of different degrees and thus form classificatory systems, but to find relations of dependence between changes in the world and express them in definite laws. Explanation steps in place of mere classification, and quantitative description forms the transition from the latter to the former. Logic and mathematics enter as tools into the work of real cognition; they no longer form a world subsisting by itself, but their principles and laws are woven into the world of thought that builds on the ground of experience. Platonism and Positivism strive to enter into as close a connection as possible.

49. Purely logically, the change that now takes place in thought's task and mode of working shows itself in the fact that judgment, not the concept, becomes the most important thought-form. As we have earlier seen, the opposition between judgment and concept means only a different weight on two sides of the same thought-process (24). But the opposition between ancient and modern thinking can very well be expressed thus, that the former uses the formation of judgment to win concepts, the latter uses concept-formation to reach judgments. The necessity of forming concepts was Socrates' great discovery, which gave Plato enough to think about throughout his whole life, and which led to the formation of a world of Ideas as the expression of true reality. But for modern thinking the point is to find judgments that express definite relations between definite changes. Through such judgments—by virtue of the relation of reciprocal action between concept and judgment—concept-formation will in turn be able to attain greater perfection.

In a characteristic way this tendency appears in the fact that two of the Renaissance's thinkers interpret Plato's Ideas as the laws of things. *Bruno* teaches that the task is not merely to form general concepts that express what is common to several phenomena, but to find expression for the real connection between different phenomena (e.g. the parts of an organism) among themselves, since only by such a connection can one understand the nature of the individual phenomena. *Bacon* expresses himself in a similar direction when, for the understanding of a phenomenon (e.g. heat or organic life), he demands that one must not remain standing at abstract concepts, but shall try to find the continuous, often hidden process (latens processus—per minima) by which things get their definite constitution[68].

With this there now also follows a change in the conception of inference. As long as one had merely to do with a system of super- and subordinate concepts, it was natural to build the theory of inference on the principle that what belongs under a concept also belongs under the concepts under which this concept belongs. Inference then stands as

[68] *Den nyere Filosofis Historie*^{2}. p. 126; 194–196.

a process of subsumption, culminating in a highest concept. But when weight is laid on the fact that concepts as well as judgments are to enter as members in the ever-continued thought-movement, it will be natural—as has also (see 33) shown itself to accord best with the essence of judgment—to conceive judgment as the expression of a relation of identity, so that inference can take place by different judgments being combined by means of substitution. The purely logical thought-movement thereby approaches the mathematical, which proceeds through equations and substitutions. *Leibniz*'s identity-logic is thus a characteristic expression of the more recent time's thinking—even though a later time's logic has in several respects had to strike out on other paths[69]. It must especially be emphasized that Leibniz did not bring his logic into connection with a doctrine of categories, and thereby came to overlook that there are many other relations than identity that can be of great significance for the thought-work (even though they can all be purely formally expressed by predicates in identical judgments). In a later section we shall have occasion to discuss the relation of identity in comparison with other relations. But it is extraordinarily characteristic that identity for Leibniz does not stand as an object of mystical beholding, like the "Idea of Likeness" for *Plato* (*Phaedo* p. 74), but as a tool of thought. This shows itself already in his definition of identity as the possibility of substitution: identical are the concepts that can be put in place of one another without the validity being violated (salva veritate). There lies in this a constant summons to test identity experimentally, to see whether it can be maintained in the individual special connections in which the concepts occur. It is in the world of multiplicity that the thought-work is to be exercised, and the question is how far relations of identity can there be demonstrated.—

While Platonism was inclined to confuse conceptual determination with causal explanation and could thereby lead to assuming just as many "forces" as there were kinds of objects, in modern thinking the concept law stepped more and more in place of the concept force, or rather this concept is subordinated to the former. In the Aristotelian doctrine of nature, nature formed a series of stages, within which every stage contained the possibility or the force (δύναμις) for the following one. But about the possibility or the force one then learned nothing other than precisely this, that it was—possibility or force for the following one. There were then just as many possibilities as there were realities. The qualitative differences were explained by just as many qualitates occultæ. In scholasticism, which in many domains maintained itself far beyond the Middle Ages, one lived high on that kind of word-explanation. When it is first and foremost a matter of

[69]Cf. on this *Couturat*: *La Logique de Leibniz.* p. 386 f; 433.—Couturat overlooks, however, that all relations can be expressed as predicates.

finding a law, that is, a judgment that states one phenomenon's dependence on another, one need not indeed give up the concept force, but this gets the definite meaning of denoting that in the preceding phenomenon which contains the ground for the following phenomenon's setting-in—the peculiarity of a phenomenon that from it one can infer another's setting-in. Leibniz is clear that the concept force contains no explanation, but only a summons to seek why the transition to a new phenomenon takes place precisely under these definite conditions. The concept of force is constructed in each single case on the basis of the law that has shown itself to hold for such a transition[70].

50. The new concept of science could not immediately be applied in all domains, and it became of decisive significance for the philosophical treatment of problems in the centuries immediately following the Renaissance that there were some domains that were especially suited to be treated according to its demands. These domains then naturally came to stand in the foreground in the scientific conception of the world. It was the doctrine of movement and mechanics that determined the conception of nature in the seventeenth century's great systems. In these domains the differences of quality and individuality stepped back before the universal laws. *Leibniz* here forms the transition to a new period, in that he so strongly brings out the concept of individuality, and demands not only a continuous connection and law-governedness demonstrated between all existing beings, but also within the little world that every single individual being constitutes. Within the individual beings each by itself too there take place changes, and if we could find the law for these changes, we would thereby have determined the individual being's peculiarity: La loi du changement fait l'individualité de chaque substance particulière (Lettre à Basnage 1698).

By degrees, as we in the series of the sciences pass over from the formal to the real, the more difficult the cognition of law becomes, the more we stop at mere experience without being able to carry through the reciprocal action of deduction and induction. The logical law of the inverse relation between concepts' content and extension shows itself here in its great significance: the simpler the objects we face, the easier it is to form concepts and laws; the more concrete and composite the objects we have before us, the more we stop at mere descriptions and at characterizations of individual peculiarities. Biology, psychology, and sociology present in increasing degree difficulties for the development of a cognition of law.

Modern thinking has not disowned Platonism. As the Greeks sought universal Ideas,

[70] *Den nyere Filosofis Historie*[2]. I. p. 346.—In a fragment brought forward by *Couturat* (see Bulletin de la Société Française de Philosophie. II. p. 88 f) Leibniz says: Vis activa nihil aliud dicit quam id, quo sequitur transitio, sed non explicat in quo consistat.

so modern science seeks universal laws. And herein shows itself the unity of human thought under all forms. Yet it would not be right, with *Henri Bergson*, to assert that the modern concept of science is only different in degree from the ancient. Bergson grounds this conception on the fact that, just as the ancient "Ideas" meant general concepts about things, so the modern "laws" mean general concepts about changes. Modern thinking has, therein one finds the advance, taken up time as an element. But the main thing for it is nonetheless always to be the universal and abstract, not the single and individual[71]. To this it must be remarked that modern science, the more it becomes conscious of its proper goal, sees it as its task to use the general concepts and laws as means and ways to understand the individual and historical totalities. It is the individual totalities whose understanding stands as science's last goal, and the poorer-in-content, but at the same time more comprehensive, thoughts are to serve to lead in the direction of this goal. The "law of change that constitutes the individuality of the single being" is the highest ideal of modern thinking. Therein lies a difference of principle between the ancient and the modern conception of science, at the same time that the world of problems becomes far greater than it could be for ancient thinking. The difference thus lies not only in the way, but also in the goal.—The doctrine of categories will give occasion to take up this question again.

[71] *Bergson*: *L'évolution créatrice.* Paris 1907. p. 359 f.

III. The Forms of Thought (The Categories)

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A. The History and Method of the Doctrine of Categories

51. Through the preceding psychological and historical presentation we have come to know the different stages by which human thought works its way forward. Through the work we here, as everywhere, come to know the energy. Sensation, intuiting (with association), and judging stood, psychologically regarded, as the most important stages, and the rise from stage to stage meant an ever clearer consciousness of objects and elements and their mutual relation. In a historical respect the relation between the objects and their interpretation was the main thing, and the ideal here was that interpretation takes place through the objects' mutual illumination of one another, an ideal that, however, in turn pointed back to definite presuppositions.

It will now be the task to give a survey of the fundamental forms with which thought works, and which condition presuppositions to which it is by its nature bound. Such fundamental forms have from old times been called categories.

The word category denotes, according to a usage established by *Aristotle*, assertion, predicate. There are therefore, in the widest sense of the word, infinitely many categories. But in philosophy one understands by category a fundamental predicate, a fundamental concept, and the doctrine of categories sets itself the task of gathering and ordering the predicates that with greater or lesser degree of consciousness are applied by thinking in the different domains where it works, and tracing them back to certain mutually kindred and yet each individually peculiar fundamental forms.

The doctrine of categories occupies a central position in philosophy. It points on the one side back to the study of the cognition won, as the psychology of cognition on the one hand and the history of science on the other present that development, on the other side forward to the philosophical treatment of problems, which investigates how far the assertions we have at our disposal can lead us.

The categories come into being involuntarily during thinking's work at satisfying the objects' demands. They arise sporadically, to meet individual special demands or answer momentary questions, and only later can the question be set whether their significance is limited to a single domain or a single situation, or can extend further. The point is then to gather them, to demonstrate their connection among themselves and with the nature and tasks of human thinking, and to trace them back to as few and simple forms as possible.

The most significant attempts in this direction have, at intervals of millennia, been made by *Aristotle* and *Kant*. All other attempts that exist, also the attempt to be made here, relate to these models as modifications that strive to give closer grounding and more exact connection. Both masters of the doctrine of categories started from an investigation of judgments—quite naturally, since predicates appear only in judgments. The fundamental predicates must be the determinations that constantly recur in the judgments that objects give us occasion to pass. An analysis of the different assertions a judgment can contain, or of the different kinds of judgment that can be passed, was necessary in order to find the fundamental predicates. Aristotle goes the one way, Kant the other.

52. Judgment is, as we have seen, not the first form in which objects appear to us. Judgment presupposes intuiting—sensation, memory, or imagination. In intuition the objects are involuntarily ordered in us and for us, and already at the coming-into-being of intuition there operates the psychic energy that can later appear in the formation of judgment. In intuition, whether it is sensation, memory, or imagination, a multiplicity is involuntarily ordered; a form of wholeness is brought about. The arising of every new sensation is an interruption of consciousness's given state, and work is now done involuntarily at ordering the new object or element (the new distinguishable thing) into the whole connection. As long as possible consciousness dwells in intuiting—in the three named forms, which, for that matter, can only at a later stage be distinctly distinguished from one another. But when there appear changes that can no longer be equalized in the form of resting intuition, there begins a work at a new kind of connection. Reflection is awakened, attention moves from object to object, from element to element, brings them out each by itself in its peculiarity, and seeks through such analysis to find a new ordering of the objects. By this way judgment can step in place of intuiting.

During the formation of judgment, as we have seen, the thought-movement goes from a given and presupposed, more or less indefinite basis, the subject, over to a more definite determination, the predicate. We think in predicates, or rather by predicating (asserting). The formation of judgment begins with the one leg of thought's compass standing in or being set on a single point, and a place then being sought where the other leg can be set. A doctrine of categories becomes possible through all special determinations or predicates being able to be subordinated under a few fundamental predicates that recur again and again. The categories are the chief forms in which decision- and determination-questions (cf. 26 and 31) are preserved.

When an object emerges in my sensation, in my memory, or in my imagination, I will involuntarily be able to raise the following questions:

Is it one with something I know?

Does it resemble something I know?

If it is different from something I know, is it then a difference in kind, in time, in number, in degree, or in place?

Does it contradict something I know?

Can it be derived from something I know, or can something I know be derived from it?

Is it cause of something I know, or is something I know cause of it?

Is it part of something I know, or is something I know comprised in it as part in whole?

Through what different intermediate forms has it become what it is?

What value does it have? Is it a means for something valuable that I know, or does it perhaps denote a higher purpose than the valuable I have hitherto known?

In the preceding psychological and historical investigations we have in various places come upon objects that can give occasion to such questions. But whether the list of questions gives us a direction to all categories must remain open. It will on the whole be impossible to demonstrate that any list of categories is complete, although both Aristotle and Kant and most of their successors have considered it possible to set up concluded systems of categories. We form our judgments on the basis of intuiting, and within intuition new objects and elements can always appear, which can require new judgments and give occasion to new predicates. The involuntary, often obscure, basis for the analyzing thought-work can shift, without our being able to say in advance how far this shifting can go—and wholly new questions can be called forth thereby. And even if the basis does not shift, proof can never be given that it is exhausted through the judgments we have with full consciousness and exactness formulated. Reflection constantly draws from a source whose spring is hidden[72]. History shows that categories can arise and pass away by reason of the shifting of the basis, or because hitherto unheeded objects or elements draw attention to themselves. A presentation of the doctrine of categories can then have validity only for a certain stage in the development of cognition. It has its essential significance in being a stocktaking that can lead to a clear understanding of the problems.

There must (cf. 3–4) indeed be a distinction made between the basis and the reflection that is directed toward it. Reflection can be compelled to make distinctions and establish differences that cannot without further ado, or in the same way as they lie before reflection, be said to be present in or "lie in" the basis. Of course there must in the latter be occasion and justification for setting up those distinctions; but they can there make themselves felt in another form. There is analogy, not identity, between intuition and judgment. They are both forms of the psychic energy that at all stages strives for connection and equilibrium; but they are two different forms of this energy.

Only in the clearer conscious life can there, e.g., be distinguished, as *Kant* does it, between matter and form, or, as *Stumpf* in the most recent time does it, between phenomena and functions. (Cf. 5–6.) In the involuntary mental life this difference does not come forth. One will perhaps say that the functions are present, even if they are not specially noticed and not set in opposition to the phenomena. But it must in any case make an essential difference whether a state appears as an undivided whole, or whether analysis has brought out differences—whether it (to use Stumpf's expression) appears "unzergliedert" or "zergliedert." There is a difference of quality between these two forms, even though we can understand them as forms of one and the same energy at different stages.

It is this difference that makes it that the doctrine of categories, however important its connection with psychology (as well as with the history of science) is, cannot without further ado be called a part of psychology. It stands between this and epistemology proper.—In sharp opposition to the psychologism that effaces the difference between the two disciplines, there appears in the history of thinking a constant tendency to give the doctrine of categories without further ado a metaphysical significance. One then sees in the fundamental concepts of the human spirit an immediate revelation of existence's innermost essence. The Platonic doctrine of Ideas is the greatest example of this assumption, which later appears under various forms, not only in philosophers like Spinoza and Hegel, but also in natural scientists like Newton (especially in his concept

[72] Cf. my treatise *Filosofien og Livet.* (Mindre Arbejder. Anden Række.)

of spatium absolutum as sensorium dei).—The doctrine of categories can be called a descriptive epistemology, an introduction to the special discussion of the problem of cognition, and it takes up as such other points of view than psychology and metaphysics.

53. Some features of the history of the doctrine of categories will serve for the illumination of its significance. I begin with the two great masters of Greek thinking. The opposition between Plato and Aristotle appears here in a characteristic and instructive way.

For *Plato* it was the greatest fact of thought and life that it was possible to form concepts that hold for multiplicities of objects. His philosophizing was a constant ascent to the world of general concepts, of Ideas. (Cf. 44.) Starting from the objects' obscure, chaotic basis, he formed a world of predicates. In the predicates, not in the subjects, the world of objects, his thought found its home. There expresses itself here the correct insight that the predicates are thought's proper work and property. Only, Plato overlooks that the objects are the constant presuppositions, and that during the formation of judgment a constant reciprocal action between subject and predicate takes place, so that in the judgment passed they steadily determine each other. (See 33–35.)

Not all Ideas stand for Plato on the same line. Repeatedly he comes back, in his dialogues, to a class of Ideas that show themselves to be inseparable from thought's activity, whatever special objects this concerns. He is, as we have earlier seen, one of the predecessors of the synthesis-theory. The soul must be able to exercise a combining, as surely as it can have sensations of different kinds and compare them among themselves. And by this very combining and comparing certain predicates are applied, which express the relations that are ascertained: Being and Not-being, likeness and unlikeness, identity and difference, number-concepts. It is by the application of the forms expressed in these concepts that thought ascends from the objects to the Ideas, and the named concepts are therefore themselves the most fundamental Ideas. (*Theaetetus* p. 185–187.) And he especially emphasizes judgment as a connection (συμπλοκή) by which the Ideas are determined in their mutual relation (*Sophist* p. 259).

In his last, Pythagoreanizing period he occupied himself with ordering the fundamental concepts in a series analogous to the number-series. But of this attempt little certain is known.—A doctrine of categories proper is first found in the thinker who introduced the very word category.

Here and there in *Aristotle*'s different writings some concepts are applied that denote the most general points of view that can be used in more special investigations, no matter what these concern. Nowhere, however, has he—as Plato did—brought them into inner connection with thought-activity itself, and in the writing (*Categories*) that (whether it directly derives from Aristotle's hand, or only indirectly) especially treats them, they are placed—ten in number—in a very external way, in a purely catalogue-like manner side by side. Probably, however, he came to them through analysis of a number of judgments.

First in the series of categories stands the concept thing (essence, substance: οὐσία; τὸ ὑποκείμενον). And yet it is really no category—for it cannot be used as predicate, cannot be asserted of anything. It denotes the basis, whose closer determination is sought through the other nine categories. In itself it can only be denoted negatively, namely as that which is not asserted of anything—and then one must, for that matter, point back to intuition. Aristotle has here aptly characterized thought's constant basis, that which is first to be determined through the thought-work itself and therefore prior to this cannot have any determination. Every positive determination would already be a predicate, and the "essence" was to be precisely that which was to be determined by predicates. In opposition to Plato, Aristotle lays great weight on this constant basis, while Plato forgets it in his enthusiasm for the Ideas, which were yet really its predicates. Here shows itself the realistic character of Aristotle's philosophy. He has a sharp eye for the fact that thought always works on objects at hand.

The object, the given, is for Aristotle always an individual (ἄτομον καὶ ἓν ἀριθμῷ. *Categ.* c. 5). Only the individual exists. Thought has here one of its limits; it needs a starting-point that cannot itself be used to determine anything else! Further back we cannot come.—Analogies to this concept appear in the absolute atomic theory, in Spinoza's Substance, in Leibniz's Monads, in Kant's "Ding an sich," as well as in the higher religions' concepts of God. (Cf. 34–35.) Aristotle demands absolute conclusion of thinking. The one limit thus lies in "Substance"; the other lies in the principles (ἀρχαί) that condition all judging and inferring and therefore cannot themselves be proved (*Analytica* post. I, 22).

The (other) nine categories are: Quantity, Quality, Relation (πρός τι), Acting, Being-acted-upon, Place, Time, Condition, and Having (the possession of a property). This list Aristotle considers complete, although he does not demonstrate the necessity that precisely these, neither more nor fewer, exist. He considers them irreducible, independent points of view that cannot in turn be brought under a common point of view, but only stand in a relation of analogy among themselves. More fully he discusses only the first three (Quantity, Quality, and Relation).

Of special interest, especially in comparison with Plato's doctrine of categories, is the category of Relation (relational determinacy). As examples he names equality and inequality of magnitude, numerical relations, cause and effect, cognition and the cognized. These are relations that have their root precisely in the combining and comparing thought-activity taken as basis by Plato[73]. Both Quality, Quantity, and the other nine categories could, on closer consideration, be brought under the category of relational determinacy, and it will later show itself that in the more recent time attempts have been made to set up Relation as the first, most comprehensive of all categories. Such an attempt Aristotle did not make, however much most of his examples point in that direction. And the ground for this must surely be sought in the fact that he did not (at least not in the writings we have) attempt to bring the individual categories into natural connection with thought's mode of operation as a whole.

The changes that the doctrine of categories underwent in antiquity after Aristotle it is not necessary for our purpose to enter into. For spiritual development it was, however, a characteristic sign of the times that the last eminent thinker of antiquity, *Plotinus*, limited the Aristotelian categories to holding only for the sensory world. In the supersensory world the differences and oppositions that the categories denote fell away, as well as the very difference between the essence and its predicates. All is there in inner unity and harmonious fullness. Since Plotinus nonetheless wishes to think about this supersensory world, he is compelled to use some of the categories, but they then get a more or less mystical significance[74]. In a similar way Christian theology placed itself. *Augustine* reproaches himself in his *Confessiones* (IV, 16) for having in his youth been so occupied with Aristotle's doctrine of the categories (decem prædicamenta), since they cannot, after all, serve for thinking God, in whom no distinction can be made between essence and properties. God cannot be subordinate (subjectum) to his own properties; one cannot, e.g., distinguish between him and his goodness. The difficulty that all theological thinking has to struggle with here came clearly forth. And it was the Neoplatonists who had first felt it.—

Apart from this, the Aristotelian doctrine of categories prevailed for millennia, and its after-effects can still be traced.

54. The next great and epoch-making attempt at a doctrine of categories was made by *Kant*. For details of his theory I must refer to my treatise *Om Kontinuiteten i Kant's filosofiske Udviklingsgang* (Vid. Selsk. Skr. VI) and to my work *Den nyere Filosofis Historie*. Here I shall only bring out some chief points.

With full consciousness Kant derives the fundamental predicates or categories from

[73]See more closely on the concept of relation in Aristotle: *Trendelenburg*: *Geschichte der Kategorienlehre*. p. 117–129.—G. *Grote* (*Aristotle*. I. p. 115–122) shows that relation lies at the basis of all nine categories.

[74]Cf. *Arthur Drews*: *Plotin und der Untergang der antiken Weltanschauung*. (1907). p. 158–173.

the judgment-function. And in that he conceives judgment as arising from a connecting operation, a synthesis, and furthermore sees in synthesis an operation that lies at the basis of all consciousness, whether we notice it or not, he has brought the doctrine of categories into inner connection with the whole nature of human consciousness.

Therefore Kant is also clear that the categories are provisionally only forms of our thought, while Aristotle immediately conceived them as predicates of existence (κατηγορίαι τοῦ ὄντος. *Categ.* c. 2). After having set up the categories, Kant then passes over to investigating how, with what right, and to what extent they can be said to lead to valid cognition. He thus distinguishes definitely between the fact that we work with certain forms of thought, and the validity of the cognition that can be won by their help.

Kant was convinced that there was a definite number of categories, which follow with necessity from the nature of thought. He supports this conviction on the fact that there is a definite number of mutually kind-different classes of judgments, and that to each such class a category must correspond. Thus, e.g., the negating judgments were to form a class of their own, likewise the quantitative judgments. According to what we have earlier developed (26; 32), there can here be no question of a difference of kind. As regards the other kinds of judgment (e.g. the problematic and hypothetical) too, it can be shown that it is linguistic, not purely logical, grounds that lead to distinguishing between them. Kant's number twelve is therefore just as little grounded as Aristotle's number ten[75].

Of great interest, on the other hand, is that Kant himself demonstrates that there is a common characteristic mark for all categories, namely the concept of continuity, which in turn appears in two forms, in the concept of magnitude and in the concept of cause. Under the concept of magnitude belong again concepts concerning extension in time and space, and the concept of intensive magnitude (degree), while the concept of cause, besides the cause-relation itself, also comprises the concepts of possibility, actuality, and necessity.

Kant's doctrine of categories can accordingly be schematized in the following way:

Synthesis (Continuity)

Magnitude	Causality.
a) Unity, Plurality, Totality. (Extension).	a) Subsistence and Change. Cause and Effect. Reciprocity.
b) Reality, Negation, Limitation. (Intensity).	b) Possibility, Actuality, Necessity.

[75]Cf. already G. E. *Schulze*'s critique in *Kritik der Theoretischen Philosophie* (1801). II. p. 291–333.

The concept of continuity hangs closely together with the concept of synthesis. For the closer the connection between objects or elements, the more intermediate links can be found between them, the more complete and exact the synthesis becomes. From his doctrine of categories Kant therefore derives the following principle: "To allow nothing in the empirical combining that could hinder the continuous connection of all phenomena; for only thereby does that unity of experience become possible within which all observations have their place."[76] Experience is for Kant the connection within which every single object is a member determined by all other members. And experience in this sense becomes possible only by reason of consciousness's original combining energy. The fruitful thing in his way of considering lies in the summons it contains to ever-renewed inquiry in order to find new members in the connection of experience and to determine this connection in an ever more exact way.

Besides continuity there is yet a concept that consistently ought to have had a place in Kant's doctrine of categories, namely the concept of Relation (relational determinacy). It is seen from Kant's notes to the "Kritik der reinen Vernunft" (the "Reflexionen" published by Benno Erdmann) that it was by occupying himself with the concept of relation and its significance for thinking that he became clear about synthesis as the fundamental form of all consciousness and all thinking. He states thus not only: "Our reason contains nothing other than relations," but also: "The category of relation (of the unity of consciousness) is the most important of all. For unity really concerns only relation."[77] He has thus assumed a very close connection between the concepts synthesis and relation. When he has not entered them among his categories, the ground has been that he regarded them as presuppositions for all the special categories. Therefore the significant utterance cited has not gained influence on his systematic presentation.

An essential objection to Kant's doctrine of categories is that he narrows the concept of category too much.—In the first place, space and time are supposed to be not categories, but "forms of intuition." But in our cognition we do not stop at the mere staring at what spreads out in space and time; it is the definite relations of space and time that we have to do with, relations of a peculiar kind that do not coincide with any of the relations that Kant's categories denote, but that must nonetheless be reckoned among the fundamental relations we work with in our thinking. Kant on this point makes too sharp a distinction between "intuition" and "understanding."—In the second place, he does not reckon the purely logical relations identity, difference, and likeness among the categories, although they, as Plato already clearly saw, are the simplest and most necessary of the forms with

[76] *Kritik der reinen Vernunft*^{2}. p. 282.
[77] See *Om Kontinuiteten i Kants filosofiske Udvikling*. (Vid. Selsk. Skr. 6. Række. p. 53. kfr. p. 27).

which our thought works, and although we must ask about the validity of these forms just as much as about the validity of concepts like magnitude and cause.—In the third place, he distinguishes sharply between categories, which are always applied to finite, limited relations, and “Ideas” (partly in another sense than the Platonic), which demand absolute conclusion of the thought-work in a comprehension of the whole. He overlooks that the categories too really make ideal demands on our thought-work. When are the concepts of magnitude and causality completely satisfied? Do they not also, just as much as, e.g., the “Idea” of the world, require a comprehension of the whole, if they are to be carried through completely? Continuity, indeed experience itself (in the sense in which Kant takes this word), set up great ideals for our cognition. Already above (37–38) it showed itself that the concept of reality is an ideal concept.

55. While Plato and Kant started from an analysis of thought-activity, as it expresses itself in judgments, *Hegel* begins his logical system directly with the simplest, most abstract concepts, relying on the fact that every concept, when only it is thought through rightly, must lead over to other concepts. For a contradiction will come forth when a single concept is set as the sole, all-comprising expression of existence. And on the forward-driving force of self-contradiction rests the train of thought in Hegel’s logic. Self-contradiction is only a testimony that no thought can stand isolated—that the world of thought constitutes a whole against which one sins by holding fast a single point of view. But precisely for this reason one could, Hegel maintains, safely begin with a single thought, even the poorest of all; it will with inner necessity lead over to the others, so that ever richer thoughts are won.

Hegel has, however, in fact conceded that the doctrine of categories presupposes a psychological-historical preparation. In his *Phänomenologie des Geistes* he has presented the stages by which the human spirit—and the “world-spirit” through it—raises itself to a standpoint on which the thoughts can unfold according to their own law, constantly through contradictions working their way forward to higher unities encompassing the conflicting oppositions.

The category that is all-determining in Hegel is one that in Aristotle was only one among many, but that will show itself to be the one that most immediately hangs together with the nature of thought: relational determinacy, relation. Contradiction comes forth in Hegel only through one’s treating a concept’s content without regard to the relations that determine and limit it. It is a contradiction to remain standing at the abstract concept Being, since it can get no content at all and is so far a Being of Nothing—a Not-being. It is a contradiction to remain standing at the concept Identity, for it presupposes members that are identical, and these members must be different, if there can be any distinguishing between them at all; the concept Identity then points over to the concept Difference, and is moreover itself different from it!—It is Relation's law, the law of relation, or perhaps better the law of relational determinacy, that Hegelian dialectic impresses.

What Hegel overlooks is that every one of the concepts that can thus be taken up and tested as examples of our thoughts' "dialectical" character have themselves grown up in a definite, given connection. That they, when they have been tested and determined, can be ordered in series, proceeding from the simplest and most abstract to the most concrete and individual, is a matter in itself; the question in this respect will only be whether Hegel's ordering by triads—position, opposition, and higher unity of both—is really the natural ordering. The main thing is that what in Hegel stands as thought's self-development, a kind of logical generatio æquivoca, is in reality a fruit of a constant reciprocal action between thought's forms and its objects. Our thought is a reflection—both in the sense that there is always something it seeks after, and in the sense that at every stage it presupposes objects and so far comes after these (cf. 2–4). Hegel constantly draws from the memory of experiences, the race's and his own; from there he has the objects that his reflection presupposes.

Reflection itself (and the concepts or Ideas in which it deposits its work) comes in Hegel's world of thought not first but last—precisely because it is the richest, most concrete category, the one that in its law of relational determinacy contains all other categories in itself. His logical system looks, by virtue of the law of the triad, thus:

I. Being. a. Quality. b. Quantity. c. Degree.
II. Essence. a. Ground. b. Phenomenon. c. Actuality.
III. Concept. a. Subjective Concept. b. Objective Concept. c. Idea.

From the simplest thoughts (I) one proceeds to thoughts that are distinctly relationally determined (II), and from these to thoughts that express wholes (III). By Idea Hegel understands, like Kant, a concept of totality. The Idea is liberated from all the limitations and contradictions under which all preceding members in the system more or less suffer.

But Hegel definitely rejects Kant's limitation of the categories and Ideas to being human fundamental thoughts whose applicability was first to be tested. His doctrine of categories is, like Plato's and Aristotle's, a metaphysics. The categories are laws of existence, not merely human laws of thought, and what drives forward from stage to stage, what operates in the dialectical method, is existence's own innermost essence. Hegel could say with Goethe: "Ist nicht der Kern der Natur Menschen im Herzen?" It is existence's essence as ideal and absolute whole that, because it cannot be exhausted in any single thought, compels thought to proceed to ever higher unities. Thought's law is one with existence's own law.

Even if one would accept this conception, there would be no ground to assume that contradiction should be the sole motive that drives thought forward. New objects can arise that demand their right, even if they do not stand in contradiction with other objects, and that can bring about the downfall of some categories and the transformation of others, or demand wholly new categories. And reflection can also be driven forward to conjectures or new seeking by a tendency to expansion, to extend won insight beyond the hitherto given, a tendency that is of such great significance for the development of the life of the spirit.—

The thinkers who in our day stand or think they stand nearest to Hegel in the doctrine of categories have, however, modified his philosophy more or less in the direction of Kant's conception. The dogmatic certainty is lacking. What animates them is the conviction that no thought can have validity when it is held fast isolated and absolute. Truth must constitute a totality, an infinite connection. "The Absolute" would be the completed, all-comprising experience, but such a one stands as a constant ideal. We work constantly at supplementing and integrating the fragmentary in our cognition. While Hegel in his logic lets the concepts develop out of one another with inner necessity and prefers to dwell up in this ethereal world of thought, his modern disciples are most occupied with ascending from the world of finite and limited experiences to that standpoint of completion. Their works remind one more of Hegel's "Phänomenologie des Geistes" than of his "System der Logik." Their thinking is really only an attempt to carry through the criterion of reality, and it is characteristic that F. H. *Bradley*, who can be regarded as the type for this group of thinkers, uses the word total experience (experience entire) for the ideal of cognition. (Cf. above 37–38.)

56. An interesting attempt at a doctrine of categories with the emphasis on Relation (relational determinacy) as the chief concept was made, in essential agreement with Kant, by *William Hamilton.* Every act of thought establishes a condition and a limitation: to think is to condition. All the fundamental concepts our thought works with indicate relations and conditions. Subject and object, thing and property, cause and effect, relations of time, space, and degree are examples of this. And with the law of condition or of relation it accords that comparison and judging cooperate in every apprehension, in every function of cognition. What is peculiar to Hamilton's conception, as to that of other Kantians, is that the law of relation is conceived especially as entailing a limitation—that it is the expression of our finitude. He does not emphasize that every

positive cognition and understanding, every advance in our knowledge, rests on more relations being demonstrated and drawn into consideration, and on each single relation being determined with greater exactness[78].

In *Comte*, the founder of the so-called positivist school, we find a doctrine of categories intimated that in its chief features reminds one of Kant's. For Comte connection is thought's most general form: tout se réduit toujours à lier. Cognition springs from an urge for unity. And connection takes place in two ways, partly by reason of the objects' mutual likeness, partly by reason of the order in which they follow one another in time. Herewith relation is already given as an essential determination. Besides the relations between the objects among themselves, Comte also emphasizes—and in express agreement with Kant—the relation between subject and object.

The difference between likeness and succession lies at the basis of the difference between relations of equilibrium and relations of movement, statics and dynamics, which Comte seeks to demonstrate in the different scientific domains. Where equilibrium and rest prevail, there are only relations of likeness to demonstrate; where there is succession and movement, the task becomes to demonstrate the dependence of the following on the preceding. Where the relation of likeness prevails, general propositions can more easily be set up and deduction applied; where the relation of succession prevails, so that the objects get a historical character, we do not so easily get beyond the special, and induction becomes predominant. According to this principle (généralité décroissante—complication croissante) Comte has laid out his ordering of the special sciences—from mathematics through astronomy, physics, chemistry, and biology to sociology. In this last, most concrete domain the relation between statics and dynamics appears as a relation between order and progress[79].

In *Charles Renouvier*, the most important representative of critical philosophy in France, both Comte's and Kant's influence is traced in the doctrine of categories. In his list of categories Relation stands first. It makes itself known in thought's distinguishing, identifying, and determining activity, an activity that is also decisive for all other categories, although Renouvier sets them up alongside the first. A definite principle for the ordering of the categories cannot be seen. The ordering is the following: 1) relation; 2) nombre; 3) position; 4) succession; 5) qualité; 6) devenir; 7) causalité; 8) finalité; 9) personnalité. Within each of them Renouvier seeks to demonstrate a triad of thesis, antithesis, and synthesis, which Kant had already intimated, but which cannot be carried through without artifice. A closer critique of Renouvier's arrangement I shall

[78] See *Den nyere Filosofis Historie*^{2}. II, p. 388 f.

[79] *Den nyere Filosofis Historie*^{2}. II. p. 333 f; 352–355.

not give; such a one will indirectly emerge from the ordering I shall later attempt to demonstrate among the categories I find. Renouvier is clear that a deduction of the categories such as Kant had attempted is impossible, and that only experience can decide whether our system of categories is the right one. Experience itself is a synthesis, and the categories are the constant relations that make themselves felt in experience—relations that presuppose just as many special forms of synthesis[80].

In *Eduard v. Hartmann* too Relation is a chief category (Urkategorie), because our conscious thinking is a reflecting, a positing of relations; it brings out the relations to which everything that becomes an object of consciousness is subject. All Being is relational Being. And this holds also for all sensation, although this has its special categories (quality, quantity, extension in time and in space). Within thinking Hartmann distinguishes between the reflecting thinking, which expresses itself in comparison, distinguishing and connecting, measuring, inferring, and the speculative thinking, whose categories are causality, finality, and substantiality[81].

Here too I cannot enter into a closer critique. It was in this survey especially a matter for me of bringing out the chief points, at which there also shows itself a certain convergence between the different catalogues. The convergence shows itself especially in the weight that is laid on consciousness's, especially thinking's, combining or connecting character and, in close connection therewith, on relational determinacy, relation as necessary consequence thereof and moreover as common characteristic feature for the more special concepts. It is of no slight interest that in a chief point of the doctrine of categories, this central part of philosophy, there is found such an agreement. The more special categories do not hang together so closely with the general nature of thought that they could be derived from this without further ado. They get a more historical character, and the order in which they are placed is conditioned in many ways by more special philosophical problems and views, something that is especially clearly seen in Renouvier and Hartmann.—

When some authors (Lotze, Sigwart, Wundt), following *Herbart*, place the categories object and property before the category relation, the objection must be made that a thing's concept contains only relations within itself and to other things, and that properties only express such relations. And to this is added, as the most essential objection, that the object is known only as thought's object; it cannot then itself be the first category. Aristotle indeed placed it first; but he added that it was really no category, since it was

[80] Cf. besides my book *Moderne Filosofer.* p. 72 f.: *G. Séailles*: *La philosophie de Charles Renouvier.* (1905). p. 84–133.—*L. Dauriac* in Revue de Métaphysique et de Morale. 1909. p. 487–496.

[81] Cf. *Arthur Drews*: *Hartmanns philosophisches System.* (1902). p. 766–846.

precisely that which was to have its determination through all categories.—

57. As regards the method of the doctrine of categories, it has already been said that the categories must be found by reflection's seeking to separate out the forms in which the life of thought involuntarily moves in reciprocal action with objects and tasks. The individual categories arise at different times and in different connections. But the presentation of them can be laid out in such a way that we begin with the simplest and proceed to the more rich in content. The simplest will probably be those that hang together most closely with the nature of thinking and therefore express the most necessary presuppositions that the objects must satisfy; they must therefore also hold in the greatest extent. They can merely indicate the demands that must be made at all on every object that is to be able to be thought. The more rich in content a category is, the less comprehensive a circle of phenomena it will be able to be applied to. It is a rule that holds for all our concepts, that content and extension—in concepts that lie in the same ascending series—stand in inverse relation to each other. From this springs, as will later show itself, one of the greatest problems for our cognition.

The transition from the one category to the other, that is, from the simpler to the more rich in content, is necessitated by the appearance of objects or elements that are not satisfied by the simpler, more comprehensive categories. Existence is then richer than thought. There must then be sought more special categories, forms in which the new objects or elements can come into their own. The more special categories will not be able to be derived from the more general; but it will perhaps be possible to show that they are forms in which the more general categories must appear under definite given conditions, or to find intermediate links between the old categories and the new. At every point thought's own nature stands in reciprocal action with its objects and its tasks. There will be a constant breaking between the two sides, a breaking that in our day has called forth a dispute between Criticism and Pragmatism. The doctrine of categories need not enter into this dispute, which belongs under the discussion of the problem of cognition itself. But it will perhaps be able to give a contribution to the illumination of the disputed question.

B. Fundamental Categories

58. What is given as an object for reflection is always a whole, whether this be loose and changing, or firm and lasting. If the object were an absolute single, thought would become one with a staring and lead to ecstasy. Even in the moment consciousness awakens from

the unconscious by the abolition of the equilibrium that the unconscious presupposes, the state is determined by the opposition to the preceding state and constitutes a whole with this. Phenomena of consciousness are always determined by a relation between background and foreground, a relation that can be successive or simultaneous, or both. In the simplest case it is only the background that determines the foreground; but when the background does not immediately vanish, but is preserved or recalled, it can itself in turn undergo change, become foreground in relation to what was before foreground. In all sensation, memory, and imagination this reciprocal relation lays itself bare. Especially clearly, however, it comes forth, as we have seen (33), in judgment, where the predicate not only determines the subject, but is also itself determined by it. In all judging, as well as in sensation, intuiting, and association, the psychic energy makes itself known as a combining, and the law of synthesis therefore becomes the fundamental law of cognition. *Synthesis* becomes the first category. It is applied involuntarily as long as consciousness subsists, and it is in analogy with it that we apprehend everything. The type for all cognition, and therefore also for everything cognized, is connection of a multiplicity into unity. Neither a wholly connectionless multiplicity nor a differenceless unity (singleness) can be cognized. Already purely psychologically, states of these kinds are impossible, although there can be strong approximations to them.

In this first category now lies at once contained the second. When a multiplicity of objects or elements is connected or combined, they are brought into relation with one another. This happens in every sensation, every memory, and every imagination, as well as in every judgment. When a definite connection is dissolved, as when intuiting is replaced by judging, this dissolution is only a conversion into other relations. The wholly relationless can be neither sensed, remembered, represented, nor judged. To think is, as *Fichte* has rightly said, to set something in relation to another.

Relation, relational determinacy, then becomes the second category. It denotes both something that appears involuntarily and an ideal. Involuntarily, in a consciousness all objects and elements are set in a certain relation to one another. But our reflection seeks to determine this relation as exactly as possible and to reshape it in such a way that all objects, earlier emerged and possible new ones, can come into their own. Wherein this "own" consists is decided by the more special categories.

The often misunderstood proposition that everything is relative does not mean that everything is equally right or equally wrong, but precisely the opposite, that everything that is cognized stands and must stand in definite relations, and is known the better the more exactly these objects are determined. Within the definite relation, cognition is absolute.

59. There are, as said, two limits for consciousness and therefore also for thinking, absolute unity and connectionless multiplicity. At neither of these limits does any relation hold, and therefore no synthesis either. Between both limits lie relations that can be designated as *Continuity and Discontinuity.*

These two categories must be named together, since they mutually presuppose each other. An abolition of equilibrium is required for consciousness to replace unconsciousness—for an intuition to be able to be articulated (17)—for judging to replace intuition and association. An interruption is required, both for the simplest sensation to come to be, and for reflection to be awakened. So far *Søren Kierkegaard* is right that thought begins with a leap, and that a special interest is required to make this leap; in any case the presupposition is an involuntary interest in preserving or restoring the equilibrium and the connection. This leap expresses a discontinuity; but it reveals at the same time that there was previously a continuity, and it may perhaps itself later show itself to be a member in a deeper-lying continuity.

It might perhaps be disputed that continuity is in itself a relation. For insofar as it subsists, no distinction can be made between members that could stand in relation to one another. When it prevails, consciousness glides imperceptibly forward, seems constantly identical with itself, and it is only the reflection awakened by discontinuity that becomes clear that changes have uninterruptedly taken place, even if these were so small that they could be set as near to unchangeableness as one likes. One can say with *Poincaré* that in all continuity a contradiction is contained. If a continuous series of the members A B C subsists, then $A = B$ and $B = C$, but $A \gtrless C$[82]. But this contradiction is first discovered when experience or memory by a leap leads to an immediate juxtaposition of A and C. Then comes the awakening. One can then ask: "whither are we gliding?" As long as one is absorbed in the gliding, one does not ask; the contradiction does not exist as long as continuity prevails. Continuity is a relation that is not discovered until it is past. It is only when discontinuity has come into force that one can apprehend continuity as a connection of objects or elements that lie infinitely near one another.

We stand here again at the earlier-mentioned psychological antinomy. When we say that in continuity, in the continuous states of consciousness, a multiplicity of members is contained, this can hold only for reflection, which looks back and makes distinctions. In the states themselves the multiplicity did not come forth as such. Both in our apprehension of sensation, of intuiting, and of thinking, this must indeed be noted, so that

[82] *La Science et l'Hypothèse.* p. 35.

one does not read something into the more immediate and involuntary states that first comes forth in reflection. In any case one has no right to identify without further ado the differences that reflection sets up with those that can already be said to "lie in" the immediate processes. (Cf. 5.) Or, as one can also express it, the rational continuum, the continuum formed by reflection, is different from the empirical continuum. When reflection is awakened, a closer determination is sought of the empirical continuum that the involuntary mental life constitutes, and this can only take place by apprehending it as discontinuous multiplicity. Discontinuity is thus not merely a condition for the awakening of reflection; but we seek, when reflection is awakened, to apprehend continuity in analogy with discontinuity. And only now can one define continuity positively; at first it stands as a negative concept, as absence of multiplicity as well as of unity. But one now assumes within continuity differences of a lower degree than the clear oppositions that discontinuity presupposes. Often great problems are hereby set. Thus when a new mental peculiarity, a new psychic tendency, suddenly reveals itself: can there previously have been such a continuum as one formerly assumed? In the material domain the discovery of radioactivity, in the biological domain the demonstration of mutations, leads to analogous questions.

There is, however, no ground to deny that discontinuity is just as original a property of conscious life as continuity. It would be incorrect, with some philosophers in our day (cf. 3), to conceive discontinuity as a more or less artificial result of attention and reflection, set in motion by practical motives. Consciousness subsists under a constant breaking of continuity and discontinuity, and this holds especially for the life of thought. The experience of difference, opposition, and interruption is just as elementary as the experience of the gliding transitions. And the experience of discontinuity is indeed, biologically-practically regarded, even a more significant experience, since it is only by an awakening that energy can be released. Experience shows us strife and tension between different tendencies in conscious life, no less than harmony and continuity, and it is only thereby that the psychic energy gets its tasks, its resistance to overcome. The differences belong to life, not merely to reflection. And since they present the clearest relations, reflection uses them to make the continuous states and objects intelligible to itself. There expresses itself here a feature that constantly recurs in our cognition—that a part or side in our experience is used to illuminate other parts or sides by way of analogy.

Conversely, continuity too is then used to illuminate discontinuity. Given differences, oppositions, and relations of tension are made distinct by the finding-out or assumption of intermediate links and transitions in as great a number as possible, so that the differences finally become infinitely small. Objects or elements that stand in pronounced opposition to one another are then sought to be apprehended as extreme points of connected series. The very impulse or leap with which both sensation and thinking begin is then sought to be brought into a relation of continuity to other objects and elements in conscious life or outside it. The apparently accidental thing in the starting-point thereby falls away. In this way the principle of continuity has constantly shown itself, in the history of science, as a great principle of discovery, in that one tries to give a once-discovered relation as great an extension as possible. The scientific imagination works according to this principle until reflection anew demonstrates discontinuity. Thus Newton asked whether the force that lets the stone fall to the earth should not also operate on the moon, and further extended this question to hold for all bodies in relation to one another. In all representations and inferences concerning the human past, life's past, nature's past as a whole, we infer from the forces and laws we now know and of whose validity we have convinced ourselves. The principle lying in this, the principle of actuality, is only a peculiar form of the principle of continuity. There makes itself known here an expansion that psychology demonstrates not only for the life of thought, but also for the life of feeling. Thus a mood called forth by a special experience will have a tendency to spread over the whole feeling-state, as long as it is not met by a mood called forth from another side.

Thinking thus gets a task, both when there are too many intermediate links, and when there are too few.

It seeks to make continuity discontinuous and discontinuity continuous, and both these apparently opposite processes stand in the closest connection with each other. In both of them thought moves forward through combining in definite relations.

60. If a closer determination of Relation is sought—besides its being able to be continuous or discontinuous—it shows itself that in the individual cases it is either a relation of likeness or a relation of difference. This is already presupposed in the foregoing. For continuity is a relation of likeness within a multiplicity that is not apprehended as multiplicity, because the difference within it is not noticed, and discontinuity presupposes a certain difference within a multiplicity.

That all thinking is a positing of relations can then be more closely expressed thus, that all thinking is a comparison. (See already 2.) It can also be said to hold for all sensation and all intuiting that it is relations of difference and likeness that are determinative for the individual objects' or elements' appearance and for the way in which they are connected. (See 11; 14; 17.) There is an analogy between these more elementary functions and judgment. But we keep from now on especially to this latter, because the relations are more easily accessible here, and because judgment is of decisive significance for the further development of cognition. Every judgment expresses both a relation of likeness and a relation of difference. Despite the difference between subject and predicate, we nonetheless have in the predicate the subject itself from a new side or in a new form. It is the relation of difference that makes the judgment possible and gives it a content; but the judgment is at the same time precisely a victory over the difference, in that this is either wholly abolished or gets a limitation.

We have earlier had occasion to mention the different kinds or degrees of likeness: identity, congruence (sameness), composite likeness, qualitative likeness, and relational likeness (analogy). (21.) The different kinds or degrees of difference we shall in what immediately follows have occasion to mention, since it is they that give occasion to the appearance of the more special categories.—

That every relation is either a relation of likeness or of difference cannot be proved in any other way than by one's finding in each single case, by analysis, that the said relations lie at the basis. It is not, however, conceded from all sides that all combining and positing of relations is a comparison. Thus Th. *Lipps*[83] maintains that a distinction must be made between combination-judgments and comparison-judgments: in the former a multiplicity is connected into a whole; in the latter different objects are compared among themselves. A comparison-judgment we have when it is said that the one of two tones is higher than the other. A combination-judgment is, e.g., the judgment: "this is a tree." In this judgment, says Lipps, "the given manifold—'this'—is by the name tree combined into an objectively necessary unity: this naming-judgment includes in itself a consciousness that, when this designation is to hold, I must think that manifold as one."—To this it must in the first place be remarked that only the different can be called manifold and connected into a whole. If the elements in the given object were absolutely identical in quality, time, and place, they would not be manifold: the object would then be an absolute unity. The relation of difference is thus presupposed. Next, there lies in the naming itself a presupposition of likeness between the object ("this") and other objects, and it is this likeness that the predicate expresses. Finally, the object, in order for its elements to be able to be gathered under a common predicate, must already be distinguished from other objects as something apart (cf. 30 ff.). Here again the relation of difference lies at the basis.

The breaking between unity and multiplicity will always show itself to be a breaking

[83] *Grundzüge der Logik.* (1893). p. 93–95.

between likenesses and differences. In this breaking there constantly intervenes the interest or the purpose that guides thought. It cooperates in the individual cases in determining which group is formed—which object is separated out to be determined—hence which difference is made; and it cooperates also in the point of view that is taken toward the object. The separated-out object (the group) corresponds to the judgment's subject, the point of view to its predicate. Instead of saying "this is a tree," I could under other conditions, toward the same "this," come to say "this is hard," or "this is something I must go around." And likewise I could under other conditions have formed another group, so that "this," e.g., became the tree with the earth it grows in, with the birds that build a nest in it, the insects in its bark, and so on. The object then became a whole little world. That already the judgment's subject presupposes a distinguishing is easily overlooked, since it can stand in a certain indefiniteness—since it is presupposed known and therefore not always expressed—and since the distinguishing act is undertaken beforehand and perhaps wholly involuntarily.

61. We involuntarily order given multiplicities according to likenesses and differences, and this naturally becomes the first step in our cognition. The question is what a complete ordering would be, and whether such is always possible. Here, in the doctrine of categories as descriptive epistemology, we have to do only with the possible orderings; the question of validity belongs under the problem of cognition.

Instead of the word ordering I shall in what follows for simplicity's sake use the word series, although this word is as a rule used in logic in a narrower sense, namely of such orderings as make inference and calculation possible (cf. 33). Epistemology must investigate whether there are not other possibilities than those that make science possible. It is no matter of course that everywhere, at all times, and in all domains, orderings can be formed that make inference and thereby science possible. We have no right to establish in advance the rationality of objects. It is, as we have seen (35; 52), not certain that thought can exhaust the whole content of sensation, intuition, and association in the clear form of judgment. And there is also no certainty that our sensations and our memories are exhaustive. There is every ground to believe that they are always fragmentary. Our objects are always limited. Already for this reason orderings or series of an imperfect kind could be thought of, at which thought must nonetheless stop. The doctrine of categories then needs an investigation, an orientation with respect to the in-themselves-possible series of difference and likeness.

When I try to order a multiplicity of objects or elements according to likeness and difference, it shows itself that the members in the series that is formed can stand in very different relations to one another. There are here especially three relations that become of significance. There can possibly be formed a series of such a constitution that any member whatever can be pushed out, and that the found relation of likeness or difference nonetheless continues to hold between the remaining members. Such a series *Augustus de Morgan*[84] called a *transitive* series. *William James* has later called it a series that makes mediate comparison possible[85]. And there can possibly be formed a series of such a constitution that any member whatever in it can be exchanged with its neighbor-member without the relation of likeness or difference thereby being changed. Such a series I will call *convertible*. In a transitive series ABCD, B can fall away, and the relation between A and C is then the same as between A and B, and as between B and C. In a convertible series ABCD … B and C, e.g., can be exchanged, and the relation between A and C, C and B, B and D is then the same as between A and B, B and C, C and D. A convertible series (in the indicated sense of the word) must at the same time be transitive. The exchange operates in part as a pushing-out. Finally, a series can be formed in which the relation continues to be the same, whether one goes through the series from the front or from behind. Such a series can be called *symmetric*. The series ABCD… is symmetric when the relation between D and C is the same as between C and D, likewise the relation between C and B the same as between B and C, and so on. Such a series can be intransitive, namely when neighbor-members cannot be exchanged. Because the relation D–C is the same as the relation C–D, it is not said that I can put D in place of C, since it depends on how the relation between B and D is. In an intransitive series every member is determined by its relation to other members, and exchange is therefore not possible. While a symmetric series can be both transitive and intransitive, a convertible series can only be transitive.

De Morgan did not distinguish between convertible and symmetric series. And neither later has that distinction, as far as I know, been made. One has as a rule used the expressions synonymously and thought only of whether the whole series could be reversed, not of whether the members in the series could be reversed among themselves.

Finally, it will show itself that a series can often be read from behind, when one constantly reverses the relation. It is an inversely symmetric series.—

I now go through these possibilities in detail and seek to find examples of them. We shall thereby come upon more special kinds of relations of likeness and difference and

[84] *Cambridge Philosophical Transactions* IX. (1860). p. 104.

[85] *Principles of Psychology*. II. p. 646: The principle of mediate comparison is only one form of a law which holds in many series of homogeneously related terms, the law that skipping intermediary terms leaves relation the same. Cf. Vol. I. p. 190.

thus be able to make the transition from the fundamental categories to the more special ones.

C. Formal Categories

a. Identity

b. Quality-Relations

1) Time

2) Number

3) Degree

4) Place

c. Negation

d. Rationality

D. Real Categories

a. Causality

b. Totality

c. Development

E. Ideal Categories

1) Formal Differences

2) Real Differences

IV. The Tasks of Thought (Problems)

Introduction

A. Cognition

a. Forms and Principles

b. Forms and Objects

α. Quality and Quantity

β. Succession and Causality

γ. Subject and Object

B. World-View

a. Science and World-View

b. Unity and Multiplicity

c. Spirit and Matter

d. Subsistence and Development

C. Valuation

a. The Ethical Problem

α. *Ethical Work.* (Formal Ethics)

β. *The Rationality of Ethical Valuation.* (Real Ethics)

b. The Religious Problem

α. *The Psychological Place of Religion*

β. *The Historical and the Philosophical Consideration*